All-phase Clifford atlas

24 projective classes / eight exact scalar phases / each distinct local alphabet

Each alphabet occupies eight pages. A page fixes k and lists all 24 operators $e^{ik\pi/4}C_j$. The representative matrices C_j are the exact named representatives in the catalog. Every displayed circuit equals its target literally; no scalar-phase repair or physical GP is included. The names identify actual catalog matrices, including the legacy distinction between SX and square-root X (Sx).
Shortest-path costs use the audited parameter-sensitive metric with nonnegative denominator penalties. A fixed Clifford alphabet such as H/S exhausts its allowed words. H/S/T/Tdg also has an exhaustive exact-word proof, including non-Clifford intermediate paths. For continuous alphabets the certificate covers the listed specializations and macro words. Missing paths retain that scope.
All presets with the same one-qubit alphabet share the same table. Entanglers and ancillary qubits are unavailable in this one-wire reference.

Exact representative matrices p. 2
GPi|GPi2 p. 3
H|P p. 11
H|S p. 19
H|S|T|Tdg p. 27
PhasedXZ p. 35
PhasedX|Rz p. 43
R p. 51
Rz|Rx p. 59
Rz|Ry p. 67
Rz|Sx|X p. 75
R|Rz p. 83
U3 p. 91

The 24 matrix representatives

These exact matrices fix the global-phase convention for every table.

C_0 / **I**

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

C_1 / **X**

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

C_2 / **Y**

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

C_3 / **Z**

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

C_4 / **H**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

C_5 / **S**

$$\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

C_6 / **Sdg**

$$\begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}$$

C_7 / **Sx**

$$\begin{pmatrix} \frac{1+i}{2} & \frac{1-i}{2} \\ \frac{1-i}{2} & \frac{1+i}{2} \end{pmatrix}$$

C_8 / **Sxdg**

$$\begin{pmatrix} \frac{1-i}{2} & \frac{1+i}{2} \\ \frac{1+i}{2} & \frac{1-i}{2} \end{pmatrix}$$

C_9 / **HX**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

C_{10} / **HY_q2**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

C_{11} / **HZ**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

C_{12} / **SX_q3**

$$\begin{pmatrix} 0 & -\frac{\sqrt{2}(1+i)}{2} \\ \frac{\sqrt{2}(-1+i)}{2} & 0 \end{pmatrix}$$

C_{13} / **SY_q3**

$$\begin{pmatrix} 0 & \frac{\sqrt{2}(-1+i)}{2} \\ -\frac{\sqrt{2}(1+i)}{2} & 0 \end{pmatrix}$$

C_{14} / **HS**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}i}{2} & -\frac{\sqrt{2}i}{2} \end{pmatrix}$$

C_{15} / **HSX_q6**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}i}{2} & -\frac{\sqrt{2}i}{2} \end{pmatrix}$$

C_{16} / **HSY**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}i}{2} & \frac{\sqrt{2}i}{2} \end{pmatrix}$$

C_{17} / **HSdg**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}i}{2} & \frac{\sqrt{2}i}{2} \end{pmatrix}$$

C_{18} / **SH**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}i}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}i}{2} \end{pmatrix}$$

C_{19} / **SHX**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}i}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}i}{2} \end{pmatrix}$$

C_{20} / **SHY_q2**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}i}{2} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}i}{2} \end{pmatrix}$$

C_{21} / **SHZ**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}i}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}i}{2} \end{pmatrix}$$

C_{22} / **SxY_q3**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}i}{2} \\ -\frac{\sqrt{2}i}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

C_{23} / **SxZ_q3**

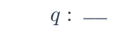
$$\begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}i}{2} \\ -\frac{\sqrt{2}i}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

GPi|GPi2 / $k = 0$

ionq_aria, ionq_aria_full, ionq_forte

$e^{i0\pi/4}$ I

Score 0 / 0 gates



$e^{i0\pi/4}$ X

Score $\frac{1001}{500}$ / 1 gates



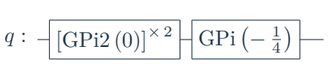
$e^{i0\pi/4}$ Y

Score $\frac{1201}{500}$ / 1 gates



$e^{i0\pi/4}$ Z

Score $\frac{3203}{500}$ / 3 gates



$e^{i0\pi/4}$ H

Score $\frac{1101}{250}$ / 2 gates



$e^{i0\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

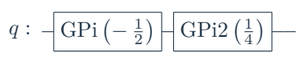
$e^{i0\pi/4}$ HX

Score $\frac{1201}{500}$ / 1 gates



$e^{i0\pi/4}$ HY_q2

Score $\frac{1151}{250}$ / 2 gates



$e^{i0\pi/4}$ HZ

Score $\frac{1201}{500}$ / 1 gates



$e^{i0\pi/4}$ SX_q3

Score $\frac{1301}{500}$ / 1 gates



$e^{i0\pi/4}$ SY_q3

Score $\frac{1301}{500}$ / 1 gates



$e^{i0\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SHY_q2

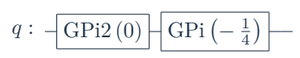
Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

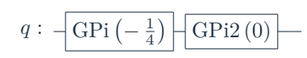
$e^{i0\pi/4}$ SxY_q3

Score $\frac{1101}{250}$ / 2 gates



$e^{i0\pi/4}$ SxZ_q3

Score $\frac{1101}{250}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

GPi|GPi2 / $k = 1$

ionq_aria, ionq_aria_full, ionq_forte

$e^{i1\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Z

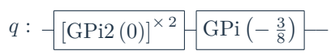
Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ S

Score $\frac{3303}{500}$ / 3 gates



$e^{i1\pi/4}$ Sdg

Score $\frac{1151}{250}$ / 2 gates



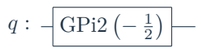
$e^{i1\pi/4}$ Sx

Score $\frac{1051}{250}$ / 2 gates



$e^{i1\pi/4}$ Sxdg

Score $\frac{1101}{500}$ / 1 gates



$e^{i1\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HS

Score $\frac{1151}{250}$ / 2 gates



$e^{i1\pi/4}$ HSX_q6

Score $\frac{1101}{250}$ / 2 gates



$e^{i1\pi/4}$ HSY

Score $\frac{1151}{250}$ / 2 gates



$e^{i1\pi/4}$ HSdg

Score $\frac{1201}{250}$ / 2 gates



$e^{i1\pi/4}$ SH

Score $\frac{1151}{250}$ / 2 gates



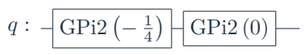
$e^{i1\pi/4}$ SHX

Score $\frac{1201}{250}$ / 2 gates



$e^{i1\pi/4}$ SHY_q2

Score $\frac{1101}{250}$ / 2 gates



$e^{i1\pi/4}$ SHZ

Score $\frac{1151}{250}$ / 2 gates



$e^{i1\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

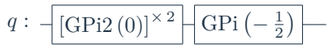
Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

GPi|GPi2 / $k = 2$

ionq_aria, ionq_aria_full, ionq_forte

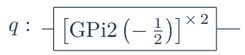
$e^{i2\pi/4}$ I

Score $\frac{3103}{500}$ / 3 gates



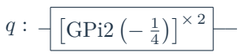
$e^{i2\pi/4}$ X

Score $\frac{1101}{250}$ / 2 gates



$e^{i2\pi/4}$ Y

Score $\frac{1201}{250}$ / 2 gates



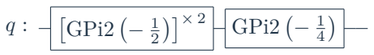
$e^{i2\pi/4}$ Z

Score $\frac{1101}{250}$ / 2 gates



$e^{i2\pi/4}$ H

Score $\frac{3403}{500}$ / 3 gates



$e^{i2\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

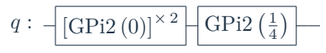
$e^{i2\pi/4}$ HX

Score $\frac{1201}{250}$ / 2 gates



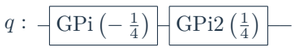
$e^{i2\pi/4}$ HY_q2

Score $\frac{3203}{500}$ / 3 gates



$e^{i2\pi/4}$ HZ

Score $\frac{1201}{250}$ / 2 gates



$e^{i2\pi/4}$ SX_q3

Score $\frac{3203}{500}$ / 3 gates



$e^{i2\pi/4}$ SY_q3

Score $\frac{3203}{500}$ / 3 gates



$e^{i2\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SxY_q3

Score $\frac{3203}{500}$ / 3 gates



$e^{i2\pi/4}$ SxZ_q3

Score $\frac{3203}{500}$ / 3 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

GPi|GPi2 / $k = 3$

ionq_aria, ionq_aria_full, ionq_forte

$e^{i3\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ H

Impossible: phase outside the local determinant group.

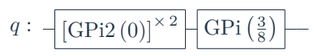
$e^{i3\pi/4}$ S

Score $\frac{1151}{250}$ / 2 gates



$e^{i3\pi/4}$ Sdg

Score $\frac{3303}{500}$ / 3 gates



$e^{i3\pi/4}$ Sx

Score $\frac{3103}{500}$ / 3 gates



$e^{i3\pi/4}$ Sxdg

Score $\frac{1051}{250}$ / 2 gates



$e^{i3\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SX_q3

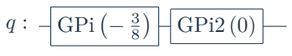
Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HS

Score $\frac{1151}{250}$ / 2 gates



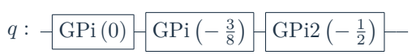
$e^{i3\pi/4}$ HSX_q6

Score $\frac{1201}{250}$ / 2 gates



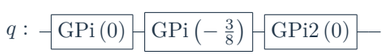
$e^{i3\pi/4}$ HSY

Score $\frac{3403}{500}$ / 3 gates



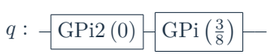
$e^{i3\pi/4}$ HSdg

Score $\frac{3303}{500}$ / 3 gates



$e^{i3\pi/4}$ SH

Score $\frac{1151}{250}$ / 2 gates



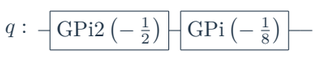
$e^{i3\pi/4}$ SHX

Score $\frac{3303}{500}$ / 3 gates



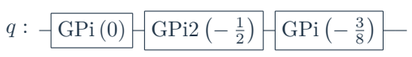
$e^{i3\pi/4}$ SHY_q2

Score $\frac{1201}{250}$ / 2 gates



$e^{i3\pi/4}$ SHZ

Score $\frac{3403}{500}$ / 3 gates



$e^{i3\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

GPi|GPi2 / $k = 4$

ionq_aria, ionq_aria_full, ionq_forte

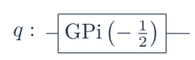
$e^{i4\pi/4}$ I

Score $\frac{1051}{250}$ / 2 gates



$e^{i4\pi/4}$ X

Score $\frac{1101}{500}$ / 1 gates



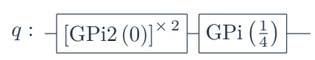
$e^{i4\pi/4}$ Y

Score $\frac{1201}{500}$ / 1 gates



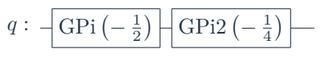
$e^{i4\pi/4}$ Z

Score $\frac{3203}{500}$ / 3 gates



$e^{i4\pi/4}$ H

Score $\frac{1151}{250}$ / 2 gates



$e^{i4\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Sx

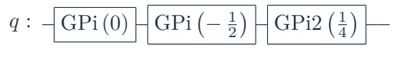
Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

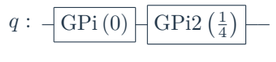
$e^{i4\pi/4}$ HX

Score $\frac{3303}{500}$ / 3 gates



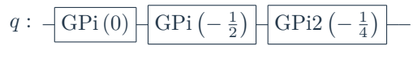
$e^{i4\pi/4}$ HY_q2

Score $\frac{1101}{250}$ / 2 gates



$e^{i4\pi/4}$ HZ

Score $\frac{3303}{500}$ / 3 gates



$e^{i4\pi/4}$ SX_q3

Score $\frac{1301}{500}$ / 1 gates



$e^{i4\pi/4}$ SY_q3

Score $\frac{1301}{500}$ / 1 gates



$e^{i4\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SHY_q2

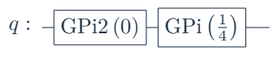
Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

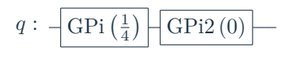
$e^{i4\pi/4}$ SxY_q3

Score $\frac{1101}{250}$ / 2 gates



$e^{i4\pi/4}$ SxZ_q3

Score $\frac{1101}{250}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

GPi|GPi2 / $k = 5$

ionq_aria, ionq_aria_full, ionq_forte

$e^{i5\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Z

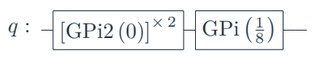
Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ S

Score $\frac{3303}{500}$ / 3 gates



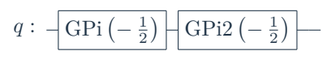
$e^{i5\pi/4}$ Sdg

Score $\frac{1151}{250}$ / 2 gates



$e^{i5\pi/4}$ Sx

Score $\frac{1101}{250}$ / 2 gates



$e^{i5\pi/4}$ Sxdg

Score $\frac{3003}{500}$ / 3 gates



$e^{i5\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SX_q3

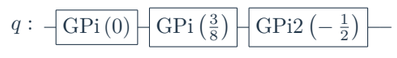
Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HS

Score $\frac{3403}{500}$ / 3 gates



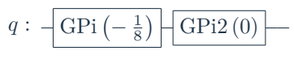
$e^{i5\pi/4}$ HSX_q6

Score $\frac{3303}{500}$ / 3 gates



$e^{i5\pi/4}$ HSY

Score $\frac{1151}{250}$ / 2 gates



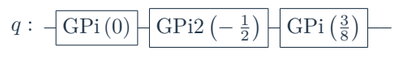
$e^{i5\pi/4}$ HSdg

Score $\frac{1201}{250}$ / 2 gates



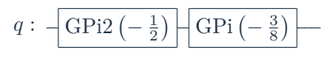
$e^{i5\pi/4}$ SH

Score $\frac{3403}{500}$ / 3 gates



$e^{i5\pi/4}$ SHX

Score $\frac{1201}{250}$ / 2 gates



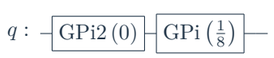
$e^{i5\pi/4}$ SHY_q2

Score $\frac{3303}{500}$ / 3 gates



$e^{i5\pi/4}$ SHZ

Score $\frac{1151}{250}$ / 2 gates



$e^{i5\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

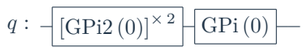
Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

GPi|GPi2 / $k = 6$

ionq_aria, ionq_aria_full, ionq_forte

$e^{i6\pi/4}$ I

Score $\frac{3003}{500}$ / 3 gates



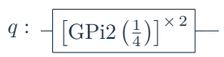
$e^{i6\pi/4}$ X

Score $\frac{1001}{250}$ / 2 gates



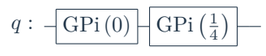
$e^{i6\pi/4}$ Y

Score $\frac{1201}{250}$ / 2 gates



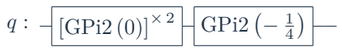
$e^{i6\pi/4}$ Z

Score $\frac{1101}{250}$ / 2 gates



$e^{i6\pi/4}$ H

Score $\frac{3203}{500}$ / 3 gates



$e^{i6\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ Sx

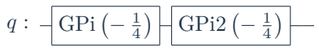
Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

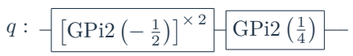
$e^{i6\pi/4}$ HX

Score $\frac{1201}{250}$ / 2 gates



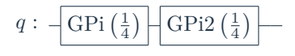
$e^{i6\pi/4}$ HY_q2

Score $\frac{3403}{500}$ / 3 gates



$e^{i6\pi/4}$ HZ

Score $\frac{1201}{250}$ / 2 gates



$e^{i6\pi/4}$ SX_q3

Score $\frac{3403}{500}$ / 3 gates



$e^{i6\pi/4}$ SY_q3

Score $\frac{3403}{500}$ / 3 gates



$e^{i6\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SxY_q3

Score $\frac{3203}{500}$ / 3 gates



$e^{i6\pi/4}$ SxZ_q3

Score $\frac{3203}{500}$ / 3 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

GPi|GPi2 / $k = 7$

ionq_aria, ionq_aria_full, ionq_forte

$e^{i7\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ S

Score $\frac{1151}{250}$ / 2 gates

$$q: \boxed{\mathrm{GPi}\,(0)}\boxed{\mathrm{GPi}\left(\tfrac{1}{8}\right)}$$

$e^{i7\pi/4}$ Sdg

Score $\frac{3303}{500}$ / 3 gates

$$q: \boxed{[\mathrm{GPi2}\,(0)]^{\times 2}}\boxed{\mathrm{GPi}\left(-\tfrac{1}{8}\right)}$$

$e^{i7\pi/4}$ Sx

Score $\frac{1001}{500}$ / 1 gates

$$q: \boxed{\mathrm{GPi2}\,(0)}$$

$e^{i7\pi/4}$ Sxdg

Score $\frac{1001}{250}$ / 2 gates

$$q: \boxed{\mathrm{GPi}\,(0)}\boxed{\mathrm{GPi2}\,(0)}$$

$e^{i7\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HS

Score $\frac{1151}{250}$ / 2 gates

$$q: \boxed{\mathrm{GPi}\left(\tfrac{1}{8}\right)}\boxed{\mathrm{GPi2}\,(0)}$$

$e^{i7\pi/4}$ HSX_q6

Score $\frac{1201}{250}$ / 2 gates

$$q: \boxed{\mathrm{GPi}\left(-\tfrac{3}{8}\right)}\boxed{\mathrm{GPi2}\left(-\tfrac{1}{2}\right)}$$

$e^{i7\pi/4}$ HSY

Score $\frac{1151}{250}$ / 2 gates

$$q: \boxed{\mathrm{GPi2}\left(-\tfrac{1}{2}\right)}\boxed{\mathrm{GPi2}\left(\tfrac{1}{4}\right)}$$

$e^{i7\pi/4}$ HSdg

Score $\frac{1101}{250}$ / 2 gates

$$q: \boxed{\mathrm{GPi2}\,(0)}\boxed{\mathrm{GPi2}\left(-\tfrac{1}{4}\right)}$$

$e^{i7\pi/4}$ SH

Score $\frac{1151}{250}$ / 2 gates

$$q: \boxed{\mathrm{GPi2}\,(0)}\boxed{\mathrm{GPi}\left(-\tfrac{1}{8}\right)}$$

$e^{i7\pi/4}$ SHX

Score $\frac{1101}{250}$ / 2 gates

$$q: \boxed{\mathrm{GPi2}\left(\tfrac{1}{4}\right)}\boxed{\mathrm{GPi2}\,(0)}$$

$e^{i7\pi/4}$ SHY_q2

Score $\frac{1201}{250}$ / 2 gates

$$q: \boxed{\mathrm{GPi2}\left(-\tfrac{1}{2}\right)}\boxed{\mathrm{GPi}\left(\tfrac{3}{8}\right)}$$

$e^{i7\pi/4}$ SHZ

Score $\frac{1151}{250}$ / 2 gates

$$q: \boxed{\mathrm{GPi2}\left(-\tfrac{1}{4}\right)}\boxed{\mathrm{GPi2}\left(-\tfrac{1}{2}\right)}$$

$e^{i7\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

H|P / $k = 0$

cx_h_phase

$e^{i0\pi/4}$ **I**

Score 0 / 0 gates



$e^{i0\pi/4}$ **X**

Score $\frac{3001}{500}$ / 3 gates



$e^{i0\pi/4}$ **Y**

Score $\frac{5203}{500}$ / 5 gates



$e^{i0\pi/4}$ **Z**

Score $\frac{1001}{500}$ / 1 gates



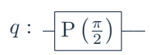
$e^{i0\pi/4}$ **H**

Score 2 / 1 gates



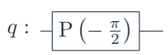
$e^{i0\pi/4}$ **S**

Score $\frac{1101}{500}$ / 1 gates



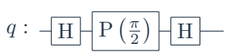
$e^{i0\pi/4}$ **Sdg**

Score $\frac{1101}{500}$ / 1 gates



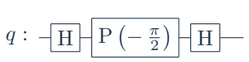
$e^{i0\pi/4}$ **Sx**

Score $\frac{3101}{500}$ / 3 gates



$e^{i0\pi/4}$ **Sxdg**

Score $\frac{3101}{500}$ / 3 gates



$e^{i0\pi/4}$ **HX**

Score $\frac{2001}{500}$ / 2 gates



$e^{i0\pi/4}$ **HY_q2**

Score $\frac{1501}{250}$ / 3 gates



$e^{i0\pi/4}$ **HZ**

Score $\frac{2001}{500}$ / 2 gates



$e^{i0\pi/4}$ **SX_q3**

Score $\frac{5403}{500}$ / 5 gates



$e^{i0\pi/4}$ **SY_q3**

Score $\frac{5403}{500}$ / 5 gates



$e^{i0\pi/4}$ **HS**

Score $\frac{2101}{500}$ / 2 gates



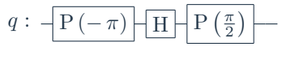
$e^{i0\pi/4}$ **HSX_q6**

Score $\frac{1551}{250}$ / 3 gates



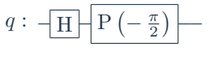
$e^{i0\pi/4}$ **HSY**

Score $\frac{1551}{250}$ / 3 gates



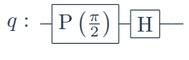
$e^{i0\pi/4}$ **HSdg**

Score $\frac{2101}{500}$ / 2 gates



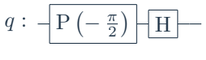
$e^{i0\pi/4}$ **SH**

Score $\frac{2101}{500}$ / 2 gates



$e^{i0\pi/4}$ **SHX**

Score $\frac{2101}{500}$ / 2 gates



$e^{i0\pi/4}$ **SHY_q2**

Score $\frac{1551}{250}$ / 3 gates



$e^{i0\pi/4}$ **SHZ**

Score $\frac{1551}{250}$ / 3 gates



$e^{i0\pi/4}$ **SxY_q3**

Score $\frac{1601}{250}$ / 3 gates



$e^{i0\pi/4}$ **SxZ_q3**

Score $\frac{1801}{125}$ / 7 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

H|P / $k = 1$

cx_h_phase

$e^{i1\pi/4}$ I

Score $\frac{6303}{500}$ / 6 gates



$e^{i1\pi/4}$ X

Score $\frac{5403}{500}$ / 5 gates



$e^{i1\pi/4}$ Y

Score $\frac{5403}{500}$ / 5 gates



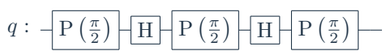
$e^{i1\pi/4}$ Z

Score $\frac{6303}{500}$ / 6 gates



$e^{i1\pi/4}$ H

Score $\frac{5303}{500}$ / 5 gates



$e^{i1\pi/4}$ S

Score $\frac{6203}{500}$ / 6 gates



$e^{i1\pi/4}$ Sdg

Score $\frac{2601}{250}$ / 5 gates



$e^{i1\pi/4}$ Sx

Score $\frac{6203}{500}$ / 6 gates



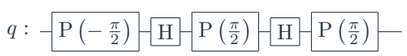
$e^{i1\pi/4}$ Sxdg

Score $\frac{1601}{250}$ / 3 gates



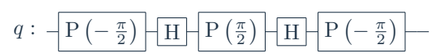
$e^{i1\pi/4}$ HX

Score $\frac{5303}{500}$ / 5 gates



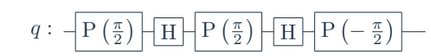
$e^{i1\pi/4}$ HY_q2

Score $\frac{5303}{500}$ / 5 gates



$e^{i1\pi/4}$ HZ

Score $\frac{5303}{500}$ / 5 gates



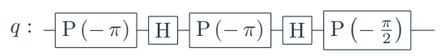
$e^{i1\pi/4}$ SX_q3

Score $\frac{5103}{500}$ / 5 gates



$e^{i1\pi/4}$ SY_q3

Score $\frac{5103}{500}$ / 5 gates



$e^{i1\pi/4}$ HS

Score $\frac{5203}{500}$ / 5 gates



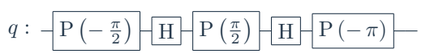
$e^{i1\pi/4}$ HSX_q6

Score $\frac{2101}{250}$ / 4 gates



$e^{i1\pi/4}$ HSY

Score $\frac{5203}{500}$ / 5 gates



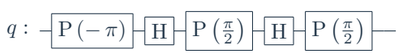
$e^{i1\pi/4}$ HSdg

Score $\frac{2101}{250}$ / 4 gates



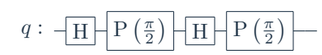
$e^{i1\pi/4}$ SH

Score $\frac{5203}{500}$ / 5 gates



$e^{i1\pi/4}$ SHX

Score $\frac{2101}{250}$ / 4 gates



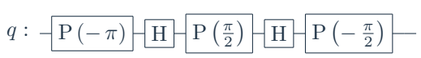
$e^{i1\pi/4}$ SHY_q2

Score $\frac{2101}{250}$ / 4 gates



$e^{i1\pi/4}$ SHZ

Score $\frac{5203}{500}$ / 5 gates



$e^{i1\pi/4}$ SxY_q3

Score $\frac{2051}{250}$ / 4 gates



$e^{i1\pi/4}$ SxZ_q3

Score $\frac{7103}{500}$ / 7 gates



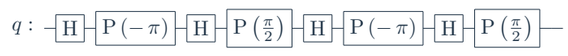
Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

H|P / $k = 2$

cx_h_phase

$e^{i2\pi/4}$ I

Score $\frac{2051}{125}$ / 8 gates



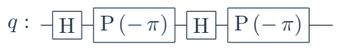
$e^{i2\pi/4}$ X

Score $\frac{5203}{500}$ / 5 gates



$e^{i2\pi/4}$ Y

Score $\frac{2001}{250}$ / 4 gates



$e^{i2\pi/4}$ Z

Score $\frac{7203}{500}$ / 7 gates



$e^{i2\pi/4}$ H

Score $\frac{1801}{125}$ / 7 gates



$e^{i2\pi/4}$ S

Score $\frac{2026}{125}$ / 8 gates



$e^{i2\pi/4}$ Sdg

Score $\frac{7103}{500}$ / 7 gates



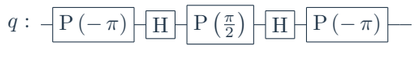
$e^{i2\pi/4}$ Sx

Score $\frac{1876}{125}$ / 7 gates



$e^{i2\pi/4}$ Sxdg

Score $\frac{5103}{500}$ / 5 gates



$e^{i2\pi/4}$ HX

Score $\frac{6203}{500}$ / 6 gates



$e^{i2\pi/4}$ HY_q2

Score $\frac{6203}{500}$ / 6 gates



$e^{i2\pi/4}$ HZ

Score $\frac{6203}{500}$ / 6 gates



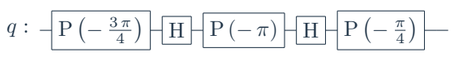
$e^{i2\pi/4}$ SX_q3

Score $\frac{5403}{500}$ / 5 gates



$e^{i2\pi/4}$ SY_q3

Score $\frac{5403}{500}$ / 5 gates



$e^{i2\pi/4}$ HS

Score $\frac{1776}{125}$ / 7 gates



$e^{i2\pi/4}$ HSX_q6

Score $\frac{2551}{250}$ / 5 gates



$e^{i2\pi/4}$ HSY

Score $\frac{6103}{500}$ / 6 gates



$e^{i2\pi/4}$ HSdg

Score $\frac{6103}{500}$ / 6 gates



$e^{i2\pi/4}$ SH

Score $\frac{1776}{125}$ / 7 gates



$e^{i2\pi/4}$ SHX

Score $\frac{6103}{500}$ / 6 gates



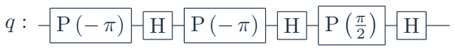
$e^{i2\pi/4}$ SHY_q2

Score $\frac{2551}{250}$ / 5 gates



$e^{i2\pi/4}$ SHZ

Score $\frac{6103}{500}$ / 6 gates



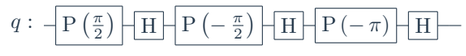
$e^{i2\pi/4}$ SxY_q3

Score $\frac{6203}{500}$ / 6 gates



$e^{i2\pi/4}$ SxZ_q3

Score $\frac{6203}{500}$ / 6 gates



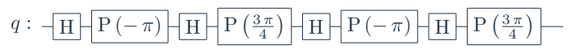
Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

H|P / $k = 3$

cx_h_phase

$e^{i3\pi/4}$ I

Score $\frac{2101}{125}$ / 8 gates



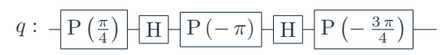
$e^{i3\pi/4}$ X

Score $\frac{5403}{500}$ / 5 gates



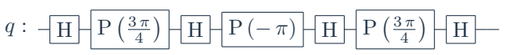
$e^{i3\pi/4}$ Y

Score $\frac{5403}{500}$ / 5 gates



$e^{i3\pi/4}$ Z

Score $\frac{7403}{500}$ / 7 gates



$e^{i3\pi/4}$ H

Score $\frac{1851}{125}$ / 7 gates



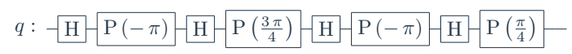
$e^{i3\pi/4}$ S

Score $\frac{2101}{125}$ / 8 gates



$e^{i3\pi/4}$ Sdg

Score $\frac{2101}{125}$ / 8 gates



$e^{i3\pi/4}$ Sx

Score $\frac{1801}{125}$ / 7 gates



$e^{i3\pi/4}$ Sxdg

Score $\frac{1801}{125}$ / 7 gates



$e^{i3\pi/4}$ HX

Score $\frac{6403}{500}$ / 6 gates



$e^{i3\pi/4}$ HY_q2

Score $\frac{6403}{500}$ / 6 gates



$e^{i3\pi/4}$ HZ

Score $\frac{6403}{500}$ / 6 gates



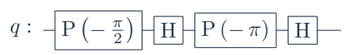
$e^{i3\pi/4}$ SX_q3

Score $\frac{2051}{250}$ / 4 gates



$e^{i3\pi/4}$ SY_q3

Score $\frac{2051}{250}$ / 4 gates



$e^{i3\pi/4}$ HS

Score $\frac{1851}{125}$ / 7 gates



$e^{i3\pi/4}$ HSX_q6

Score $\frac{6403}{500}$ / 6 gates



$e^{i3\pi/4}$ HSY

Score $\frac{6403}{500}$ / 6 gates



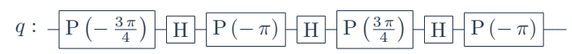
$e^{i3\pi/4}$ HSdg

Score $\frac{1851}{125}$ / 7 gates



$e^{i3\pi/4}$ SH

Score $\frac{1851}{125}$ / 7 gates



$e^{i3\pi/4}$ SHX

Score $\frac{1851}{125}$ / 7 gates



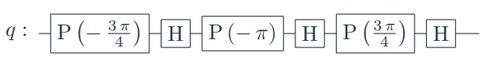
$e^{i3\pi/4}$ SHY_q2

Score $\frac{6403}{500}$ / 6 gates



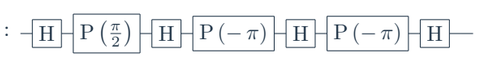
$e^{i3\pi/4}$ SHZ

Score $\frac{6403}{500}$ / 6 gates



$e^{i3\pi/4}$ SxY_q3

Score $\frac{7103}{500}$ / 7 gates



$e^{i3\pi/4}$ SxZ_q3

Score $\frac{2051}{250}$ / 4 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

H|P / $k = 4$

cx_h_phase

$e^{i4\pi/4}$ I

Score $\frac{2001}{125}$ / 8 gates



$e^{i4\pi/4}$ X

Score $\frac{5003}{500}$ / 5 gates



$e^{i4\pi/4}$ Y

Score $\frac{5203}{500}$ / 5 gates



$e^{i4\pi/4}$ Z

Score $\frac{7003}{500}$ / 7 gates



$e^{i4\pi/4}$ H

Score $\frac{1751}{125}$ / 7 gates



$e^{i4\pi/4}$ S

Score $\frac{2026}{125}$ / 8 gates



$e^{i4\pi/4}$ Sdg

Score $\frac{2026}{125}$ / 8 gates



$e^{i4\pi/4}$ Sx

Score $\frac{1876}{125}$ / 7 gates



$e^{i4\pi/4}$ Sxdg

Score $\frac{1876}{125}$ / 7 gates



$e^{i4\pi/4}$ HX

Score $\frac{6003}{500}$ / 6 gates



$e^{i4\pi/4}$ HY_q2

Score $\frac{2501}{250}$ / 5 gates



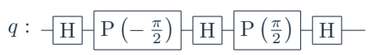
$e^{i4\pi/4}$ HZ

Score $\frac{6003}{500}$ / 6 gates



$e^{i4\pi/4}$ SX_q3

Score $\frac{2601}{250}$ / 5 gates



$e^{i4\pi/4}$ SY_q3

Score $\frac{2601}{250}$ / 5 gates



$e^{i4\pi/4}$ HS

Score $\frac{1776}{125}$ / 7 gates



$e^{i4\pi/4}$ HSX_q6

Score $\frac{6103}{500}$ / 6 gates



$e^{i4\pi/4}$ HSY

Score $\frac{6103}{500}$ / 6 gates



$e^{i4\pi/4}$ HSdg

Score $\frac{1776}{125}$ / 7 gates



$e^{i4\pi/4}$ SH

Score $\frac{1776}{125}$ / 7 gates



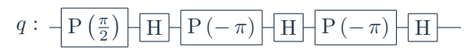
$e^{i4\pi/4}$ SHX

Score $\frac{1776}{125}$ / 7 gates



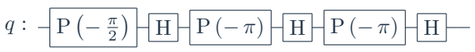
$e^{i4\pi/4}$ SHY_q2

Score $\frac{6103}{500}$ / 6 gates



$e^{i4\pi/4}$ SHZ

Score $\frac{6103}{500}$ / 6 gates



$e^{i4\pi/4}$ SxY_q3

Score $\frac{1801}{125}$ / 7 gates



$e^{i4\pi/4}$ SxZ_q3

Score $\frac{1601}{250}$ / 3 gates



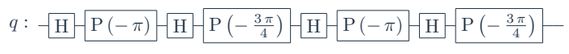
Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

H|P / $k = 5$

cx_h_phase

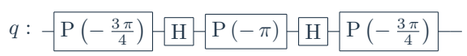
$e^{i5\pi/4}$ I

Score $\frac{2101}{125}$ / 8 gates



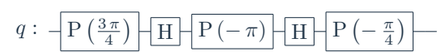
$e^{i5\pi/4}$ X

Score $\frac{5403}{500}$ / 5 gates



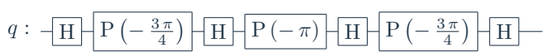
$e^{i5\pi/4}$ Y

Score $\frac{5403}{500}$ / 5 gates



$e^{i5\pi/4}$ Z

Score $\frac{7403}{500}$ / 7 gates



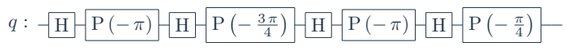
$e^{i5\pi/4}$ H

Score $\frac{1851}{125}$ / 7 gates



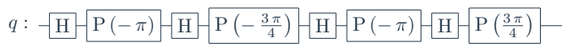
$e^{i5\pi/4}$ S

Score $\frac{2101}{125}$ / 8 gates



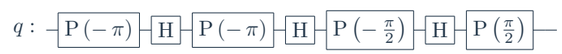
$e^{i5\pi/4}$ Sdg

Score $\frac{2101}{125}$ / 8 gates



$e^{i5\pi/4}$ Sx

Score $\frac{1801}{125}$ / 7 gates



$e^{i5\pi/4}$ Sxdg

Score $\frac{1801}{125}$ / 7 gates



$e^{i5\pi/4}$ HX

Score $\frac{6403}{500}$ / 6 gates



$e^{i5\pi/4}$ HY_q2

Score $\frac{6403}{500}$ / 6 gates



$e^{i5\pi/4}$ HZ

Score $\frac{6403}{500}$ / 6 gates



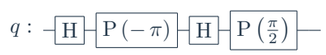
$e^{i5\pi/4}$ SX_q3

Score $\frac{2051}{250}$ / 4 gates



$e^{i5\pi/4}$ SY_q3

Score $\frac{2051}{250}$ / 4 gates



$e^{i5\pi/4}$ HS

Score $\frac{1851}{125}$ / 7 gates



$e^{i5\pi/4}$ HSX_q6

Score $\frac{6403}{500}$ / 6 gates



$e^{i5\pi/4}$ HSY

Score $\frac{6403}{500}$ / 6 gates



$e^{i5\pi/4}$ HSdg

Score $\frac{1851}{125}$ / 7 gates



$e^{i5\pi/4}$ SH

Score $\frac{1851}{125}$ / 7 gates



$e^{i5\pi/4}$ SHX

Score $\frac{1851}{125}$ / 7 gates



$e^{i5\pi/4}$ SHY_q2

Score $\frac{6403}{500}$ / 6 gates



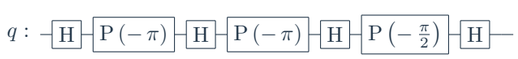
$e^{i5\pi/4}$ SHZ

Score $\frac{6403}{500}$ / 6 gates



$e^{i5\pi/4}$ SxY_q3

Score $\frac{7103}{500}$ / 7 gates



$e^{i5\pi/4}$ SxZ_q3

Score $\frac{2051}{250}$ / 4 gates



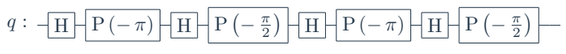
Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

H|P / $k = 6$

cx_h_phase

$e^{i6\pi/4}$ I

Score $\frac{2051}{125}$ / 8 gates



$e^{i6\pi/4}$ X

Score $\frac{5203}{500}$ / 5 gates



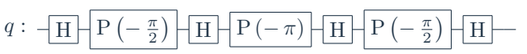
$e^{i6\pi/4}$ Y

Score $\frac{2001}{250}$ / 4 gates



$e^{i6\pi/4}$ Z

Score $\frac{7203}{500}$ / 7 gates



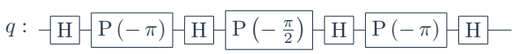
$e^{i6\pi/4}$ H

Score $\frac{1801}{125}$ / 7 gates



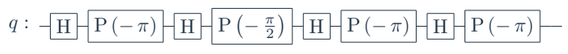
$e^{i6\pi/4}$ S

Score $\frac{7103}{500}$ / 7 gates



$e^{i6\pi/4}$ Sdg

Score $\frac{2026}{125}$ / 8 gates



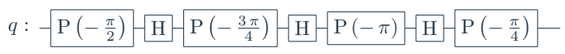
$e^{i6\pi/4}$ Sx

Score $\frac{5103}{500}$ / 5 gates



$e^{i6\pi/4}$ Sxdg

Score $\frac{1876}{125}$ / 7 gates



$e^{i6\pi/4}$ HX

Score $\frac{6203}{500}$ / 6 gates



$e^{i6\pi/4}$ HY_q2

Score $\frac{6203}{500}$ / 6 gates



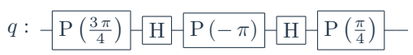
$e^{i6\pi/4}$ HZ

Score $\frac{6203}{500}$ / 6 gates



$e^{i6\pi/4}$ SX_q3

Score $\frac{5403}{500}$ / 5 gates



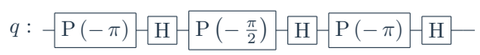
$e^{i6\pi/4}$ SY_q3

Score $\frac{5403}{500}$ / 5 gates



$e^{i6\pi/4}$ HS

Score $\frac{6103}{500}$ / 6 gates



$e^{i6\pi/4}$ HSX_q6

Score $\frac{6103}{500}$ / 6 gates



$e^{i6\pi/4}$ HSY

Score $\frac{2551}{250}$ / 5 gates



$e^{i6\pi/4}$ HSdg

Score $\frac{1776}{125}$ / 7 gates



$e^{i6\pi/4}$ SH

Score $\frac{6103}{500}$ / 6 gates



$e^{i6\pi/4}$ SHX

Score $\frac{1776}{125}$ / 7 gates



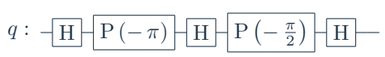
$e^{i6\pi/4}$ SHY_q2

Score $\frac{6103}{500}$ / 6 gates



$e^{i6\pi/4}$ SHZ

Score $\frac{2551}{250}$ / 5 gates



$e^{i6\pi/4}$ SxY_q3

Score $\frac{6203}{500}$ / 6 gates



$e^{i6\pi/4}$ SxZ_q3

Score $\frac{6203}{500}$ / 6 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

H|P / $k = 7$

cx_h_phase

$e^{i7\pi/4}$ I

Score $\frac{6303}{500}$ / 6 gates



$e^{i7\pi/4}$ X

Score $\frac{5403}{500}$ / 5 gates



$e^{i7\pi/4}$ Y

Score $\frac{5403}{500}$ / 5 gates



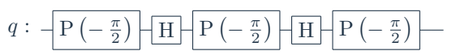
$e^{i7\pi/4}$ Z

Score $\frac{6303}{500}$ / 6 gates



$e^{i7\pi/4}$ H

Score $\frac{5303}{500}$ / 5 gates



$e^{i7\pi/4}$ S

Score $\frac{2601}{250}$ / 5 gates



$e^{i7\pi/4}$ Sdg

Score $\frac{6203}{500}$ / 6 gates



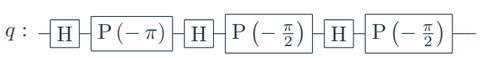
$e^{i7\pi/4}$ Sx

Score $\frac{1601}{250}$ / 3 gates



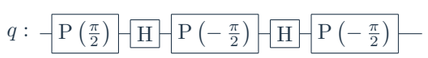
$e^{i7\pi/4}$ Sxdg

Score $\frac{6203}{500}$ / 6 gates



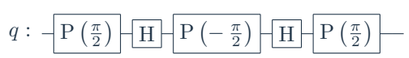
$e^{i7\pi/4}$ HX

Score $\frac{5303}{500}$ / 5 gates



$e^{i7\pi/4}$ HY_q2

Score $\frac{5303}{500}$ / 5 gates



$e^{i7\pi/4}$ HZ

Score $\frac{5303}{500}$ / 5 gates



$e^{i7\pi/4}$ SX_q3

Score $\frac{5103}{500}$ / 5 gates



$e^{i7\pi/4}$ SY_q3

Score $\frac{5103}{500}$ / 5 gates



$e^{i7\pi/4}$ HS

Score $\frac{2101}{250}$ / 4 gates



$e^{i7\pi/4}$ HSX_q6

Score $\frac{5203}{500}$ / 5 gates



$e^{i7\pi/4}$ HSY

Score $\frac{2101}{250}$ / 4 gates



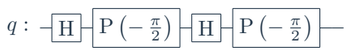
$e^{i7\pi/4}$ HSdg

Score $\frac{5203}{500}$ / 5 gates



$e^{i7\pi/4}$ SH

Score $\frac{2101}{250}$ / 4 gates



$e^{i7\pi/4}$ SHX

Score $\frac{5203}{500}$ / 5 gates



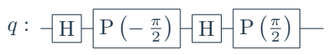
$e^{i7\pi/4}$ SHY_q2

Score $\frac{5203}{500}$ / 5 gates



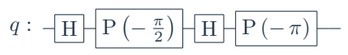
$e^{i7\pi/4}$ SHZ

Score $\frac{2101}{250}$ / 4 gates



$e^{i7\pi/4}$ SxY_q3

Score $\frac{2051}{250}$ / 4 gates



$e^{i7\pi/4}$ SxZ_q3

Score $\frac{7103}{500}$ / 7 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

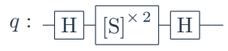
H|S / $k = 0$

clifford_only

$e^{i0\pi/4}$ **I**
Score 0 / 0 gates



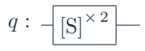
$e^{i0\pi/4}$ **X**
Score 8 / 4 gates



$e^{i0\pi/4}$ **Y**
Score 16 / 8 gates



$e^{i0\pi/4}$ **Z**
Score 4 / 2 gates



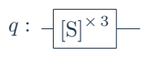
$e^{i0\pi/4}$ **H**
Score 2 / 1 gates



$e^{i0\pi/4}$ **S**
Score 2 / 1 gates



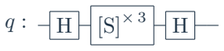
$e^{i0\pi/4}$ **Sdg**
Score 6 / 3 gates



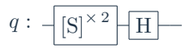
$e^{i0\pi/4}$ **Sx**
Score 6 / 3 gates



$e^{i0\pi/4}$ **Sxdg**
Score 10 / 5 gates



$e^{i0\pi/4}$ **HX**
Score 6 / 3 gates



$e^{i0\pi/4}$ **HY_q2**
Score 10 / 5 gates



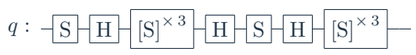
$e^{i0\pi/4}$ **HZ**
Score 6 / 3 gates



$e^{i0\pi/4}$ **SX_q3**
Score 22 / 11 gates



$e^{i0\pi/4}$ **SY_q3**
Score 22 / 11 gates



$e^{i0\pi/4}$ **HS**
Score 4 / 2 gates



$e^{i0\pi/4}$ **HSX_q6**
Score 12 / 6 gates



$e^{i0\pi/4}$ **HSY**
Score 8 / 4 gates



$e^{i0\pi/4}$ **HSdg**
Score 8 / 4 gates



$e^{i0\pi/4}$ **SH**
Score 4 / 2 gates



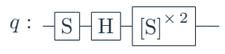
$e^{i0\pi/4}$ **SHX**
Score 8 / 4 gates



$e^{i0\pi/4}$ **SHY_q2**
Score 12 / 6 gates



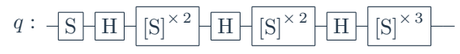
$e^{i0\pi/4}$ **SHZ**
Score 8 / 4 gates



$e^{i0\pi/4}$ **SxY_q3**
Score 10 / 5 gates



$e^{i0\pi/4}$ **SxZ_q3**
Score 22 / 11 gates



Certificate scope: all_circuits_in_supplied_fixed_clifford_basis. Phase is part of the target matrix.

H|S / $k = 1$

clifford_only

$e^{i1\pi/4}$ **I**
Score 12 / 6 gates



$e^{i1\pi/4}$ **X**
Score 16 / 8 gates



$e^{i1\pi/4}$ **Y**
Score 16 / 8 gates



$e^{i1\pi/4}$ **Z**
Score 16 / 8 gates



$e^{i1\pi/4}$ **H**
Score 10 / 5 gates



$e^{i1\pi/4}$ **S**
Score 14 / 7 gates



$e^{i1\pi/4}$ **Sdg**
Score 10 / 5 gates



$e^{i1\pi/4}$ **Sx**
Score 14 / 7 gates



$e^{i1\pi/4}$ **Sxdg**
Score 6 / 3 gates



$e^{i1\pi/4}$ **HX**
Score 14 / 7 gates



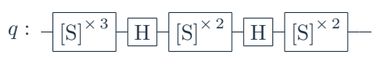
$e^{i1\pi/4}$ **HY_q2**
Score 18 / 9 gates



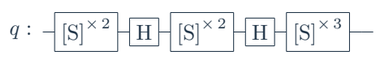
$e^{i1\pi/4}$ **HZ**
Score 14 / 7 gates



$e^{i1\pi/4}$ **SX_q3**
Score 18 / 9 gates



$e^{i1\pi/4}$ **SY_q3**
Score 18 / 9 gates



$e^{i1\pi/4}$ **HS**
Score 12 / 6 gates



$e^{i1\pi/4}$ **HSX_q6**
Score 12 / 6 gates



$e^{i1\pi/4}$ **HSY**
Score 16 / 8 gates



$e^{i1\pi/4}$ **HSdg**
Score 8 / 4 gates



$e^{i1\pi/4}$ **SH**
Score 12 / 6 gates



$e^{i1\pi/4}$ **SHX**
Score 8 / 4 gates



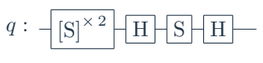
$e^{i1\pi/4}$ **SHY_q2**
Score 12 / 6 gates



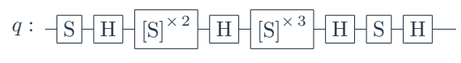
$e^{i1\pi/4}$ **SHZ**
Score 16 / 8 gates



$e^{i1\pi/4}$ **SxY_q3**
Score 10 / 5 gates



$e^{i1\pi/4}$ **SxZ_q3**
Score 22 / 11 gates



Certificate scope: all_circuits_in_supplied_fixed_clifford_basis. Phase is part of the target matrix.

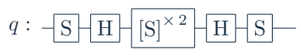
H|S / $k = 2$

clifford_only

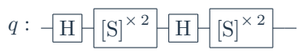
$e^{i2\pi/4}$ **I**
Score 20 / 10 gates



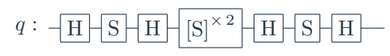
$e^{i2\pi/4}$ **X**
Score 12 / 6 gates



$e^{i2\pi/4}$ **Y**
Score 12 / 6 gates



$e^{i2\pi/4}$ **Z**
Score 16 / 8 gates



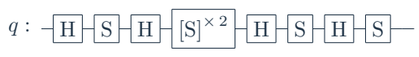
$e^{i2\pi/4}$ **H**
Score 18 / 9 gates



$e^{i2\pi/4}$ **S**
Score 22 / 11 gates



$e^{i2\pi/4}$ **Sdg**
Score 18 / 9 gates



$e^{i2\pi/4}$ **Sx**
Score 22 / 11 gates



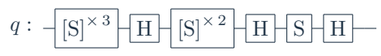
$e^{i2\pi/4}$ **Sxdg**
Score 14 / 7 gates



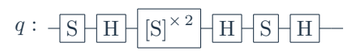
$e^{i2\pi/4}$ **HX**
Score 14 / 7 gates



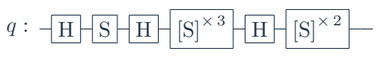
$e^{i2\pi/4}$ **HY_q2**
Score 18 / 9 gates



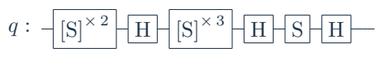
$e^{i2\pi/4}$ **HZ**
Score 14 / 7 gates



$e^{i2\pi/4}$ **SX_q3**
Score 18 / 9 gates



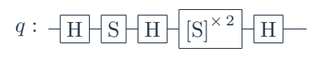
$e^{i2\pi/4}$ **SY_q3**
Score 18 / 9 gates



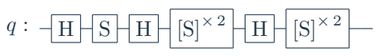
$e^{i2\pi/4}$ **HS**
Score 20 / 10 gates



$e^{i2\pi/4}$ **HSX_q6**
Score 12 / 6 gates



$e^{i2\pi/4}$ **HSY**
Score 16 / 8 gates



$e^{i2\pi/4}$ **HSdg**
Score 16 / 8 gates



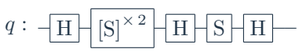
$e^{i2\pi/4}$ **SH**
Score 20 / 10 gates



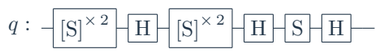
$e^{i2\pi/4}$ **SHX**
Score 16 / 8 gates



$e^{i2\pi/4}$ **SHY_q2**
Score 12 / 6 gates



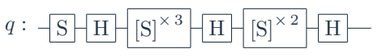
$e^{i2\pi/4}$ **SHZ**
Score 16 / 8 gates



$e^{i2\pi/4}$ **SxY_q3**
Score 18 / 9 gates



$e^{i2\pi/4}$ **SxZ_q3**
Score 18 / 9 gates

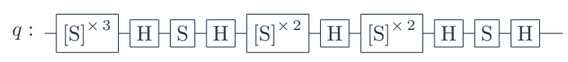


Certificate scope: all_circuits_in_supplied_fixed_clifford_basis. Phase is part of the target matrix.

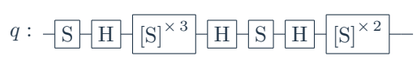
H|S / $k = 3$

clifford_only

$e^{i3\pi/4}$ **I**
Score 28 / 14 gates



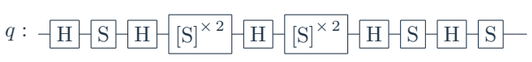
$e^{i3\pi/4}$ **X**
Score 20 / 10 gates



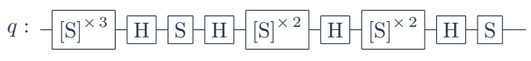
$e^{i3\pi/4}$ **Y**
Score 20 / 10 gates



$e^{i3\pi/4}$ **Z**
Score 24 / 12 gates



$e^{i3\pi/4}$ **H**
Score 26 / 13 gates



$e^{i3\pi/4}$ **S**
Score 22 / 11 gates



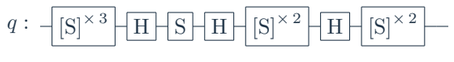
$e^{i3\pi/4}$ **Sdg**
Score 26 / 13 gates



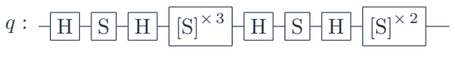
$e^{i3\pi/4}$ **Sx**
Score 18 / 9 gates



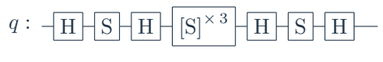
$e^{i3\pi/4}$ **Sxdg**
Score 22 / 11 gates



$e^{i3\pi/4}$ **HX**
Score 22 / 11 gates



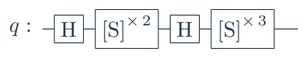
$e^{i3\pi/4}$ **HY_q2**
Score 18 / 9 gates



$e^{i3\pi/4}$ **HZ**
Score 22 / 11 gates



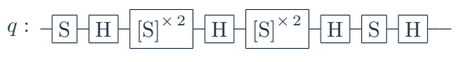
$e^{i3\pi/4}$ **SX_q3**
Score 14 / 7 gates



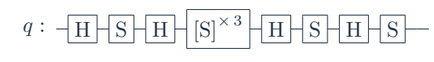
$e^{i3\pi/4}$ **SY_q3**
Score 14 / 7 gates



$e^{i3\pi/4}$ **HS**
Score 20 / 10 gates



$e^{i3\pi/4}$ **HSX_q6**
Score 20 / 10 gates



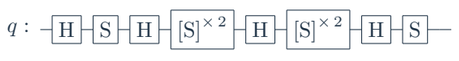
$e^{i3\pi/4}$ **HSY**
Score 24 / 12 gates



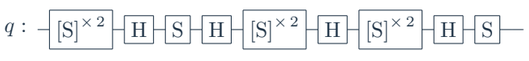
$e^{i3\pi/4}$ **HSdg**
Score 24 / 12 gates



$e^{i3\pi/4}$ **SH**
Score 20 / 10 gates



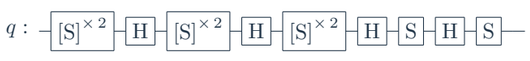
$e^{i3\pi/4}$ **SHX**
Score 24 / 12 gates



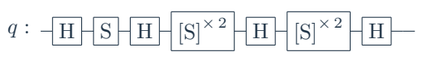
$e^{i3\pi/4}$ **SHY_q2**
Score 20 / 10 gates



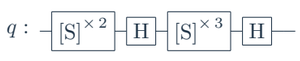
$e^{i3\pi/4}$ **SHZ**
Score 24 / 12 gates



$e^{i3\pi/4}$ **SxY_q3**
Score 18 / 9 gates



$e^{i3\pi/4}$ **SxZ_q3**
Score 14 / 7 gates



Certificate scope: all_circuits_in_supplied_fixed_clifford_basis. Phase is part of the target matrix.

H|S / $k = 4$

clifford_only

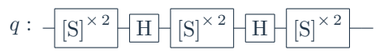
$e^{i4\pi/4}$ I

Score 24 / 12 gates



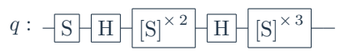
$e^{i4\pi/4}$ X

Score 16 / 8 gates



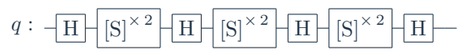
$e^{i4\pi/4}$ Y

Score 16 / 8 gates



$e^{i4\pi/4}$ Z

Score 20 / 10 gates



$e^{i4\pi/4}$ H

Score 22 / 11 gates



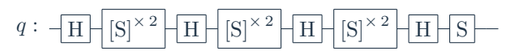
$e^{i4\pi/4}$ S

Score 26 / 13 gates



$e^{i4\pi/4}$ Sdg

Score 22 / 11 gates



$e^{i4\pi/4}$ Sx

Score 26 / 13 gates



$e^{i4\pi/4}$ Sxdg

Score 22 / 11 gates



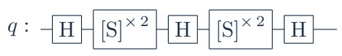
$e^{i4\pi/4}$ HX

Score 18 / 9 gates



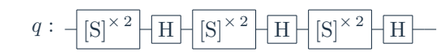
$e^{i4\pi/4}$ HY_q2

Score 14 / 7 gates



$e^{i4\pi/4}$ HZ

Score 18 / 9 gates



$e^{i4\pi/4}$ SX_q3

Score 14 / 7 gates



$e^{i4\pi/4}$ SY_q3

Score 14 / 7 gates



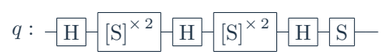
$e^{i4\pi/4}$ HS

Score 24 / 12 gates



$e^{i4\pi/4}$ HSX_q6

Score 16 / 8 gates



$e^{i4\pi/4}$ HSY

Score 20 / 10 gates



$e^{i4\pi/4}$ HSdg

Score 20 / 10 gates



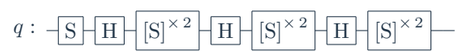
$e^{i4\pi/4}$ SH

Score 24 / 12 gates



$e^{i4\pi/4}$ SHX

Score 20 / 10 gates



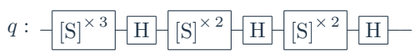
$e^{i4\pi/4}$ SHY_q2

Score 16 / 8 gates



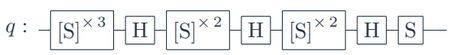
$e^{i4\pi/4}$ SHZ

Score 20 / 10 gates



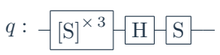
$e^{i4\pi/4}$ SxY_q3

Score 22 / 11 gates



$e^{i4\pi/4}$ SxZ_q3

Score 10 / 5 gates

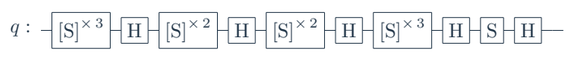


Certificate scope: all_circuits_in_supplied_fixed_clifford_basis. Phase is part of the target matrix.

H|S / $k = 5$

clifford_only

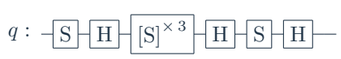
$e^{i5\pi/4}$ **I**
Score 32 / 16 gates



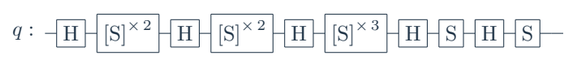
$e^{i5\pi/4}$ **X**
Score 24 / 12 gates



$e^{i5\pi/4}$ **Y**
Score 16 / 8 gates



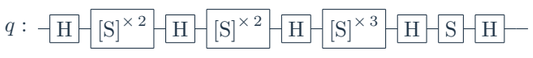
$e^{i5\pi/4}$ **Z**
Score 28 / 14 gates



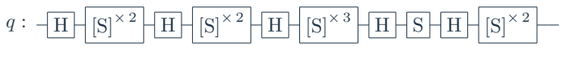
$e^{i5\pi/4}$ **H**
Score 30 / 15 gates



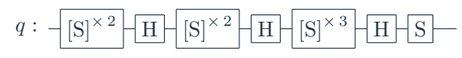
$e^{i5\pi/4}$ **S**
Score 26 / 13 gates



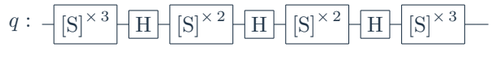
$e^{i5\pi/4}$ **Sdg**
Score 30 / 15 gates



$e^{i5\pi/4}$ **Sx**
Score 22 / 11 gates



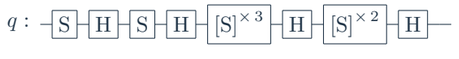
$e^{i5\pi/4}$ **Sxdg**
Score 26 / 13 gates



$e^{i5\pi/4}$ **HX**
Score 26 / 13 gates



$e^{i5\pi/4}$ **HY_q2**
Score 22 / 11 gates



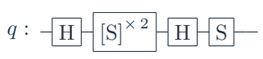
$e^{i5\pi/4}$ **HZ**
Score 26 / 13 gates



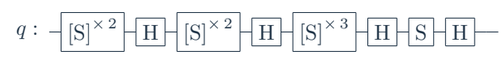
$e^{i5\pi/4}$ **SX_q3**
Score 10 / 5 gates



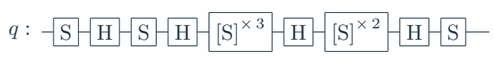
$e^{i5\pi/4}$ **SY_q3**
Score 10 / 5 gates



$e^{i5\pi/4}$ **HS**
Score 24 / 12 gates



$e^{i5\pi/4}$ **HSX_q6**
Score 24 / 12 gates



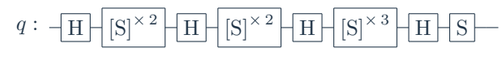
$e^{i5\pi/4}$ **HSY**
Score 20 / 10 gates



$e^{i5\pi/4}$ **HSdg**
Score 28 / 14 gates



$e^{i5\pi/4}$ **SH**
Score 24 / 12 gates



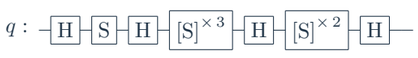
$e^{i5\pi/4}$ **SHX**
Score 28 / 14 gates



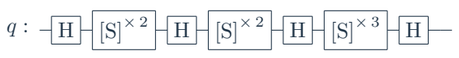
$e^{i5\pi/4}$ **SHY_q2**
Score 24 / 12 gates



$e^{i5\pi/4}$ **SHZ**
Score 20 / 10 gates



$e^{i5\pi/4}$ **SxY_q3**
Score 22 / 11 gates



$e^{i5\pi/4}$ **SxZ_q3**
Score 10 / 5 gates



Certificate scope: all_circuits_in_supplied_fixed_clifford_basis. Phase is part of the target matrix.

H|S / $k = 6$

clifford_only

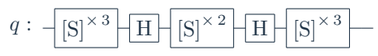
$e^{i6\pi/4}$ I

Score 28 / 14 gates



$e^{i6\pi/4}$ X

Score 20 / 10 gates



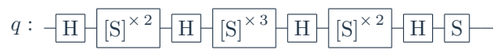
$e^{i6\pi/4}$ Y

Score 12 / 6 gates



$e^{i6\pi/4}$ Z

Score 24 / 12 gates



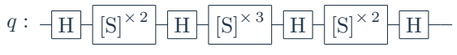
$e^{i6\pi/4}$ H

Score 26 / 13 gates



$e^{i6\pi/4}$ S

Score 22 / 11 gates



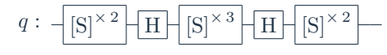
$e^{i6\pi/4}$ Sdg

Score 26 / 13 gates



$e^{i6\pi/4}$ Sx

Score 18 / 9 gates



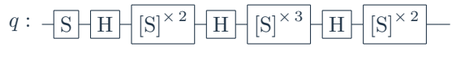
$e^{i6\pi/4}$ Sxdg

Score 26 / 13 gates



$e^{i6\pi/4}$ HX

Score 22 / 11 gates



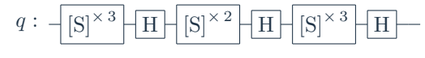
$e^{i6\pi/4}$ HY_q2

Score 18 / 9 gates



$e^{i6\pi/4}$ HZ

Score 22 / 11 gates



$e^{i6\pi/4}$ SX_q3

Score 18 / 9 gates



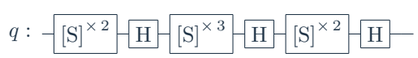
$e^{i6\pi/4}$ SY_q3

Score 18 / 9 gates



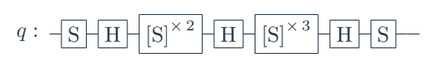
$e^{i6\pi/4}$ HS

Score 20 / 10 gates



$e^{i6\pi/4}$ HSX_q6

Score 20 / 10 gates



$e^{i6\pi/4}$ HSY

Score 16 / 8 gates



$e^{i6\pi/4}$ HSdg

Score 24 / 12 gates



$e^{i6\pi/4}$ SH

Score 20 / 10 gates



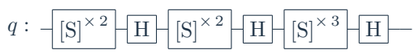
$e^{i6\pi/4}$ SHX

Score 24 / 12 gates



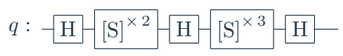
$e^{i6\pi/4}$ SHY_q2

Score 20 / 10 gates



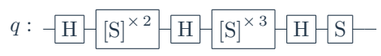
$e^{i6\pi/4}$ SHZ

Score 16 / 8 gates



$e^{i6\pi/4}$ SxY_q3

Score 18 / 9 gates



$e^{i6\pi/4}$ SxZ_q3

Score 18 / 9 gates



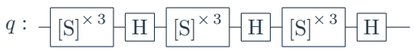
Certificate scope: all_circuits_in_supplied_fixed_clifford_basis. Phase is part of the target matrix.

H|S / $k = 7$

clifford_only

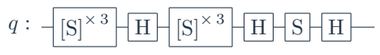
$e^{i7\pi/4}$ I

Score 24 / 12 gates



$e^{i7\pi/4}$ X

Score 20 / 10 gates



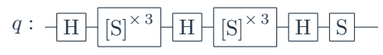
$e^{i7\pi/4}$ Y

Score 20 / 10 gates



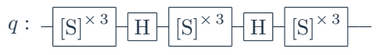
$e^{i7\pi/4}$ Z

Score 20 / 10 gates



$e^{i7\pi/4}$ H

Score 22 / 11 gates



$e^{i7\pi/4}$ S

Score 18 / 9 gates



$e^{i7\pi/4}$ Sdg

Score 22 / 11 gates



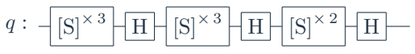
$e^{i7\pi/4}$ Sx

Score 14 / 7 gates



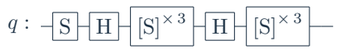
$e^{i7\pi/4}$ Sxdg

Score 22 / 11 gates



$e^{i7\pi/4}$ HX

Score 18 / 9 gates



$e^{i7\pi/4}$ HY_q2

Score 14 / 7 gates



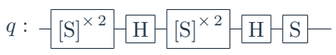
$e^{i7\pi/4}$ HZ

Score 18 / 9 gates



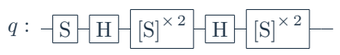
$e^{i7\pi/4}$ SX_q3

Score 14 / 7 gates



$e^{i7\pi/4}$ SY_q3

Score 14 / 7 gates



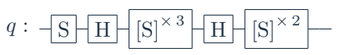
$e^{i7\pi/4}$ HS

Score 16 / 8 gates



$e^{i7\pi/4}$ HSX_q6

Score 16 / 8 gates



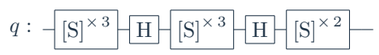
$e^{i7\pi/4}$ HSY

Score 12 / 6 gates



$e^{i7\pi/4}$ HSdg

Score 20 / 10 gates



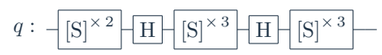
$e^{i7\pi/4}$ SH

Score 16 / 8 gates



$e^{i7\pi/4}$ SHX

Score 20 / 10 gates



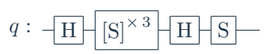
$e^{i7\pi/4}$ SHY_q2

Score 16 / 8 gates



$e^{i7\pi/4}$ SHZ

Score 12 / 6 gates



$e^{i7\pi/4}$ SxY_q3

Score 14 / 7 gates



$e^{i7\pi/4}$ SxZ_q3

Score 18 / 9 gates



Certificate scope: all_circuits_in_supplied_fixed_clifford_basis. Phase is part of the target matrix.

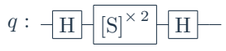
H|S|T|Tdg / $k = 0$

clifford_t_cx, clifford_t_cz

$e^{i0\pi/4}$ **I**
Score 0 / 0 gates



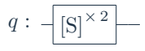
$e^{i0\pi/4}$ **X**
Score 8 / 4 gates



$e^{i0\pi/4}$ **Y**
Score 14 / 7 gates



$e^{i0\pi/4}$ **Z**
Score 4 / 2 gates



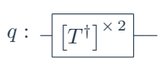
$e^{i0\pi/4}$ **H**
Score 2 / 1 gates



$e^{i0\pi/4}$ **S**
Score 2 / 1 gates



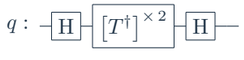
$e^{i0\pi/4}$ **Sdg**
Score 4 / 2 gates



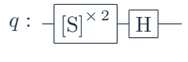
$e^{i0\pi/4}$ **Sx**
Score 6 / 3 gates



$e^{i0\pi/4}$ **Sxdg**
Score 8 / 4 gates



$e^{i0\pi/4}$ **HX**
Score 6 / 3 gates



$e^{i0\pi/4}$ **HY_q2**
Score 10 / 5 gates



$e^{i0\pi/4}$ **HZ**
Score 6 / 3 gates



$e^{i0\pi/4}$ **SX_q3**
Score 18 / 9 gates



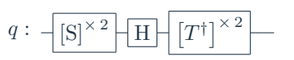
$e^{i0\pi/4}$ **SY_q3**
Score 18 / 9 gates



$e^{i0\pi/4}$ **HS**
Score 4 / 2 gates



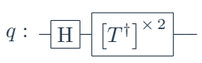
$e^{i0\pi/4}$ **HSX_q6**
Score 10 / 5 gates



$e^{i0\pi/4}$ **HSY**
Score 8 / 4 gates



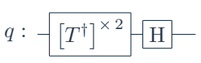
$e^{i0\pi/4}$ **HSdg**
Score 6 / 3 gates



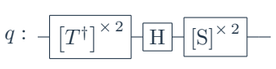
$e^{i0\pi/4}$ **SH**
Score 4 / 2 gates



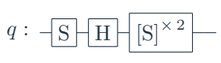
$e^{i0\pi/4}$ **SHX**
Score 6 / 3 gates



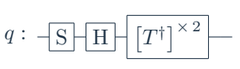
$e^{i0\pi/4}$ **SHY_q2**
Score 10 / 5 gates



$e^{i0\pi/4}$ **SHZ**
Score 8 / 4 gates



$e^{i0\pi/4}$ **SxY_q3**
Score 8 / 4 gates



$e^{i0\pi/4}$ **SxZ_q3**
Score 20 / 10 gates



Certificate scope: all_words_in_fixed_H_S_T_Tdg_alphabet_including_nonclifford_intermediates. Phase is part of the target matrix.

H|S|T|Tdg / $k = 1$

clifford_t_cx, clifford_t_cz

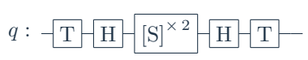
$e^{i1\pi/4}$ I

Score 12 / 6 gates



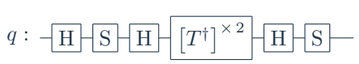
$e^{i1\pi/4}$ X

Score 12 / 6 gates



$e^{i1\pi/4}$ Y

Score 14 / 7 gates



$e^{i1\pi/4}$ Z

Score 14 / 7 gates



$e^{i1\pi/4}$ H

Score 10 / 5 gates



$e^{i1\pi/4}$ S

Score 14 / 7 gates



$e^{i1\pi/4}$ Sdg

Score 10 / 5 gates



$e^{i1\pi/4}$ Sx

Score 14 / 7 gates



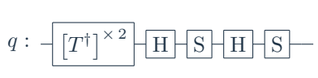
$e^{i1\pi/4}$ Sxdg

Score 6 / 3 gates



$e^{i1\pi/4}$ HX

Score 12 / 6 gates



$e^{i1\pi/4}$ HY_q2

Score 14 / 7 gates



$e^{i1\pi/4}$ HZ

Score 12 / 6 gates



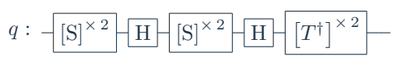
$e^{i1\pi/4}$ SX_q3

Score 16 / 8 gates



$e^{i1\pi/4}$ SY_q3

Score 16 / 8 gates



$e^{i1\pi/4}$ HS

Score 12 / 6 gates



$e^{i1\pi/4}$ HSX_q6

Score 10 / 5 gates



$e^{i1\pi/4}$ HSY

Score 14 / 7 gates



$e^{i1\pi/4}$ HSdg

Score 8 / 4 gates



$e^{i1\pi/4}$ SH

Score 12 / 6 gates



$e^{i1\pi/4}$ SHX

Score 8 / 4 gates



$e^{i1\pi/4}$ SHY_q2

Score 10 / 5 gates



$e^{i1\pi/4}$ SHZ

Score 14 / 7 gates



$e^{i1\pi/4}$ SxY_q3

Score 10 / 5 gates



$e^{i1\pi/4}$ SxZ_q3

Score 20 / 10 gates



Certificate scope: all_words_in_fixed_H_S_T_Tdg_alphabet_including_nonclifford_intermediates. Phase is part of the target matrix.

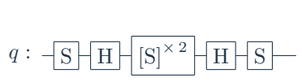
H|S|T|Tdg / $k = 2$

clifford_t_cx, clifford_t_cz

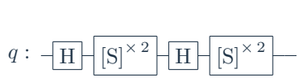
$e^{i2\pi/4}$ I
Score 20 / 10 gates



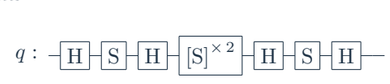
$e^{i2\pi/4}$ X
Score 12 / 6 gates



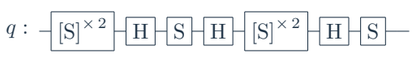
$e^{i2\pi/4}$ Y
Score 12 / 6 gates



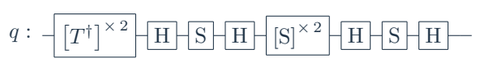
$e^{i2\pi/4}$ Z
Score 16 / 8 gates



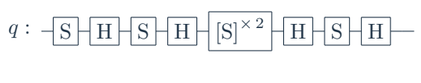
$e^{i2\pi/4}$ H
Score 18 / 9 gates



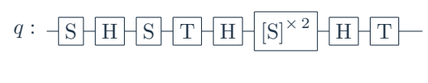
$e^{i2\pi/4}$ S
Score 20 / 10 gates



$e^{i2\pi/4}$ Sdg
Score 18 / 9 gates



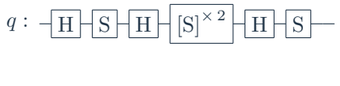
$e^{i2\pi/4}$ Sx
Score 18 / 9 gates



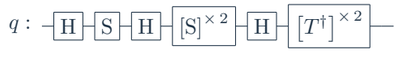
$e^{i2\pi/4}$ Sxdg
Score 14 / 7 gates



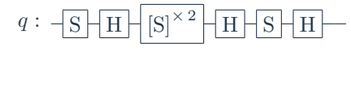
$e^{i2\pi/4}$ HX
Score 14 / 7 gates



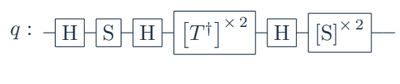
$e^{i2\pi/4}$ HY_q2
Score 16 / 8 gates



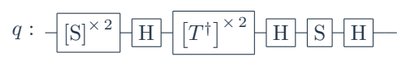
$e^{i2\pi/4}$ HZ
Score 14 / 7 gates



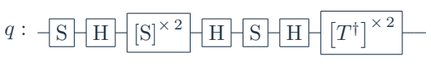
$e^{i2\pi/4}$ SX_q3
Score 16 / 8 gates



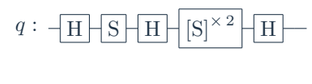
$e^{i2\pi/4}$ SY_q3
Score 16 / 8 gates



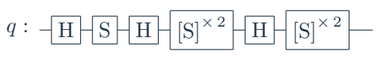
$e^{i2\pi/4}$ HS
Score 18 / 9 gates



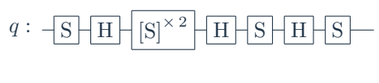
$e^{i2\pi/4}$ HSX_q6
Score 12 / 6 gates



$e^{i2\pi/4}$ HSY
Score 16 / 8 gates



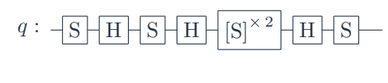
$e^{i2\pi/4}$ HSdg
Score 16 / 8 gates



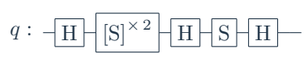
$e^{i2\pi/4}$ SH
Score 18 / 9 gates



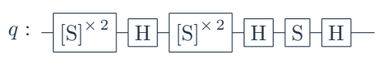
$e^{i2\pi/4}$ SHX
Score 16 / 8 gates



$e^{i2\pi/4}$ SHY_q2
Score 12 / 6 gates



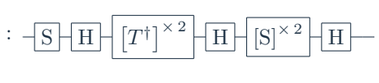
$e^{i2\pi/4}$ SHZ
Score 16 / 8 gates



$e^{i2\pi/4}$ SxY_q3
Score 16 / 8 gates



$e^{i2\pi/4}$ SxZ_q3
Score 16 / 8 gates



Certificate scope: all_words_in_fixed_H_S_T_Tdg_alphabet_including_nonclifford_intermediates. Phase is part of the target matrix.

H|S|T|Tdg / $k = 3$

clifford_t_cx, clifford_t_cz

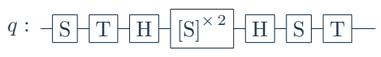
$e^{i3\pi/4}$ I

Score 24 / 12 gates



$e^{i3\pi/4}$ X

Score 16 / 8 gates



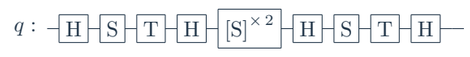
$e^{i3\pi/4}$ Y

Score 16 / 8 gates



$e^{i3\pi/4}$ Z

Score 20 / 10 gates



$e^{i3\pi/4}$ H

Score 22 / 11 gates



$e^{i3\pi/4}$ S

Score 22 / 11 gates



$e^{i3\pi/4}$ Sdg

Score 22 / 11 gates



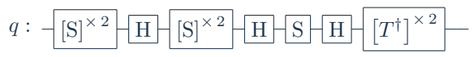
$e^{i3\pi/4}$ Sx

Score 18 / 9 gates



$e^{i3\pi/4}$ Sxdg

Score 20 / 10 gates



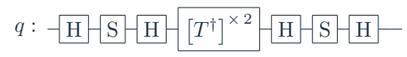
$e^{i3\pi/4}$ HX

Score 18 / 9 gates



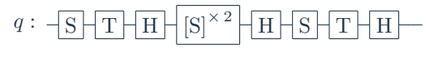
$e^{i3\pi/4}$ HY_q2

Score 16 / 8 gates



$e^{i3\pi/4}$ HZ

Score 18 / 9 gates



$e^{i3\pi/4}$ SX_q3

Score 12 / 6 gates



$e^{i3\pi/4}$ SY_q3

Score 12 / 6 gates



$e^{i3\pi/4}$ HS

Score 20 / 10 gates



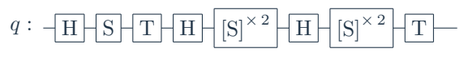
$e^{i3\pi/4}$ HSX_q6

Score 16 / 8 gates



$e^{i3\pi/4}$ HSY

Score 20 / 10 gates



$e^{i3\pi/4}$ HSdg

Score 20 / 10 gates



$e^{i3\pi/4}$ SH

Score 20 / 10 gates



$e^{i3\pi/4}$ SHX

Score 20 / 10 gates



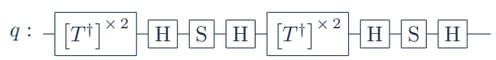
$e^{i3\pi/4}$ SHY_q2

Score 16 / 8 gates



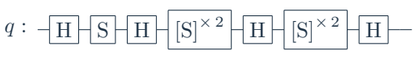
$e^{i3\pi/4}$ SHZ

Score 20 / 10 gates



$e^{i3\pi/4}$ SxY_q3

Score 18 / 9 gates



$e^{i3\pi/4}$ SxZ_q3

Score 12 / 6 gates



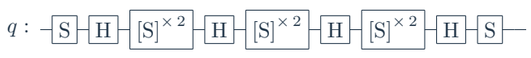
Certificate scope: all_words_in_fixed_H_S_T_Tdg_alphabet_including_nonclifford_intermediates. Phase is part of the target matrix.

H|S|T|Tdg / $k = 4$

clifford_t_cx, clifford_t_cz

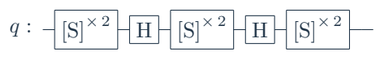
$e^{i4\pi/4}$ I

Score 24 / 12 gates



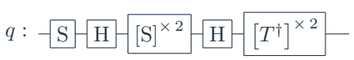
$e^{i4\pi/4}$ X

Score 16 / 8 gates



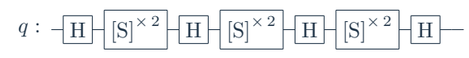
$e^{i4\pi/4}$ Y

Score 14 / 7 gates



$e^{i4\pi/4}$ Z

Score 20 / 10 gates



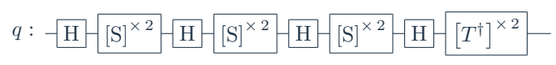
$e^{i4\pi/4}$ H

Score 22 / 11 gates



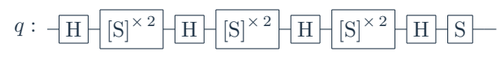
$e^{i4\pi/4}$ S

Score 24 / 12 gates



$e^{i4\pi/4}$ Sdg

Score 22 / 11 gates



$e^{i4\pi/4}$ Sx

Score 22 / 11 gates



$e^{i4\pi/4}$ Sxdg

Score 22 / 11 gates

$e^{i4\pi/4}$ HX

Score 18 / 9 gates

$e^{i4\pi/4}$ HY_q2

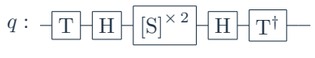
Score 14 / 7 gates

$e^{i4\pi/4}$ HZ

Score 18 / 9 gates

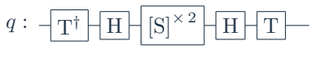
$e^{i4\pi/4}$ SX_q3

Score 12 / 6 gates



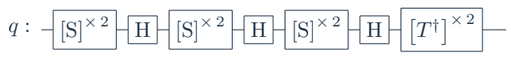
$e^{i4\pi/4}$ SY_q3

Score 12 / 6 gates



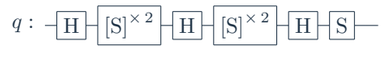
$e^{i4\pi/4}$ HS

Score 22 / 11 gates



$e^{i4\pi/4}$ HSX_q6

Score 16 / 8 gates



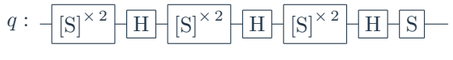
$e^{i4\pi/4}$ HSY

Score 18 / 9 gates



$e^{i4\pi/4}$ HSdg

Score 20 / 10 gates



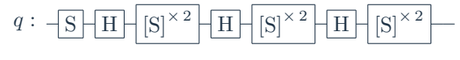
$e^{i4\pi/4}$ SH

Score 22 / 11 gates



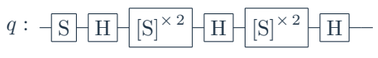
$e^{i4\pi/4}$ SHX

Score 20 / 10 gates



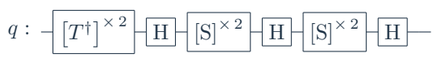
$e^{i4\pi/4}$ SHY_q2

Score 16 / 8 gates



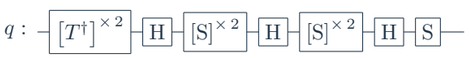
$e^{i4\pi/4}$ SHZ

Score 18 / 9 gates



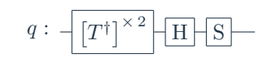
$e^{i4\pi/4}$ SxY_q3

Score 20 / 10 gates



$e^{i4\pi/4}$ SxZ_q3

Score 8 / 4 gates



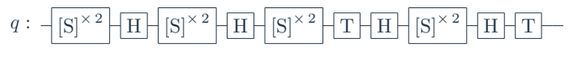
Certificate scope: all_words_in_fixed_H_S_T_Tdg_alphabet_including_nonclifford_intermediates. Phase is part of the target matrix.

H|S|T|Tdg / $k = 5$

clifford_t_cx, clifford_t_cz

$e^{i5\pi/4}$ I

Score 28 / 14 gates



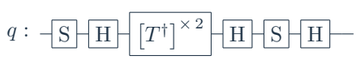
$e^{i5\pi/4}$ X

Score 20 / 10 gates



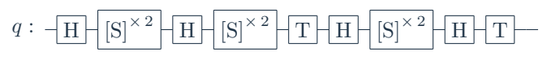
$e^{i5\pi/4}$ Y

Score 14 / 7 gates



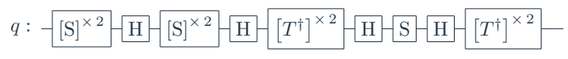
$e^{i5\pi/4}$ Z

Score 24 / 12 gates



$e^{i5\pi/4}$ H

Score 26 / 13 gates



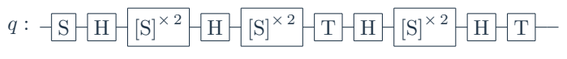
$e^{i5\pi/4}$ S

Score 24 / 12 gates



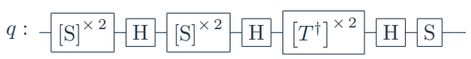
$e^{i5\pi/4}$ Sdg

Score 26 / 13 gates



$e^{i5\pi/4}$ Sx

Score 20 / 10 gates



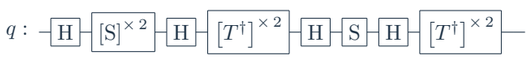
$e^{i5\pi/4}$ Sxdg

Score 22 / 11 gates



$e^{i5\pi/4}$ HX

Score 22 / 11 gates



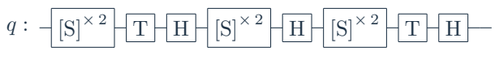
$e^{i5\pi/4}$ HY_q2

Score 18 / 9 gates



$e^{i5\pi/4}$ HZ

Score 22 / 11 gates



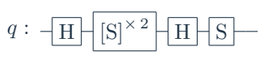
$e^{i5\pi/4}$ SX_q3

Score 10 / 5 gates



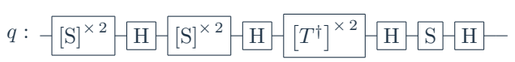
$e^{i5\pi/4}$ SY_q3

Score 10 / 5 gates



$e^{i5\pi/4}$ HS

Score 22 / 11 gates



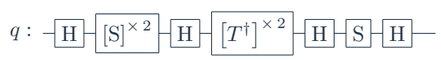
$e^{i5\pi/4}$ HSX_q6

Score 20 / 10 gates



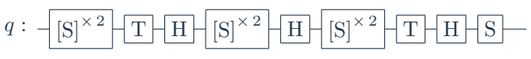
$e^{i5\pi/4}$ HSY

Score 18 / 9 gates



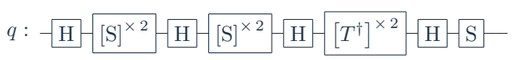
$e^{i5\pi/4}$ HSdg

Score 24 / 12 gates



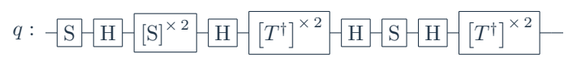
$e^{i5\pi/4}$ SH

Score 22 / 11 gates



$e^{i5\pi/4}$ SHX

Score 24 / 12 gates



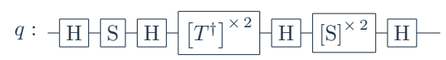
$e^{i5\pi/4}$ SHY_q2

Score 20 / 10 gates



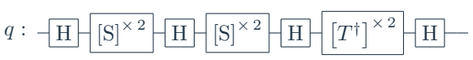
$e^{i5\pi/4}$ SHZ

Score 18 / 9 gates



$e^{i5\pi/4}$ SxY_q3

Score 20 / 10 gates



$e^{i5\pi/4}$ SxZ_q3

Score 10 / 5 gates



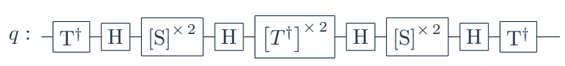
Certificate scope: all_words_in_fixed_H_S_T_Tdg_alphabet_including_nonclifford_intermediates. Phase is part of the target matrix.

H|S|T|Tdg / $k = 6$

clifford_t_cx, clifford_t_cz

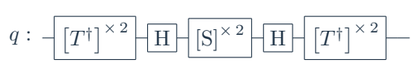
$e^{i6\pi/4}$ I

Score 24 / 12 gates



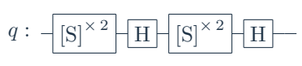
$e^{i6\pi/4}$ X

Score 16 / 8 gates



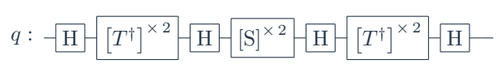
$e^{i6\pi/4}$ Y

Score 12 / 6 gates



$e^{i6\pi/4}$ Z

Score 20 / 10 gates



$e^{i6\pi/4}$ H

Score 22 / 11 gates



$e^{i6\pi/4}$ S

Score 20 / 10 gates



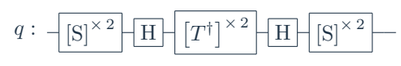
$e^{i6\pi/4}$ Sdg

Score 22 / 11 gates



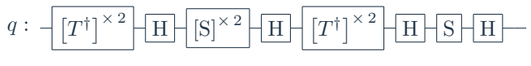
$e^{i6\pi/4}$ Sx

Score 16 / 8 gates



$e^{i6\pi/4}$ Sxdg

Score 22 / 11 gates



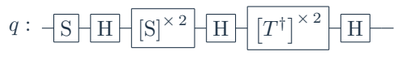
$e^{i6\pi/4}$ HX

Score 18 / 9 gates



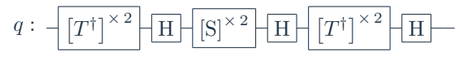
$e^{i6\pi/4}$ HY_q2

Score 16 / 8 gates



$e^{i6\pi/4}$ HZ

Score 18 / 9 gates



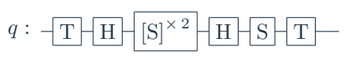
$e^{i6\pi/4}$ SX_q3

Score 14 / 7 gates



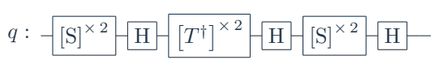
$e^{i6\pi/4}$ SY_q3

Score 14 / 7 gates



$e^{i6\pi/4}$ HS

Score 18 / 9 gates



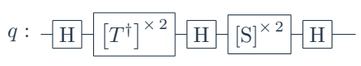
$e^{i6\pi/4}$ HSX_q6

Score 18 / 9 gates



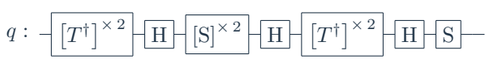
$e^{i6\pi/4}$ HSY

Score 14 / 7 gates



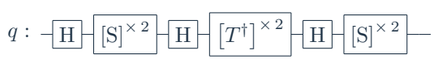
$e^{i6\pi/4}$ HSdg

Score 20 / 10 gates



$e^{i6\pi/4}$ SH

Score 18 / 9 gates



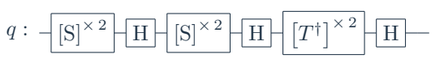
$e^{i6\pi/4}$ SHX

Score 20 / 10 gates



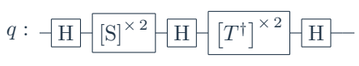
$e^{i6\pi/4}$ SHY_q2

Score 18 / 9 gates



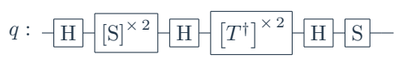
$e^{i6\pi/4}$ SHZ

Score 14 / 7 gates



$e^{i6\pi/4}$ SxY_q3

Score 16 / 8 gates



$e^{i6\pi/4}$ SxZ_q3

Score 16 / 8 gates



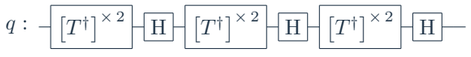
Certificate scope: all_words_in_fixed_H_S_T_Tdg_alphabet_including_nonclifford_intermediates. Phase is part of the target matrix.

H|S|T|Tdg / $k = 7$

clifford_t_cx, clifford_t_cz

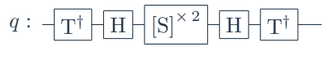
$e^{i7\pi/4}$ I

Score 18 / 9 gates



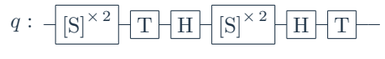
$e^{i7\pi/4}$ X

Score 12 / 6 gates



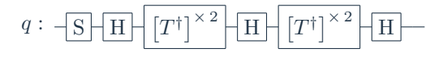
$e^{i7\pi/4}$ Y

Score 16 / 8 gates



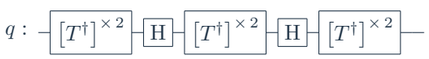
$e^{i7\pi/4}$ Z

Score 16 / 8 gates



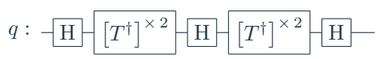
$e^{i7\pi/4}$ H

Score 16 / 8 gates



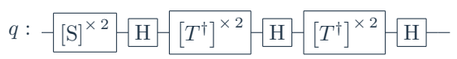
$e^{i7\pi/4}$ S

Score 14 / 7 gates



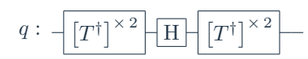
$e^{i7\pi/4}$ Sdg

Score 18 / 9 gates



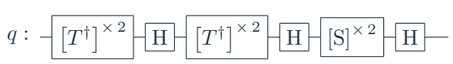
$e^{i7\pi/4}$ Sx

Score 10 / 5 gates



$e^{i7\pi/4}$ Sxdg

Score 18 / 9 gates



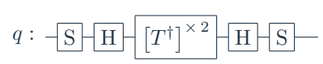
$e^{i7\pi/4}$ HX

Score 14 / 7 gates



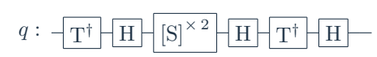
$e^{i7\pi/4}$ HY_q2

Score 12 / 6 gates



$e^{i7\pi/4}$ HZ

Score 14 / 7 gates



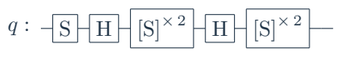
$e^{i7\pi/4}$ SX_q3

Score 14 / 7 gates



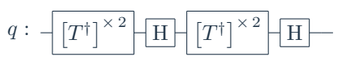
$e^{i7\pi/4}$ SY_q3

Score 14 / 7 gates



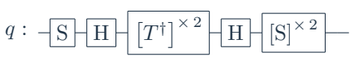
$e^{i7\pi/4}$ HS

Score 12 / 6 gates



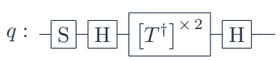
$e^{i7\pi/4}$ HSX_q6

Score 14 / 7 gates



$e^{i7\pi/4}$ HSY

Score 10 / 5 gates



$e^{i7\pi/4}$ HSdg

Score 16 / 8 gates



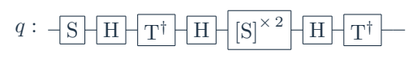
$e^{i7\pi/4}$ SH

Score 12 / 6 gates



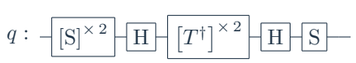
$e^{i7\pi/4}$ SHX

Score 16 / 8 gates



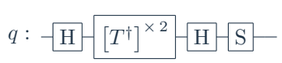
$e^{i7\pi/4}$ SHY_q2

Score 14 / 7 gates



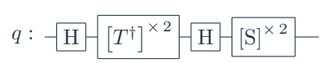
$e^{i7\pi/4}$ SHZ

Score 10 / 5 gates



$e^{i7\pi/4}$ SxY_q3

Score 12 / 6 gates



$e^{i7\pi/4}$ SxZ_q3

Score 18 / 9 gates



Certificate scope: all_words_in_fixed_H_S_T_Tdg_alphabet_including_nonclifford_intermediates. Phase is part of the target matrix.

PhasedXZ / $k = 0$

google_cz, google_fsim, google_sqrt_iswap, google_syc

$e^{i0\pi/4}$ I

Score 0 / 0 gates



$e^{i0\pi/4}$ X

Score $\frac{1003}{500}$ / 1 gates



$e^{i0\pi/4}$ Y

Score $\frac{1103}{500}$ / 1 gates



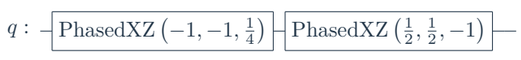
$e^{i0\pi/4}$ Z

Score $\frac{1003}{500}$ / 1 gates



$e^{i0\pi/4}$ H

Score $\frac{1203}{250}$ / 2 gates



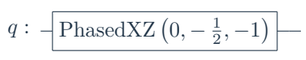
$e^{i0\pi/4}$ S

Score $\frac{1103}{500}$ / 1 gates



$e^{i0\pi/4}$ Sdg

Score $\frac{1103}{500}$ / 1 gates



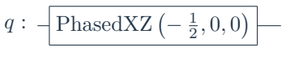
$e^{i0\pi/4}$ Sx

Score $\frac{1103}{500}$ / 1 gates



$e^{i0\pi/4}$ Sxdg

Score $\frac{1103}{500}$ / 1 gates



$e^{i0\pi/4}$ HX

Score $\frac{1203}{250}$ / 2 gates



$e^{i0\pi/4}$ HY_q2

Score $\frac{1203}{250}$ / 2 gates



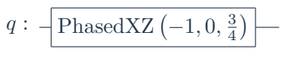
$e^{i0\pi/4}$ HZ

Score $\frac{1203}{250}$ / 2 gates



$e^{i0\pi/4}$ SX_q3

Score $\frac{1203}{500}$ / 1 gates



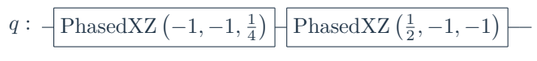
$e^{i0\pi/4}$ SY_q3

Score $\frac{1203}{500}$ / 1 gates



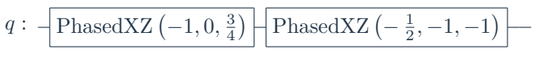
$e^{i0\pi/4}$ HS

Score $\frac{1153}{250}$ / 2 gates



$e^{i0\pi/4}$ HSX_q6

Score $\frac{1153}{250}$ / 2 gates



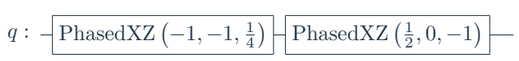
$e^{i0\pi/4}$ HSY

Score $\frac{1153}{250}$ / 2 gates



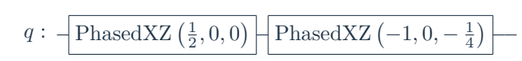
$e^{i0\pi/4}$ HSdg

Score $\frac{1153}{250}$ / 2 gates



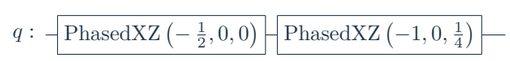
$e^{i0\pi/4}$ SH

Score $\frac{1153}{250}$ / 2 gates



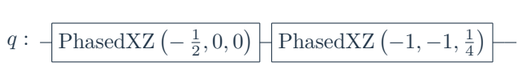
$e^{i0\pi/4}$ SHX

Score $\frac{1153}{250}$ / 2 gates



$e^{i0\pi/4}$ SHY_q2

Score $\frac{1153}{250}$ / 2 gates



$e^{i0\pi/4}$ SHZ

Score $\frac{1153}{250}$ / 2 gates



$e^{i0\pi/4}$ SxY_q3

Score $\frac{1203}{250}$ / 2 gates



$e^{i0\pi/4}$ SxZ_q3

Score $\frac{1203}{250}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

PhasedXZ / $k = 1$

google_cz, google_fsim, google_sqrt_iswap, google_syc

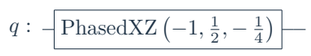
$e^{i1\pi/4}$ I

Score $\frac{1153}{250}$ / 2 gates



$e^{i1\pi/4}$ X

Score $\frac{1303}{500}$ / 1 gates



$e^{i1\pi/4}$ Y

Score $\frac{1303}{500}$ / 1 gates



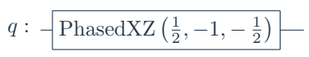
$e^{i1\pi/4}$ Z

Score $\frac{1153}{250}$ / 2 gates



$e^{i1\pi/4}$ H

Score $\frac{1203}{500}$ / 1 gates



$e^{i1\pi/4}$ S

Score $\frac{1103}{250}$ / 2 gates



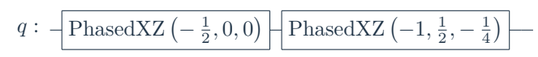
$e^{i1\pi/4}$ Sdg

Score $\frac{1103}{250}$ / 2 gates



$e^{i1\pi/4}$ Sx

Score $\frac{1203}{250}$ / 2 gates



$e^{i1\pi/4}$ Sxdg

Score $\frac{1203}{250}$ / 2 gates



$e^{i1\pi/4}$ HX

Score $\frac{1203}{500}$ / 1 gates



$e^{i1\pi/4}$ HY_q2

Score $\frac{1203}{500}$ / 1 gates



$e^{i1\pi/4}$ HZ

Score $\frac{1203}{500}$ / 1 gates



$e^{i1\pi/4}$ SX_q3

Score $\frac{1203}{500}$ / 1 gates



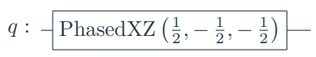
$e^{i1\pi/4}$ SY_q3

Score $\frac{1103}{500}$ / 1 gates



$e^{i1\pi/4}$ HS

Score $\frac{1303}{500}$ / 1 gates



$e^{i1\pi/4}$ HSX_q6

Score $\frac{1303}{500}$ / 1 gates



$e^{i1\pi/4}$ HSY

Score $\frac{1303}{500}$ / 1 gates



$e^{i1\pi/4}$ HSdg

Score $\frac{1303}{500}$ / 1 gates



$e^{i1\pi/4}$ SH

Score $\frac{1203}{500}$ / 1 gates



$e^{i1\pi/4}$ SHX

Score $\frac{1203}{500}$ / 1 gates



$e^{i1\pi/4}$ SHY_q2

Score $\frac{1203}{500}$ / 1 gates



$e^{i1\pi/4}$ SHZ

Score $\frac{1203}{500}$ / 1 gates



$e^{i1\pi/4}$ SxY_q3

Score $\frac{1103}{500}$ / 1 gates



$e^{i1\pi/4}$ SxZ_q3

Score $\frac{1053}{250}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

PhasedXZ / $k = 2$

google_cz, google_fsim, google_sqrt_iswap, google_syc

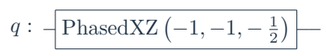
$e^{i2\pi/4}$ I

Score $\frac{1053}{250}$ / 2 gates



$e^{i2\pi/4}$ X

Score $\frac{1103}{500}$ / 1 gates



$e^{i2\pi/4}$ Y

Score $\frac{1003}{500}$ / 1 gates



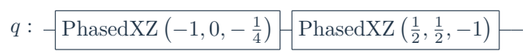
$e^{i2\pi/4}$ Z

Score $\frac{1053}{250}$ / 2 gates



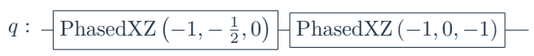
$e^{i2\pi/4}$ H

Score $\frac{1203}{250}$ / 2 gates



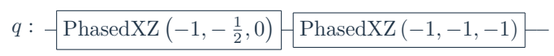
$e^{i2\pi/4}$ S

Score $\frac{1053}{250}$ / 2 gates



$e^{i2\pi/4}$ Sdg

Score $\frac{1053}{250}$ / 2 gates



$e^{i2\pi/4}$ Sx

Score $\frac{1053}{250}$ / 2 gates



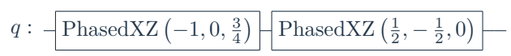
$e^{i2\pi/4}$ Sxdg

Score $\frac{1103}{500}$ / 1 gates



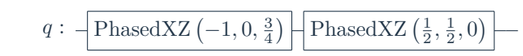
$e^{i2\pi/4}$ HX

Score $\frac{1203}{250}$ / 2 gates



$e^{i2\pi/4}$ HY_q2

Score $\frac{1203}{250}$ / 2 gates



$e^{i2\pi/4}$ HZ

Score $\frac{1203}{250}$ / 2 gates



$e^{i2\pi/4}$ SX_q3

Score $\frac{1203}{500}$ / 1 gates



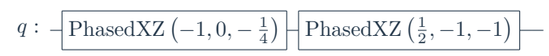
$e^{i2\pi/4}$ SY_q3

Score $\frac{1203}{500}$ / 1 gates



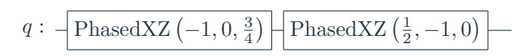
$e^{i2\pi/4}$ HS

Score $\frac{1153}{250}$ / 2 gates



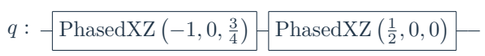
$e^{i2\pi/4}$ HSX_q6

Score $\frac{1153}{250}$ / 2 gates



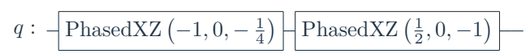
$e^{i2\pi/4}$ HSY

Score $\frac{1153}{250}$ / 2 gates



$e^{i2\pi/4}$ HSdg

Score $\frac{1153}{250}$ / 2 gates



$e^{i2\pi/4}$ SH

Score $\frac{1153}{250}$ / 2 gates



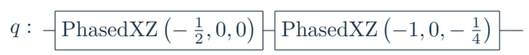
$e^{i2\pi/4}$ SHX

Score $\frac{1153}{250}$ / 2 gates



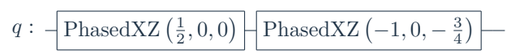
$e^{i2\pi/4}$ SHY_q2

Score $\frac{1153}{250}$ / 2 gates



$e^{i2\pi/4}$ SHZ

Score $\frac{1153}{250}$ / 2 gates



$e^{i2\pi/4}$ SxY_q3

Score $\frac{1203}{250}$ / 2 gates



$e^{i2\pi/4}$ SxZ_q3

Score $\frac{1203}{250}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

PhasedXZ / $k = 3$

google_cz, google_fsim, google_sqrt_iswap, google_syc

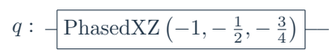
$e^{i3\pi/4}$ I

Score $\frac{1153}{250}$ / 2 gates



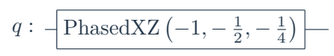
$e^{i3\pi/4}$ X

Score $\frac{1303}{500}$ / 1 gates



$e^{i3\pi/4}$ Y

Score $\frac{1303}{500}$ / 1 gates



$e^{i3\pi/4}$ Z

Score $\frac{1153}{250}$ / 2 gates



$e^{i3\pi/4}$ H

Score $\frac{1103}{250}$ / 2 gates



$e^{i3\pi/4}$ S

Score $\frac{1103}{250}$ / 2 gates



$e^{i3\pi/4}$ Sdg

Score $\frac{1103}{250}$ / 2 gates



$e^{i3\pi/4}$ Sx

Score $\frac{1203}{250}$ / 2 gates



$e^{i3\pi/4}$ Sxdg

Score $\frac{1203}{250}$ / 2 gates



$e^{i3\pi/4}$ HX

Score $\frac{1103}{250}$ / 2 gates



$e^{i3\pi/4}$ HY_q2

Score $\frac{1103}{250}$ / 2 gates



$e^{i3\pi/4}$ HZ

Score $\frac{1103}{250}$ / 2 gates



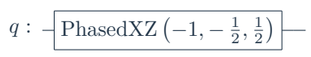
$e^{i3\pi/4}$ SX_q3

Score $\frac{1103}{500}$ / 1 gates



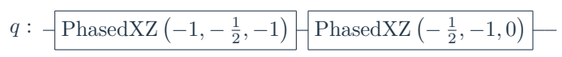
$e^{i3\pi/4}$ SY_q3

Score $\frac{1203}{500}$ / 1 gates



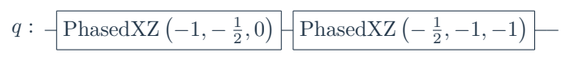
$e^{i3\pi/4}$ HS

Score $\frac{1103}{250}$ / 2 gates



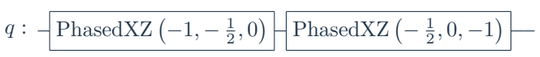
$e^{i3\pi/4}$ HSX_q6

Score $\frac{1103}{250}$ / 2 gates



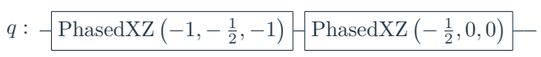
$e^{i3\pi/4}$ HSY

Score $\frac{1103}{250}$ / 2 gates



$e^{i3\pi/4}$ HSdg

Score $\frac{1103}{250}$ / 2 gates



$e^{i3\pi/4}$ SH

Score $\frac{1103}{250}$ / 2 gates



$e^{i3\pi/4}$ SHX

Score $\frac{1103}{250}$ / 2 gates



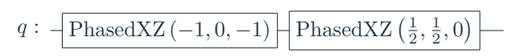
$e^{i3\pi/4}$ SHY_q2

Score $\frac{1103}{250}$ / 2 gates



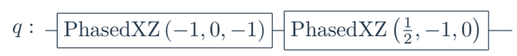
$e^{i3\pi/4}$ SHZ

Score $\frac{1103}{250}$ / 2 gates



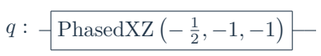
$e^{i3\pi/4}$ SxY_q3

Score $\frac{1053}{250}$ / 2 gates



$e^{i3\pi/4}$ SxZ_q3

Score $\frac{1103}{500}$ / 1 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

PhasedXZ / $k = 4$

google_cz, google_fsim, google_sqrt_iswap, google_syc

$e^{i4\pi/4}$ I

Score $\frac{1003}{250}$ / 2 gates



$e^{i4\pi/4}$ X

Score $\frac{1003}{500}$ / 1 gates



$e^{i4\pi/4}$ Y

Score $\frac{1103}{500}$ / 1 gates



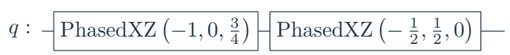
$e^{i4\pi/4}$ Z

Score $\frac{1003}{250}$ / 2 gates



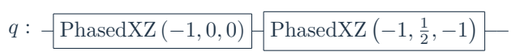
$e^{i4\pi/4}$ H

Score $\frac{1203}{250}$ / 2 gates



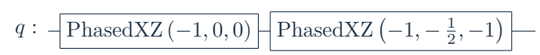
$e^{i4\pi/4}$ S

Score $\frac{1053}{250}$ / 2 gates



$e^{i4\pi/4}$ Sdg

Score $\frac{1053}{250}$ / 2 gates



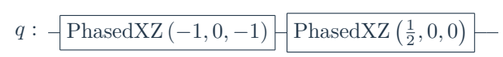
$e^{i4\pi/4}$ Sx

Score $\frac{1053}{250}$ / 2 gates



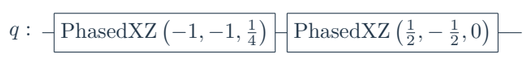
$e^{i4\pi/4}$ Sxdg

Score $\frac{1053}{250}$ / 2 gates



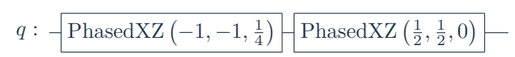
$e^{i4\pi/4}$ HX

Score $\frac{1203}{250}$ / 2 gates



$e^{i4\pi/4}$ HY_q2

Score $\frac{1203}{250}$ / 2 gates



$e^{i4\pi/4}$ HZ

Score $\frac{1203}{250}$ / 2 gates



$e^{i4\pi/4}$ SX_q3

Score $\frac{1203}{500}$ / 1 gates



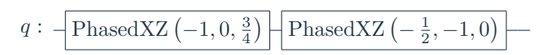
$e^{i4\pi/4}$ SY_q3

Score $\frac{1203}{500}$ / 1 gates



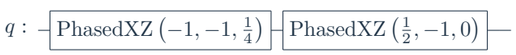
$e^{i4\pi/4}$ HS

Score $\frac{1153}{250}$ / 2 gates



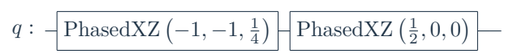
$e^{i4\pi/4}$ HSX_q6

Score $\frac{1153}{250}$ / 2 gates



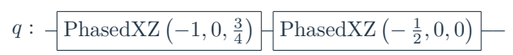
$e^{i4\pi/4}$ HSY

Score $\frac{1153}{250}$ / 2 gates



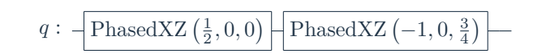
$e^{i4\pi/4}$ HSdg

Score $\frac{1153}{250}$ / 2 gates



$e^{i4\pi/4}$ SH

Score $\frac{1153}{250}$ / 2 gates



$e^{i4\pi/4}$ SHX

Score $\frac{1153}{250}$ / 2 gates



$e^{i4\pi/4}$ SHY_q2

Score $\frac{1153}{250}$ / 2 gates



$e^{i4\pi/4}$ SHZ

Score $\frac{1153}{250}$ / 2 gates



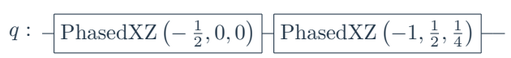
$e^{i4\pi/4}$ SxY_q3

Score $\frac{1203}{250}$ / 2 gates



$e^{i4\pi/4}$ SxZ_q3

Score $\frac{1203}{250}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

PhasedXZ / $k = 5$

google_cz, google_fsim, google_sqrt_iswap, google_syc

$e^{i5\pi/4}$ I

Score $\frac{1153}{250}$ / 2 gates



$e^{i5\pi/4}$ X

Score $\frac{1303}{500}$ / 1 gates



$e^{i5\pi/4}$ Y

Score $\frac{1303}{500}$ / 1 gates



$e^{i5\pi/4}$ Z

Score $\frac{1153}{250}$ / 2 gates



$e^{i5\pi/4}$ H

Score $\frac{1103}{250}$ / 2 gates



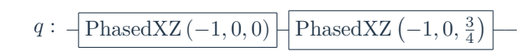
$e^{i5\pi/4}$ S

Score $\frac{1103}{250}$ / 2 gates



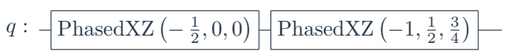
$e^{i5\pi/4}$ Sdg

Score $\frac{1103}{250}$ / 2 gates



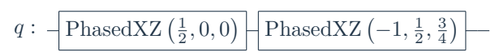
$e^{i5\pi/4}$ Sx

Score $\frac{1203}{250}$ / 2 gates



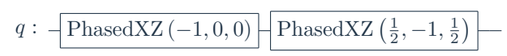
$e^{i5\pi/4}$ Sxdg

Score $\frac{1203}{250}$ / 2 gates



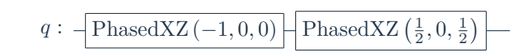
$e^{i5\pi/4}$ HX

Score $\frac{1103}{250}$ / 2 gates



$e^{i5\pi/4}$ HY_q2

Score $\frac{1103}{250}$ / 2 gates



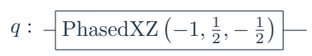
$e^{i5\pi/4}$ HZ

Score $\frac{1103}{250}$ / 2 gates



$e^{i5\pi/4}$ SX_q3

Score $\frac{1203}{500}$ / 1 gates



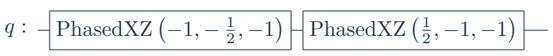
$e^{i5\pi/4}$ SY_q3

Score $\frac{1103}{500}$ / 1 gates



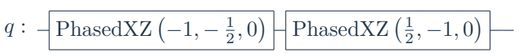
$e^{i5\pi/4}$ HS

Score $\frac{1103}{250}$ / 2 gates



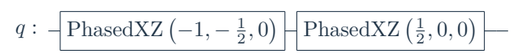
$e^{i5\pi/4}$ HSX_q6

Score $\frac{1103}{250}$ / 2 gates



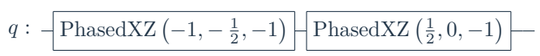
$e^{i5\pi/4}$ HSY

Score $\frac{1103}{250}$ / 2 gates



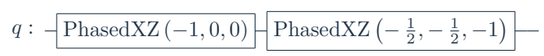
$e^{i5\pi/4}$ HSdg

Score $\frac{1103}{250}$ / 2 gates



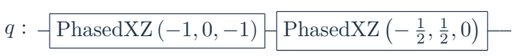
$e^{i5\pi/4}$ SH

Score $\frac{1103}{250}$ / 2 gates



$e^{i5\pi/4}$ SHX

Score $\frac{1103}{250}$ / 2 gates



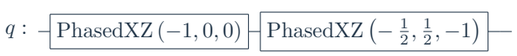
$e^{i5\pi/4}$ SHY_q2

Score $\frac{1103}{250}$ / 2 gates



$e^{i5\pi/4}$ SHZ

Score $\frac{1103}{250}$ / 2 gates



$e^{i5\pi/4}$ SxY_q3

Score $\frac{1053}{250}$ / 2 gates



$e^{i5\pi/4}$ SxZ_q3

Score $\frac{1103}{500}$ / 1 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

PhasedXZ / $k = 6$

google_cz, google_fsim, google_sqrt_iswap, google_syc

$e^{i6\pi/4}$ I

Score $\frac{1053}{250}$ / 2 gates



$e^{i6\pi/4}$ X

Score $\frac{1103}{500}$ / 1 gates



$e^{i6\pi/4}$ Y

Score $\frac{1003}{500}$ / 1 gates



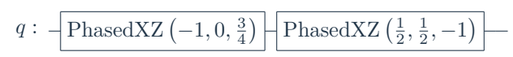
$e^{i6\pi/4}$ Z

Score $\frac{1053}{250}$ / 2 gates



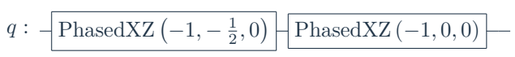
$e^{i6\pi/4}$ H

Score $\frac{1203}{250}$ / 2 gates



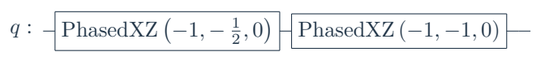
$e^{i6\pi/4}$ S

Score $\frac{1053}{250}$ / 2 gates



$e^{i6\pi/4}$ Sdg

Score $\frac{1053}{250}$ / 2 gates



$e^{i6\pi/4}$ Sx

Score $\frac{1103}{500}$ / 1 gates



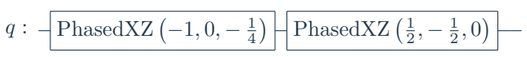
$e^{i6\pi/4}$ Sxdg

Score $\frac{1053}{250}$ / 2 gates



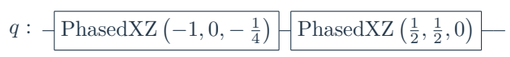
$e^{i6\pi/4}$ HX

Score $\frac{1203}{250}$ / 2 gates



$e^{i6\pi/4}$ HY_q2

Score $\frac{1203}{250}$ / 2 gates



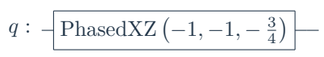
$e^{i6\pi/4}$ HZ

Score $\frac{1203}{250}$ / 2 gates



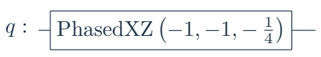
$e^{i6\pi/4}$ SX_q3

Score $\frac{1203}{500}$ / 1 gates



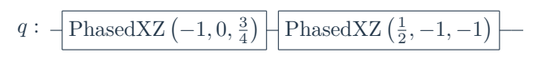
$e^{i6\pi/4}$ SY_q3

Score $\frac{1203}{500}$ / 1 gates



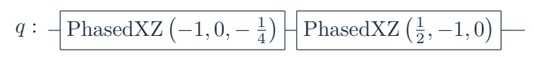
$e^{i6\pi/4}$ HS

Score $\frac{1153}{250}$ / 2 gates



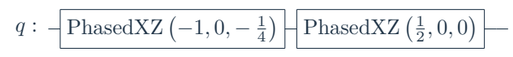
$e^{i6\pi/4}$ HSX_q6

Score $\frac{1153}{250}$ / 2 gates



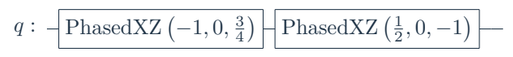
$e^{i6\pi/4}$ HSY

Score $\frac{1153}{250}$ / 2 gates



$e^{i6\pi/4}$ HSdg

Score $\frac{1153}{250}$ / 2 gates



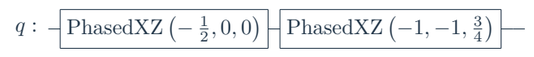
$e^{i6\pi/4}$ SH

Score $\frac{1153}{250}$ / 2 gates



$e^{i6\pi/4}$ SHX

Score $\frac{1153}{250}$ / 2 gates



$e^{i6\pi/4}$ SHY_q2

Score $\frac{1153}{250}$ / 2 gates



$e^{i6\pi/4}$ SHZ

Score $\frac{1153}{250}$ / 2 gates



$e^{i6\pi/4}$ SxY_q3

Score $\frac{1203}{250}$ / 2 gates



$e^{i6\pi/4}$ SxZ_q3

Score $\frac{1203}{250}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

PhasedXZ / $k = 7$

google_cz, google_fsim, google_sqrt_iswap, google_syc

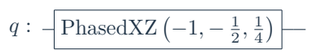
$e^{i7\pi/4}$ I

Score $\frac{1153}{250}$ / 2 gates



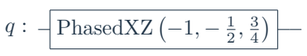
$e^{i7\pi/4}$ X

Score $\frac{1303}{500}$ / 1 gates



$e^{i7\pi/4}$ Y

Score $\frac{1303}{500}$ / 1 gates



$e^{i7\pi/4}$ Z

Score $\frac{1153}{250}$ / 2 gates



$e^{i7\pi/4}$ H

Score $\frac{1203}{500}$ / 1 gates



$e^{i7\pi/4}$ S

Score $\frac{1103}{250}$ / 2 gates



$e^{i7\pi/4}$ Sdg

Score $\frac{1103}{250}$ / 2 gates



$e^{i7\pi/4}$ Sx

Score $\frac{1203}{250}$ / 2 gates



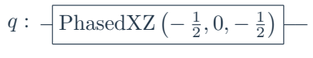
$e^{i7\pi/4}$ Sxdg

Score $\frac{1203}{250}$ / 2 gates



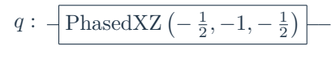
$e^{i7\pi/4}$ HX

Score $\frac{1203}{500}$ / 1 gates



$e^{i7\pi/4}$ HY_q2

Score $\frac{1203}{500}$ / 1 gates



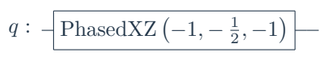
$e^{i7\pi/4}$ HZ

Score $\frac{1203}{500}$ / 1 gates



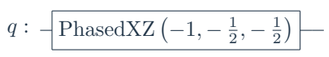
$e^{i7\pi/4}$ SX_q3

Score $\frac{1103}{500}$ / 1 gates



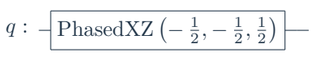
$e^{i7\pi/4}$ SY_q3

Score $\frac{1203}{500}$ / 1 gates



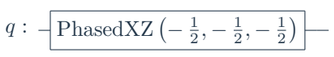
$e^{i7\pi/4}$ HS

Score $\frac{1303}{500}$ / 1 gates



$e^{i7\pi/4}$ HSX_q6

Score $\frac{1303}{500}$ / 1 gates



$e^{i7\pi/4}$ HSY

Score $\frac{1303}{500}$ / 1 gates



$e^{i7\pi/4}$ HSdg

Score $\frac{1303}{500}$ / 1 gates



$e^{i7\pi/4}$ SH

Score $\frac{1203}{500}$ / 1 gates



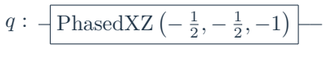
$e^{i7\pi/4}$ SHX

Score $\frac{1203}{500}$ / 1 gates



$e^{i7\pi/4}$ SHY_q2

Score $\frac{1203}{500}$ / 1 gates



$e^{i7\pi/4}$ SHZ

Score $\frac{1203}{500}$ / 1 gates



$e^{i7\pi/4}$ SxY_q3

Score $\frac{1103}{500}$ / 1 gates



$e^{i7\pi/4}$ SxZ_q3

Score $\frac{1053}{250}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

PhasedX|Rz / $k = 0$

quantinuum_zzmax, quantinuum_zzphase, tket_tk2

$e^{i0\pi/4}$ **I**

Score 0 / 0 gates

q : —

$e^{i0\pi/4}$ **X**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **Y**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **Z**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **H**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **S**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **Sdg**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **Sx**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **Sxdg**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **HX**

Score $\frac{601}{250}$ / 1 gates

q : $\boxed{\mathrm{PhasedX}\left(-\frac{1}{2}, -\frac{1}{2}\right)}$

$e^{i0\pi/4}$ **HY_q2**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **HZ**

Score $\frac{601}{250}$ / 1 gates

q : $\boxed{\mathrm{PhasedX}\left(-\frac{1}{2}, \frac{1}{2}\right)}$

$e^{i0\pi/4}$ **SX_q3**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **SY_q3**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **HS**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **HSX_q6**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **HSY**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **HSdg**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **SH**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **SHX**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **SHY_q2**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **SHZ**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **SxY_q3**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **SxZ_q3**

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

PhasedX|Rz / $k = 1$

quantinuum_zzmax, quantinuum_zzphase, tket_tk2

$e^{i1\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Sdg

Score $\frac{1101}{500}$ / 1 gates

$$q : \boxed{R_z\left(-\tfrac{\pi}{2}\right)}$$

$e^{i1\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Sxdg

Score $\frac{551}{250}$ / 1 gates

$$q : \boxed{\mathrm{PhasedX}\left(-\tfrac{1}{2}, 0\right)}$$

$e^{i1\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HS

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{R_z\left(-\tfrac{\pi}{2}\right)}\boxed{\mathrm{PhasedX}\left(-\tfrac{1}{2}, 0\right)}$$

$e^{i1\pi/4}$ HSX_q6

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{R_z\left(-\tfrac{\pi}{2}\right)}\boxed{\mathrm{PhasedX}\left(-\tfrac{1}{2}, -1\right)}$$

$e^{i1\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SH

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{\mathrm{PhasedX}\left(-\tfrac{1}{2}, 0\right)}\boxed{R_z\left(-\tfrac{\pi}{2}\right)}$$

$e^{i1\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SHY_q2

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{\mathrm{PhasedX}\left(-\tfrac{3}{2}, 0\right)}\boxed{R_z\left(\tfrac{3\pi}{2}\right)}$$

$e^{i1\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

PhasedX|Rz / $k = 2$

quantinuum_zzmax, quantinuum_zzphase, tket_tk2

$e^{i2\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ X

Score $\frac{501}{250}$ / 1 gates



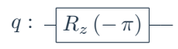
$e^{i2\pi/4}$ Y

Score $\frac{551}{250}$ / 1 gates



$e^{i2\pi/4}$ Z

Score $\frac{1001}{500}$ / 1 gates



$e^{i2\pi/4}$ H

Score $\frac{2203}{500}$ / 2 gates



$e^{i2\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HY_q2

Score $\frac{2203}{500}$ / 2 gates



$e^{i2\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SX_q3

Score $\frac{601}{250}$ / 1 gates



$e^{i2\pi/4}$ SY_q3

Score $\frac{601}{250}$ / 1 gates



$e^{i2\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SxY_q3

Score $\frac{2103}{500}$ / 2 gates



$e^{i2\pi/4}$ SxZ_q3

Score $\frac{2103}{500}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

PhasedX|Rz / $k = 3$

quantinuum_zzmax, quantinuum_zzphase, tket_tk2

$e^{i3\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ Z

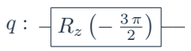
Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ S

Score $\frac{1101}{500}$ / 1 gates



$e^{i3\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ Sx

Score $\frac{551}{250}$ / 1 gates



$e^{i3\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HSY

Score $\frac{2203}{500}$ / 2 gates



$e^{i3\pi/4}$ HSdg

Score $\frac{2203}{500}$ / 2 gates

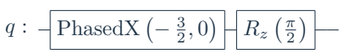


$e^{i3\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SHX

Score $\frac{2203}{500}$ / 2 gates

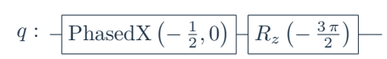


$e^{i3\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SHZ

Score $\frac{2203}{500}$ / 2 gates



$e^{i3\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

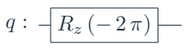
Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

PhasedX|Rz / $k = 4$

quantinuum_zzmax, quantinuum_zzphase, tket_tk2

$e^{i4\pi/4}$ I

Score $\frac{1001}{500}$ / 1 gates



$e^{i4\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HX

Score $\frac{601}{250}$ / 1 gates

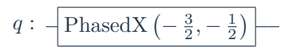


$e^{i4\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HZ

Score $\frac{601}{250}$ / 1 gates



$e^{i4\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

PhasedX|Rz / $k = 5$

quantinuum_zzmax, quantinuum_zzphase, tket_tk2

$e^{i5\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Sdg

Score $\frac{1101}{500}$ / 1 gates

$$q : \boxed{R_z\left(\frac{3\pi}{2}\right)}$$

$e^{i5\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Sxdg

Score $\frac{551}{250}$ / 1 gates

$$q : \boxed{\mathrm{PhasedX}\left(-\frac{3}{2}, -1\right)}$$

$e^{i5\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HS

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{R_z\left(-\frac{\pi}{2}\right)}\boxed{\mathrm{PhasedX}\left(-\frac{3}{2}, -1\right)}$$

$e^{i5\pi/4}$ HSX_q6

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{R_z\left(-\frac{\pi}{2}\right)}\boxed{\mathrm{PhasedX}\left(-\frac{3}{2}, 0\right)}$$

$e^{i5\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SH

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{\mathrm{PhasedX}\left(-\frac{1}{2}, 0\right)}\boxed{R_z\left(\frac{3\pi}{2}\right)}$$

$e^{i5\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SHY_q2

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{\mathrm{PhasedX}\left(-\frac{3}{2}, 0\right)}\boxed{R_z\left(-\frac{\pi}{2}\right)}$$

$e^{i5\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

PhasedX|Rz / $k = 6$

quantinuum_zzmax, quantinuum_zzphase, tket_tk2

$e^{i6\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ X

Score $\frac{501}{250}$ / 1 gates



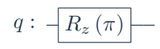
$e^{i6\pi/4}$ Y

Score $\frac{551}{250}$ / 1 gates



$e^{i6\pi/4}$ Z

Score $\frac{1001}{500}$ / 1 gates



$e^{i6\pi/4}$ H

Score $\frac{2203}{500}$ / 2 gates



$e^{i6\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HY_q2

Score $\frac{2203}{500}$ / 2 gates



$e^{i6\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SX_q3

Score $\frac{601}{250}$ / 1 gates



$e^{i6\pi/4}$ SY_q3

Score $\frac{601}{250}$ / 1 gates



$e^{i6\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SxY_q3

Score $\frac{2103}{500}$ / 2 gates



$e^{i6\pi/4}$ SxZ_q3

Score $\frac{2103}{500}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

PhasedX|Rz / $k = 7$

quantinuum_zzmax, quantinuum_zzphase, tket_tk2

$e^{i7\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ Z

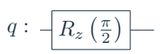
Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ S

Score $\frac{1101}{500}$ / 1 gates

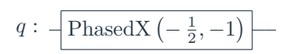


$e^{i7\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ Sx

Score $\frac{551}{250}$ / 1 gates



$e^{i7\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HSY

Score $\frac{2203}{500}$ / 2 gates



$e^{i7\pi/4}$ HSdg

Score $\frac{2203}{500}$ / 2 gates

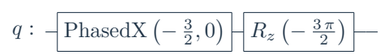


$e^{i7\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SHX

Score $\frac{2203}{500}$ / 2 gates

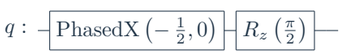


$e^{i7\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SHZ

Score $\frac{2203}{500}$ / 2 gates



$e^{i7\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

R / $k = 0$

iqm_cz

$e^{i0\pi/4}$ I

Score 0 / 0 gates



$e^{i0\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ Sx

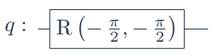
Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HX

Score $\frac{601}{250}$ / 1 gates

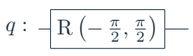


$e^{i0\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HZ

Score $\frac{601}{250}$ / 1 gates



$e^{i0\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

R / $k = 1$

iqm_cz

$e^{i1\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Sdg

Score $\frac{551}{125}$ / 2 gates

$$q : \boxed{\mathrm{R}\left(-\pi, 0\right)} \boxed{\mathrm{R}\left(-\pi, \frac{3\pi}{4}\right)}$$

$e^{i1\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Sxdg

Score $\frac{551}{250}$ / 1 gates

$$q : \boxed{\mathrm{R}\left(-\frac{\pi}{2}, 0\right)}$$

$e^{i1\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HS

Score $\frac{576}{125}$ / 2 gates

$$q : \boxed{\mathrm{R}\left(-\frac{\pi}{2}, 0\right)} \boxed{\mathrm{R}\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)}$$

$e^{i1\pi/4}$ HSX_q6

Score $\frac{576}{125}$ / 2 gates

$$q : \boxed{\mathrm{R}\left(-\frac{3\pi}{2}, 0\right)} \boxed{\mathrm{R}\left(-\frac{3\pi}{2}, \frac{\pi}{2}\right)}$$

$e^{i1\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SH

Score $\frac{576}{125}$ / 2 gates

$$q : \boxed{\mathrm{R}\left(-\frac{3\pi}{2}, 0\right)} \boxed{\mathrm{R}\left(-\pi, \frac{3\pi}{4}\right)}$$

$e^{i1\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SHY_q2

Score $\frac{576}{125}$ / 2 gates

$$q : \boxed{\mathrm{R}\left(-\frac{\pi}{2}, 0\right)} \boxed{\mathrm{R}\left(-\pi, \frac{3\pi}{4}\right)}$$

$e^{i1\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

R / $k = 2$

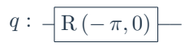
iqm_cz

$e^{i2\pi/4}$ I

Impossible: phase outside the local determinant group.

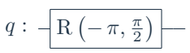
$e^{i2\pi/4}$ X

Score $\frac{501}{250}$ / 1 gates



$e^{i2\pi/4}$ Y

Score $\frac{551}{250}$ / 1 gates



$e^{i2\pi/4}$ Z

Score $\frac{526}{125}$ / 2 gates



$e^{i2\pi/4}$ H

Score $\frac{551}{125}$ / 2 gates



$e^{i2\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HY_q2

Score $\frac{551}{125}$ / 2 gates

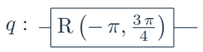


$e^{i2\pi/4}$ HZ

Impossible: phase outside the local determinant group.

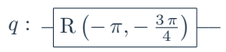
$e^{i2\pi/4}$ SX_q3

Score $\frac{601}{250}$ / 1 gates



$e^{i2\pi/4}$ SY_q3

Score $\frac{601}{250}$ / 1 gates



$e^{i2\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SxY_q3

Score $\frac{551}{125}$ / 2 gates



$e^{i2\pi/4}$ SxZ_q3

Score $\frac{551}{125}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

R / $k = 3$

iqm_cz

$e^{i3\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ S

Score $\frac{551}{125}$ / 2 gates

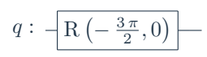


$e^{i3\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ Sx

Score $\frac{551}{250}$ / 1 gates



$e^{i3\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HS

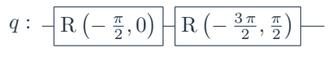
Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

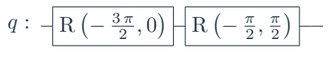
$e^{i3\pi/4}$ HSY

Score $\frac{576}{125}$ / 2 gates



$e^{i3\pi/4}$ HSdg

Score $\frac{576}{125}$ / 2 gates



$e^{i3\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SHX

Score $\frac{576}{125}$ / 2 gates

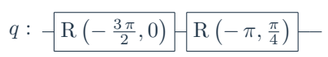


$e^{i3\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SHZ

Score $\frac{576}{125}$ / 2 gates



$e^{i3\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

R / $k = 4$

iqm_cz

$e^{i4\pi/4}$ I

Score $\frac{501}{250}$ / 1 gates

$$q : \boxed{\mathrm{R}\left(-2\,\pi, -\pi\right)}$$

$e^{i4\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HX

Score $\frac{601}{250}$ / 1 gates

$$q : \boxed{\mathrm{R}\left(-\frac{3\,\pi}{2}, \frac{\pi}{2}\right)}$$

$e^{i4\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HZ

Score $\frac{601}{250}$ / 1 gates

$$q : \boxed{\mathrm{R}\left(-\frac{3\,\pi}{2}, -\frac{\pi}{2}\right)}$$

$e^{i4\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

R / $k = 5$

iqm_cz

$e^{i5\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Sdg

Score $\frac{551}{125}$ / 2 gates

$$q : \boxed{\mathrm{R}\left(-\pi, 0\right)} \boxed{\mathrm{R}\left(-\pi, -\frac{\pi}{4}\right)}$$

$e^{i5\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Sxdg

Score $\frac{551}{250}$ / 1 gates

$$q : \boxed{\mathrm{R}\left(-\frac{3\pi}{2}, -\pi\right)}$$

$e^{i5\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HS

Score $\frac{576}{125}$ / 2 gates

$$q : \boxed{\mathrm{R}\left(-\frac{\pi}{2}, 0\right)} \boxed{\mathrm{R}\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right)}$$

$e^{i5\pi/4}$ HSX_q6

Score $\frac{576}{125}$ / 2 gates

$$q : \boxed{\mathrm{R}\left(-\frac{3\pi}{2}, 0\right)} \boxed{\mathrm{R}\left(-\frac{\pi}{2}, -\frac{\pi}{2}\right)}$$

$e^{i5\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SH

Score $\frac{576}{125}$ / 2 gates

$$q : \boxed{\mathrm{R}\left(-\frac{3\pi}{2}, 0\right)} \boxed{\mathrm{R}\left(-\pi, -\frac{\pi}{4}\right)}$$

$e^{i5\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SHY_q2

Score $\frac{576}{125}$ / 2 gates

$$q : \boxed{\mathrm{R}\left(-\frac{\pi}{2}, 0\right)} \boxed{\mathrm{R}\left(-\pi, -\frac{\pi}{4}\right)}$$

$e^{i5\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

R / $k = 6$

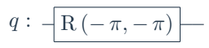
iqm_cz

$e^{i6\pi/4}$ I

Impossible: phase outside the local determinant group.

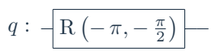
$e^{i6\pi/4}$ X

Score $\frac{501}{250}$ / 1 gates



$e^{i6\pi/4}$ Y

Score $\frac{551}{250}$ / 1 gates



$e^{i6\pi/4}$ Z

Score $\frac{526}{125}$ / 2 gates



$e^{i6\pi/4}$ H

Score $\frac{551}{125}$ / 2 gates



$e^{i6\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HY_q2

Score $\frac{551}{125}$ / 2 gates



$e^{i6\pi/4}$ HZ

Impossible: phase outside the local determinant group.

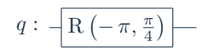
$e^{i6\pi/4}$ SX_q3

Score $\frac{601}{250}$ / 1 gates



$e^{i6\pi/4}$ SY_q3

Score $\frac{601}{250}$ / 1 gates



$e^{i6\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SHY_q2

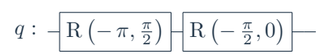
Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SxY_q3

Score $\frac{551}{125}$ / 2 gates



$e^{i6\pi/4}$ SxZ_q3

Score $\frac{551}{125}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

R / $k = 7$

iqm_cz

$e^{i7\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ S

Score $\frac{551}{125}$ / 2 gates

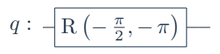


$e^{i7\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ Sx

Score $\frac{551}{250}$ / 1 gates



$e^{i7\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HSY

Score $\frac{576}{125}$ / 2 gates



$e^{i7\pi/4}$ HSdg

Score $\frac{576}{125}$ / 2 gates



$e^{i7\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SHX

Score $\frac{576}{125}$ / 2 gates

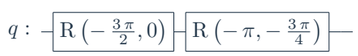


$e^{i7\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SHZ

Score $\frac{576}{125}$ / 2 gates



$e^{i7\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Rx / $k = 0$

cz_rz_rx

$e^{i0\pi/4}$ I

Score 0 / 0 gates



$e^{i0\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ Sx

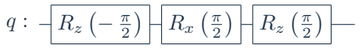
Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HX

Score $\frac{3303}{500}$ / 3 gates

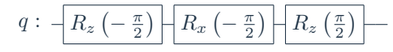


$e^{i0\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HZ

Score $\frac{3303}{500}$ / 3 gates



$e^{i0\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Rx / $k = 1$

cz_rz_rx

$e^{i1\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ H

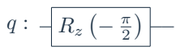
Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Sdg

Score $\frac{1101}{500}$ / 1 gates

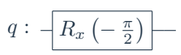


$e^{i1\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Sxdg

Score $\frac{1101}{500}$ / 1 gates



$e^{i1\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SX_q3

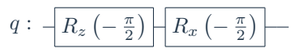
Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

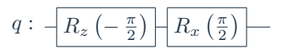
$e^{i1\pi/4}$ HS

Score $\frac{1101}{250}$ / 2 gates



$e^{i1\pi/4}$ HSX_q6

Score $\frac{1101}{250}$ / 2 gates



$e^{i1\pi/4}$ HSY

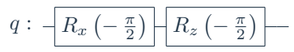
Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SH

Score $\frac{1101}{250}$ / 2 gates

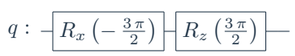


$e^{i1\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SHY_q2

Score $\frac{1101}{250}$ / 2 gates



$e^{i1\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Rx / $k = 2$

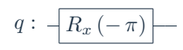
cz_rz_rx

$e^{i2\pi/4}$ I

Impossible: phase outside the local determinant group.

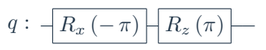
$e^{i2\pi/4}$ X

Score $\frac{1001}{500}$ / 1 gates



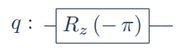
$e^{i2\pi/4}$ Y

Score $\frac{1001}{250}$ / 2 gates



$e^{i2\pi/4}$ Z

Score $\frac{1001}{500}$ / 1 gates



$e^{i2\pi/4}$ H

Score $\frac{3303}{500}$ / 3 gates



$e^{i2\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ Sxdg

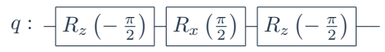
Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HY_q2

Score $\frac{3303}{500}$ / 3 gates

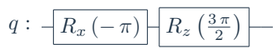


$e^{i2\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SX_q3

Score $\frac{1051}{250}$ / 2 gates



$e^{i2\pi/4}$ SY_q3

Score $\frac{1051}{250}$ / 2 gates



$e^{i2\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SHY_q2

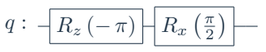
Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SxY_q3

Score $\frac{1051}{250}$ / 2 gates



$e^{i2\pi/4}$ SxZ_q3

Score $\frac{1051}{250}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Rx / $k = 3$

cz_rz_rx

$e^{i3\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ Z

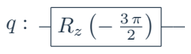
Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ S

Score $\frac{1101}{500}$ / 1 gates

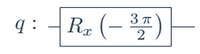


$e^{i3\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ Sx

Score $\frac{1101}{500}$ / 1 gates



$e^{i3\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HS

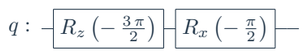
Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

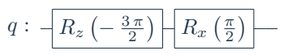
$e^{i3\pi/4}$ HSY

Score $\frac{1101}{250}$ / 2 gates



$e^{i3\pi/4}$ HSdg

Score $\frac{1101}{250}$ / 2 gates

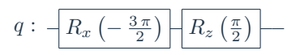


$e^{i3\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SHX

Score $\frac{1101}{250}$ / 2 gates



$e^{i3\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SHZ

Score $\frac{1101}{250}$ / 2 gates



$e^{i3\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

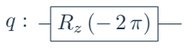
Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Rx / $k = 4$

cz_rz_rx

$e^{i4\pi/4}$ I

Score $\frac{1001}{500}$ / 1 gates



$e^{i4\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Sx

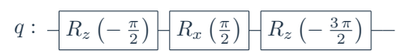
Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HX

Score $\frac{3303}{500}$ / 3 gates

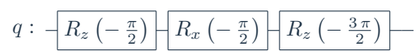


$e^{i4\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HZ

Score $\frac{3303}{500}$ / 3 gates



$e^{i4\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Rx / $k = 5$

cz_rz_rx

$e^{i5\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ H

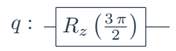
Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Sdg

Score $\frac{1101}{500}$ / 1 gates

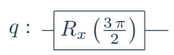


$e^{i5\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Sxdg

Score $\frac{1101}{500}$ / 1 gates



$e^{i5\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SX_q3

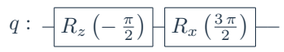
Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HS

Score $\frac{1101}{250}$ / 2 gates



$e^{i5\pi/4}$ HSX_q6

Score $\frac{1101}{250}$ / 2 gates



$e^{i5\pi/4}$ HSY

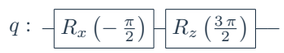
Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SH

Score $\frac{1101}{250}$ / 2 gates

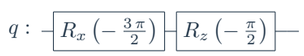


$e^{i5\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SHY_q2

Score $\frac{1101}{250}$ / 2 gates



$e^{i5\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Rx / $k = 6$

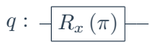
cz_rz_rx

$e^{i6\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ X

Score $\frac{1001}{500}$ / 1 gates



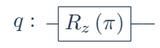
$e^{i6\pi/4}$ Y

Score $\frac{1001}{250}$ / 2 gates



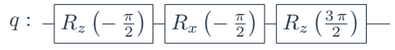
$e^{i6\pi/4}$ Z

Score $\frac{1001}{500}$ / 1 gates



$e^{i6\pi/4}$ H

Score $\frac{3303}{500}$ / 3 gates



$e^{i6\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ Sxdg

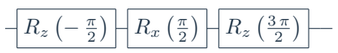
Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HY_q2

Score $\frac{3303}{500}$ / 3 gates



$e^{i6\pi/4}$ HZ

Impossible: phase outside the local determinant group.

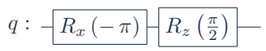
$e^{i6\pi/4}$ SX_q3

Score $\frac{1051}{250}$ / 2 gates



$e^{i6\pi/4}$ SY_q3

Score $\frac{1051}{250}$ / 2 gates



$e^{i6\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

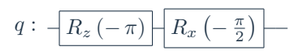
$e^{i6\pi/4}$ SxY_q3

Score $\frac{1051}{250}$ / 2 gates



$e^{i6\pi/4}$ SxZ_q3

Score $\frac{1051}{250}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Rx / $k = 7$

cz_rz_rx

$e^{i7\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ S

Score $\frac{1101}{500}$ / 1 gates

$$q : \boxed{R_z\left(\frac{\pi}{2}\right)}$$

$e^{i7\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ Sx

Score $\frac{1101}{500}$ / 1 gates

$$q : \boxed{R_x\left(\frac{\pi}{2}\right)}$$

$e^{i7\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HSY

Score $\frac{1101}{250}$ / 2 gates

$$q : \boxed{R_z\left(-\frac{3\pi}{2}\right)}\boxed{R_x\left(\frac{3\pi}{2}\right)}$$

$e^{i7\pi/4}$ HSdg

Score $\frac{1101}{250}$ / 2 gates

$$q : \boxed{R_z\left(-\frac{3\pi}{2}\right)}\boxed{R_x\left(-\frac{3\pi}{2}\right)}$$

$e^{i7\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SHX

Score $\frac{1101}{250}$ / 2 gates

$$q : \boxed{R_x\left(-\frac{3\pi}{2}\right)}\boxed{R_z\left(-\frac{3\pi}{2}\right)}$$

$e^{i7\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SHZ

Score $\frac{1101}{250}$ / 2 gates

$$q : \boxed{R_x\left(-\frac{\pi}{2}\right)}\boxed{R_z\left(\frac{\pi}{2}\right)}$$

$e^{i7\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Ry / $k = 0$

cx_rz_ry, xx_rz_ry

$e^{i0\pi/4}$ I

Score 0 / 0 gates



$e^{i0\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ Sx

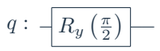
Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HX

Score $\frac{1101}{500}$ / 1 gates

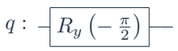


$e^{i0\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HZ

Score $\frac{1101}{500}$ / 1 gates



$e^{i0\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Ry / $k = 1$

cx_rz_ry, xx_rz_ry

$e^{i1\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Sdg

Score $\frac{1101}{500}$ / 1 gates

$$q : \boxed{R_z\left(-\tfrac{\pi}{2}\right)}$$

$e^{i1\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Sxdg

Score $\frac{3303}{500}$ / 3 gates

$$q : \boxed{R_y\left(-\tfrac{\pi}{2}\right)}\boxed{R_z\left(-\tfrac{\pi}{2}\right)}\boxed{R_y\left(\tfrac{\pi}{2}\right)}$$

$e^{i1\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HS

Score $\frac{1101}{250}$ / 2 gates

$$q : \boxed{R_y\left(-\tfrac{\pi}{2}\right)}\boxed{R_z\left(-\tfrac{\pi}{2}\right)}$$

$e^{i1\pi/4}$ HSX_q6

Score $\frac{1101}{250}$ / 2 gates

$$q : \boxed{R_y\left(\tfrac{\pi}{2}\right)}\boxed{R_z\left(-\tfrac{\pi}{2}\right)}$$

$e^{i1\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SH

Score $\frac{1101}{250}$ / 2 gates

$$q : \boxed{R_z\left(-\tfrac{\pi}{2}\right)}\boxed{R_y\left(\tfrac{\pi}{2}\right)}$$

$e^{i1\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SHY_q2

Score $\frac{1101}{250}$ / 2 gates

$$q : \boxed{R_z\left(-\tfrac{\pi}{2}\right)}\boxed{R_y\left(-\tfrac{\pi}{2}\right)}$$

$e^{i1\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Ry / $k = 2$

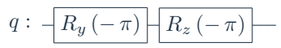
cx_rz_ry, xx_rz_ry

$e^{i2\pi/4}$ I

Impossible: phase outside the local determinant group.

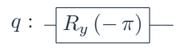
$e^{i2\pi/4}$ X

Score $\frac{1001}{250}$ / 2 gates



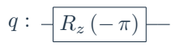
$e^{i2\pi/4}$ Y

Score $\frac{1001}{500}$ / 1 gates



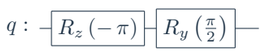
$e^{i2\pi/4}$ Z

Score $\frac{1001}{500}$ / 1 gates



$e^{i2\pi/4}$ H

Score $\frac{1051}{250}$ / 2 gates



$e^{i2\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ Sxdg

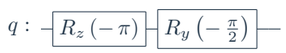
Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HY_q2

Score $\frac{1051}{250}$ / 2 gates



$e^{i2\pi/4}$ HZ

Impossible: phase outside the local determinant group.

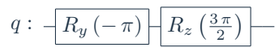
$e^{i2\pi/4}$ SX_q3

Score $\frac{1051}{250}$ / 2 gates



$e^{i2\pi/4}$ SY_q3

Score $\frac{1051}{250}$ / 2 gates



$e^{i2\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SHY_q2

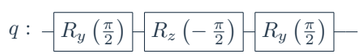
Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

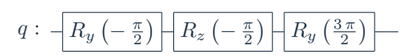
$e^{i2\pi/4}$ SxY_q3

Score $\frac{3303}{500}$ / 3 gates



$e^{i2\pi/4}$ SxZ_q3

Score $\frac{3303}{500}$ / 3 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Ry / $k = 3$

cx_rz_ry, xx_rz_ry

$e^{i3\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ Z

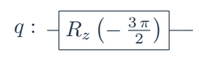
Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ S

Score $\frac{1101}{500}$ / 1 gates

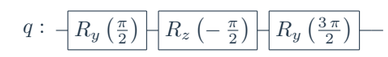


$e^{i3\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ Sx

Score $\frac{3303}{500}$ / 3 gates



$e^{i3\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

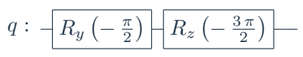
$e^{i3\pi/4}$ HSY

Score $\frac{1101}{250}$ / 2 gates



$e^{i3\pi/4}$ HSdg

Score $\frac{1101}{250}$ / 2 gates

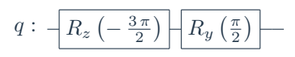


$e^{i3\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SHX

Score $\frac{1101}{250}$ / 2 gates

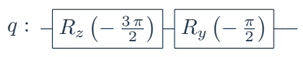


$e^{i3\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SHZ

Score $\frac{1101}{250}$ / 2 gates



$e^{i3\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

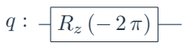
Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Ry / $k = 4$

cx_rz_ry, xx_rz_ry

$e^{i4\pi/4}$ I

Score $\frac{1001}{500}$ / 1 gates



$e^{i4\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Sx

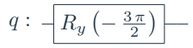
Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HX

Score $\frac{1101}{500}$ / 1 gates

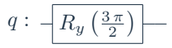


$e^{i4\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HZ

Score $\frac{1101}{500}$ / 1 gates



$e^{i4\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Ry / $k = 5$

cx_rz_ry, xx_rz_ry

$e^{i5\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Sdg

Score $\frac{1101}{500}$ / 1 gates

$$q : \boxed{R_z\left(\frac{3\pi}{2}\right)}$$

$e^{i5\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Sxdg

Score $\frac{3303}{500}$ / 3 gates

$$q : \boxed{R_y\left(-\frac{\pi}{2}\right)}\boxed{R_z\left(-\frac{\pi}{2}\right)}\boxed{R_y\left(-\frac{3\pi}{2}\right)}$$

$e^{i5\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HS

Score $\frac{1101}{250}$ / 2 gates

$$q : \boxed{R_y\left(-\frac{\pi}{2}\right)}\boxed{R_z\left(\frac{3\pi}{2}\right)}$$

$e^{i5\pi/4}$ HSX_q6

Score $\frac{1101}{250}$ / 2 gates

$$q : \boxed{R_y\left(\frac{\pi}{2}\right)}\boxed{R_z\left(\frac{3\pi}{2}\right)}$$

$e^{i5\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SH

Score $\frac{1101}{250}$ / 2 gates

$$q : \boxed{R_z\left(-\frac{\pi}{2}\right)}\boxed{R_y\left(-\frac{3\pi}{2}\right)}$$

$e^{i5\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SHY_q2

Score $\frac{1101}{250}$ / 2 gates

$$q : \boxed{R_z\left(-\frac{\pi}{2}\right)}\boxed{R_y\left(\frac{3\pi}{2}\right)}$$

$e^{i5\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Ry / $k = 6$

cx_rz_ry, xx_rz_ry

$e^{i6\pi/4}$ I

Impossible: phase outside the local determinant group.

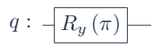
$e^{i6\pi/4}$ X

Score $\frac{1001}{250}$ / 2 gates



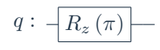
$e^{i6\pi/4}$ Y

Score $\frac{1001}{500}$ / 1 gates



$e^{i6\pi/4}$ Z

Score $\frac{1001}{500}$ / 1 gates



$e^{i6\pi/4}$ H

Score $\frac{1051}{250}$ / 2 gates



$e^{i6\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ Sxdg

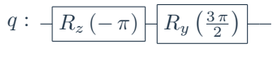
Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HY_q2

Score $\frac{1051}{250}$ / 2 gates



$e^{i6\pi/4}$ HZ

Impossible: phase outside the local determinant group.

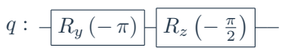
$e^{i6\pi/4}$ SX_q3

Score $\frac{1051}{250}$ / 2 gates



$e^{i6\pi/4}$ SY_q3

Score $\frac{1051}{250}$ / 2 gates



$e^{i6\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SHY_q2

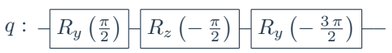
Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

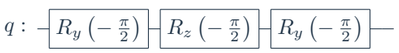
$e^{i6\pi/4}$ SxY_q3

Score $\frac{3303}{500}$ / 3 gates



$e^{i6\pi/4}$ SxZ_q3

Score $\frac{3303}{500}$ / 3 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Ry / $k = 7$

cx_rz_ry, xx_rz_ry

$e^{i7\pi/4}$ **I**

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ **X**

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ **Y**

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ **Z**

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ **H**

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ **S**

Score $\frac{1101}{500}$ / 1 gates

$$q : \boxed{R_z\left(\frac{\pi}{2}\right)}$$

$e^{i7\pi/4}$ **Sdg**

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ **Sx**

Score $\frac{3303}{500}$ / 3 gates

$$q : \boxed{R_y\left(\frac{\pi}{2}\right)}\boxed{R_z\left(-\frac{\pi}{2}\right)}\boxed{R_y\left(-\frac{\pi}{2}\right)}$$

$e^{i7\pi/4}$ **Sxdg**

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ **HX**

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ **HY_q2**

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ **HZ**

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ **SX_q3**

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ **SY_q3**

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ **HS**

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ **HSX_q6**

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ **HSY**

Score $\frac{1101}{250}$ / 2 gates

$$q : \boxed{R_y\left(\frac{\pi}{2}\right)}\boxed{R_z\left(\frac{\pi}{2}\right)}$$

$e^{i7\pi/4}$ **HSdg**

Score $\frac{1101}{250}$ / 2 gates

$$q : \boxed{R_y\left(-\frac{\pi}{2}\right)}\boxed{R_z\left(\frac{\pi}{2}\right)}$$

$e^{i7\pi/4}$ **SH**

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ **SHX**

Score $\frac{1101}{250}$ / 2 gates

$$q : \boxed{R_z\left(-\frac{3\pi}{2}\right)}\boxed{R_y\left(-\frac{3\pi}{2}\right)}$$

$e^{i7\pi/4}$ **SHY_q2**

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ **SHZ**

Score $\frac{1101}{250}$ / 2 gates

$$q : \boxed{R_z\left(-\frac{3\pi}{2}\right)}\boxed{R_y\left(\frac{3\pi}{2}\right)}$$

$e^{i7\pi/4}$ **SxY_q3**

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ **SxZ_q3**

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Sx|X / $k = 0$

ibm_cx, ibm_cz, ibm_ecr, rigetti_cz

$e^{i0\pi/4}$ **I**

Score 0 / 0 gates



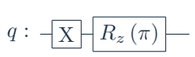
$e^{i0\pi/4}$ **X**

Score 2 / 1 gates



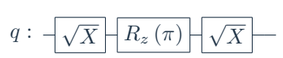
$e^{i0\pi/4}$ **Y**

Score $\frac{2001}{500}$ / 2 gates



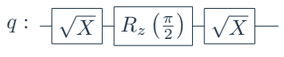
$e^{i0\pi/4}$ **Z**

Score $\frac{3001}{500}$ / 3 gates



$e^{i0\pi/4}$ **H**

Score $\frac{3101}{500}$ / 3 gates



$e^{i0\pi/4}$ **S**

Score $\frac{3101}{250}$ / 6 gates



$e^{i0\pi/4}$ **Sdg**

Score $\frac{2601}{250}$ / 5 gates



$e^{i0\pi/4}$ **Sx**

Score 2 / 1 gates



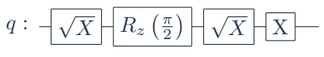
$e^{i0\pi/4}$ **Sxdg**

Score 4 / 2 gates



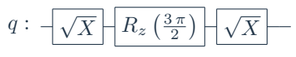
$e^{i0\pi/4}$ **HX**

Score $\frac{4101}{500}$ / 4 gates



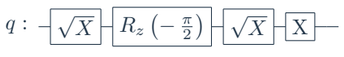
$e^{i0\pi/4}$ **HY_q2**

Score $\frac{3101}{500}$ / 3 gates



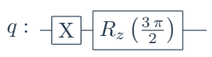
$e^{i0\pi/4}$ **HZ**

Score $\frac{4101}{500}$ / 4 gates



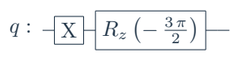
$e^{i0\pi/4}$ **SX_q3**

Score $\frac{2101}{500}$ / 2 gates



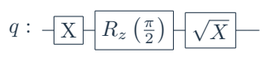
$e^{i0\pi/4}$ **SY_q3**

Score $\frac{2101}{500}$ / 2 gates



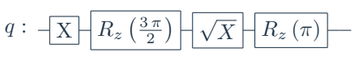
$e^{i0\pi/4}$ **HS**

Score $\frac{3101}{500}$ / 3 gates



$e^{i0\pi/4}$ **HSX_q6**

Score $\frac{2051}{250}$ / 4 gates



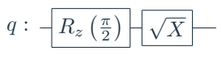
$e^{i0\pi/4}$ **HSY**

Score $\frac{1551}{250}$ / 3 gates



$e^{i0\pi/4}$ **HSdg**

Score $\frac{2101}{500}$ / 2 gates



$e^{i0\pi/4}$ **SH**

Score $\frac{3101}{500}$ / 3 gates



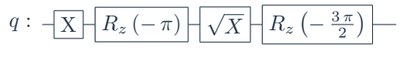
$e^{i0\pi/4}$ **SHX**

Score $\frac{2101}{500}$ / 2 gates



$e^{i0\pi/4}$ **SHY_q2**

Score $\frac{2051}{250}$ / 4 gates



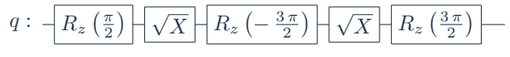
$e^{i0\pi/4}$ **SHZ**

Score $\frac{1551}{250}$ / 3 gates



$e^{i0\pi/4}$ **SxY_q3**

Score $\frac{5303}{500}$ / 5 gates



$e^{i0\pi/4}$ **SxZ_q3**

Score $\frac{5303}{500}$ / 5 gates



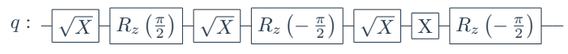
Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Sx|X / $k = 1$

ibm_cx, ibm_cz, ibm_ecr, rigetti_cz

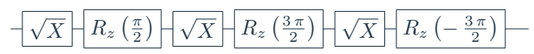
$e^{i1\pi/4}$ I

Score $\frac{7303}{500}$ / 7 gates



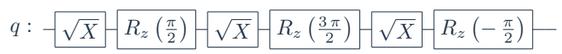
$e^{i1\pi/4}$ X

Score $\frac{6303}{500}$ / 6 gates



$e^{i1\pi/4}$ Y

Score $\frac{6303}{500}$ / 6 gates



$e^{i1\pi/4}$ Z

Score $\frac{6303}{500}$ / 6 gates



$e^{i1\pi/4}$ H

Score $\frac{2101}{250}$ / 4 gates



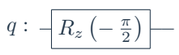
$e^{i1\pi/4}$ S

Score $\frac{2051}{250}$ / 4 gates



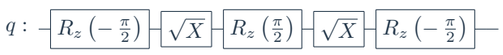
$e^{i1\pi/4}$ Sdg

Score $\frac{1101}{500}$ / 1 gates



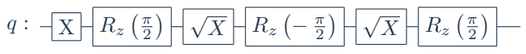
$e^{i1\pi/4}$ Sx

Score $\frac{5303}{500}$ / 5 gates



$e^{i1\pi/4}$ Sxdg

Score $\frac{6303}{500}$ / 6 gates



$e^{i1\pi/4}$ HX

Score $\frac{1601}{250}$ / 3 gates



$e^{i1\pi/4}$ HY_q2

Score $\frac{2101}{250}$ / 4 gates



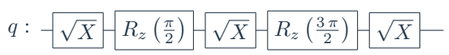
$e^{i1\pi/4}$ HZ

Score $\frac{1601}{250}$ / 3 gates



$e^{i1\pi/4}$ SX_q3

Score $\frac{2601}{250}$ / 5 gates



$e^{i1\pi/4}$ SY_q3

Score $\frac{2601}{250}$ / 5 gates



$e^{i1\pi/4}$ HS

Score $\frac{2601}{250}$ / 5 gates



$e^{i1\pi/4}$ HSX_q6

Score $\frac{2601}{250}$ / 5 gates



$e^{i1\pi/4}$ HSY

Score $\frac{2101}{250}$ / 4 gates



$e^{i1\pi/4}$ HSdg

Score $\frac{2101}{250}$ / 4 gates



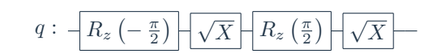
$e^{i1\pi/4}$ SH

Score $\frac{2601}{250}$ / 5 gates



$e^{i1\pi/4}$ SHX

Score $\frac{2101}{250}$ / 4 gates



$e^{i1\pi/4}$ SHY_q2

Score $\frac{2601}{250}$ / 5 gates



$e^{i1\pi/4}$ SHZ

Score $\frac{2101}{250}$ / 4 gates



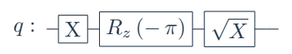
$e^{i1\pi/4}$ SxY_q3

Score $\frac{3001}{500}$ / 3 gates



$e^{i1\pi/4}$ SxZ_q3

Score $\frac{3001}{500}$ / 3 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Sx|X / $k = 2$

ibm_cx, ibm_cz, ibm_ecr, rigetti_cz

$e^{i2\pi/4}$ I

Score $\frac{2001}{250}$ / 4 gates



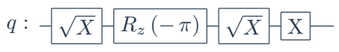
$e^{i2\pi/4}$ X

Score $\frac{2501}{250}$ / 5 gates



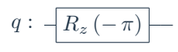
$e^{i2\pi/4}$ Y

Score $\frac{4001}{500}$ / 4 gates



$e^{i2\pi/4}$ Z

Score $\frac{1001}{500}$ / 1 gates



$e^{i2\pi/4}$ H

Score $\frac{2551}{250}$ / 5 gates



$e^{i2\pi/4}$ S

Score $\frac{2601}{250}$ / 5 gates



$e^{i2\pi/4}$ Sdg

Score $\frac{3101}{250}$ / 6 gates



$e^{i2\pi/4}$ Sx

Score $\frac{2001}{250}$ / 4 gates



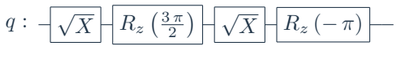
$e^{i2\pi/4}$ Sxdg

Score $\frac{1501}{250}$ / 3 gates



$e^{i2\pi/4}$ HX

Score $\frac{2051}{250}$ / 4 gates



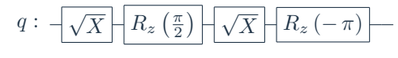
$e^{i2\pi/4}$ HY_q2

Score $\frac{2551}{250}$ / 5 gates



$e^{i2\pi/4}$ HZ

Score $\frac{2051}{250}$ / 4 gates



$e^{i2\pi/4}$ SX_q3

Score $\frac{2551}{250}$ / 5 gates



$e^{i2\pi/4}$ SY_q3

Score $\frac{2551}{250}$ / 5 gates



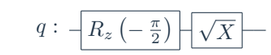
$e^{i2\pi/4}$ HS

Score $\frac{1551}{250}$ / 3 gates



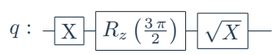
$e^{i2\pi/4}$ HSX_q6

Score $\frac{2101}{500}$ / 2 gates



$e^{i2\pi/4}$ HSY

Score $\frac{3101}{500}$ / 3 gates



$e^{i2\pi/4}$ HSdg

Score $\frac{2051}{250}$ / 4 gates



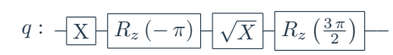
$e^{i2\pi/4}$ SH

Score $\frac{1551}{250}$ / 3 gates



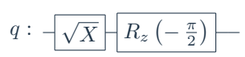
$e^{i2\pi/4}$ SHX

Score $\frac{2051}{250}$ / 4 gates



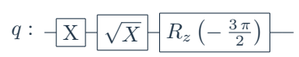
$e^{i2\pi/4}$ SHY_q2

Score $\frac{2101}{500}$ / 2 gates



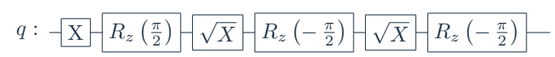
$e^{i2\pi/4}$ SHZ

Score $\frac{3101}{500}$ / 3 gates



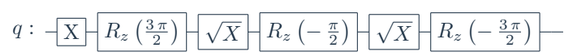
$e^{i2\pi/4}$ SxY_q3

Score $\frac{6303}{500}$ / 6 gates



$e^{i2\pi/4}$ SxZ_q3

Score $\frac{6303}{500}$ / 6 gates



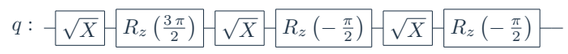
Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Sx|X / $k = 3$

ibm_cx, ibm_cz, ibm_ecr, rigetti_cz

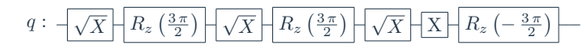
$e^{i3\pi/4}$ I

Score $\frac{6303}{500}$ / 6 gates



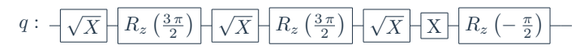
$e^{i3\pi/4}$ X

Score $\frac{7303}{500}$ / 7 gates



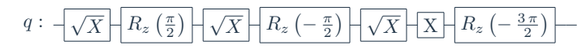
$e^{i3\pi/4}$ Y

Score $\frac{7303}{500}$ / 7 gates



$e^{i3\pi/4}$ Z

Score $\frac{7303}{500}$ / 7 gates



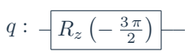
$e^{i3\pi/4}$ H

Score $\frac{1601}{250}$ / 3 gates



$e^{i3\pi/4}$ S

Score $\frac{1101}{500}$ / 1 gates



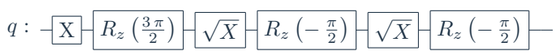
$e^{i3\pi/4}$ Sdg

Score $\frac{2051}{250}$ / 4 gates



$e^{i3\pi/4}$ Sx

Score $\frac{6303}{500}$ / 6 gates



$e^{i3\pi/4}$ Sxdg

Score $\frac{5303}{500}$ / 5 gates

$e^{i3\pi/4}$ HX

Score $\frac{2101}{250}$ / 4 gates

$e^{i3\pi/4}$ HY_q2

Score $\frac{1601}{250}$ / 3 gates

$e^{i3\pi/4}$ HZ

Score $\frac{2101}{250}$ / 4 gates

$e^{i3\pi/4}$ SX_q3

Score $\frac{3101}{250}$ / 6 gates



$e^{i3\pi/4}$ SY_q3

Score $\frac{3101}{250}$ / 6 gates



$e^{i3\pi/4}$ HS

Score $\frac{2101}{250}$ / 4 gates



$e^{i3\pi/4}$ HSX_q6

Score $\frac{2101}{250}$ / 4 gates



$e^{i3\pi/4}$ HSY

Score $\frac{2601}{250}$ / 5 gates



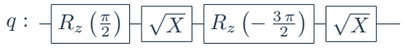
$e^{i3\pi/4}$ HSdg

Score $\frac{2601}{250}$ / 5 gates



$e^{i3\pi/4}$ SH

Score $\frac{2101}{250}$ / 4 gates



$e^{i3\pi/4}$ SHX

Score $\frac{2601}{250}$ / 5 gates



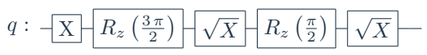
$e^{i3\pi/4}$ SHY_q2

Score $\frac{2101}{250}$ / 4 gates



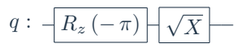
$e^{i3\pi/4}$ SHZ

Score $\frac{2601}{250}$ / 5 gates



$e^{i3\pi/4}$ SxY_q3

Score $\frac{2001}{500}$ / 2 gates



$e^{i3\pi/4}$ SxZ_q3

Score $\frac{2001}{500}$ / 2 gates



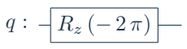
Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Sx|X / $k = 4$

ibm_cx, ibm_cz, ibm_ecr, rigetti_cz

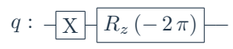
$e^{i4\pi/4}$ I

Score $\frac{1001}{500}$ / 1 gates



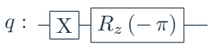
$e^{i4\pi/4}$ X

Score $\frac{2001}{500}$ / 2 gates



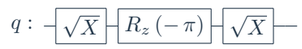
$e^{i4\pi/4}$ Y

Score $\frac{2001}{500}$ / 2 gates



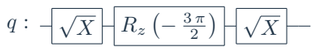
$e^{i4\pi/4}$ Z

Score $\frac{3001}{500}$ / 3 gates



$e^{i4\pi/4}$ H

Score $\frac{3101}{500}$ / 3 gates



$e^{i4\pi/4}$ S

Score $\frac{3101}{250}$ / 6 gates



$e^{i4\pi/4}$ Sdg

Score $\frac{2601}{250}$ / 5 gates



$e^{i4\pi/4}$ Sx

Score $\frac{2001}{500}$ / 2 gates



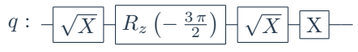
$e^{i4\pi/4}$ Sxdg

Score $\frac{3001}{500}$ / 3 gates



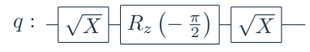
$e^{i4\pi/4}$ HX

Score $\frac{4101}{500}$ / 4 gates



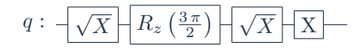
$e^{i4\pi/4}$ HY_q2

Score $\frac{3101}{500}$ / 3 gates



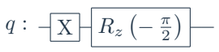
$e^{i4\pi/4}$ HZ

Score $\frac{4101}{500}$ / 4 gates



$e^{i4\pi/4}$ SX_q3

Score $\frac{2101}{500}$ / 2 gates



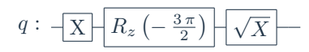
$e^{i4\pi/4}$ SY_q3

Score $\frac{2101}{500}$ / 2 gates



$e^{i4\pi/4}$ HS

Score $\frac{3101}{500}$ / 3 gates



$e^{i4\pi/4}$ HSX_q6

Score $\frac{2051}{250}$ / 4 gates



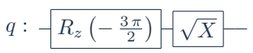
$e^{i4\pi/4}$ HSY

Score $\frac{1551}{250}$ / 3 gates



$e^{i4\pi/4}$ HSdg

Score $\frac{2101}{500}$ / 2 gates



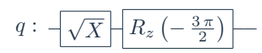
$e^{i4\pi/4}$ SH

Score $\frac{3101}{500}$ / 3 gates



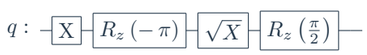
$e^{i4\pi/4}$ SHX

Score $\frac{2101}{500}$ / 2 gates



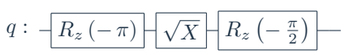
$e^{i4\pi/4}$ SHY_q2

Score $\frac{2051}{250}$ / 4 gates



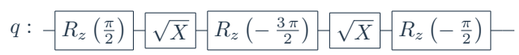
$e^{i4\pi/4}$ SHZ

Score $\frac{1551}{250}$ / 3 gates



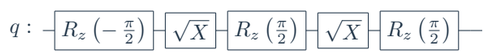
$e^{i4\pi/4}$ SxY_q3

Score $\frac{5303}{500}$ / 5 gates



$e^{i4\pi/4}$ SxZ_q3

Score $\frac{5303}{500}$ / 5 gates



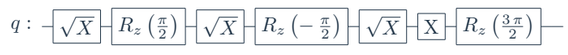
Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Sx|X / $k = 5$

ibm_cx, ibm_cz, ibm_ecr, rigetti_cz

$e^{i5\pi/4}$ I

Score $\frac{7303}{500}$ / 7 gates



$e^{i5\pi/4}$ X

Score $\frac{6303}{500}$ / 6 gates



$e^{i5\pi/4}$ Y

Score $\frac{6303}{500}$ / 6 gates



$e^{i5\pi/4}$ Z

Score $\frac{6303}{500}$ / 6 gates



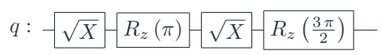
$e^{i5\pi/4}$ H

Score $\frac{2101}{250}$ / 4 gates



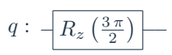
$e^{i5\pi/4}$ S

Score $\frac{2051}{250}$ / 4 gates



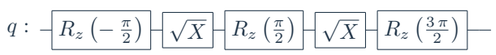
$e^{i5\pi/4}$ Sdg

Score $\frac{1101}{500}$ / 1 gates



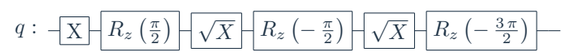
$e^{i5\pi/4}$ Sx

Score $\frac{5303}{500}$ / 5 gates



$e^{i5\pi/4}$ Sxdg

Score $\frac{6303}{500}$ / 6 gates



$e^{i5\pi/4}$ HX

Score $\frac{1601}{250}$ / 3 gates



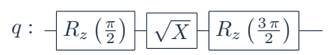
$e^{i5\pi/4}$ HY_q2

Score $\frac{2101}{250}$ / 4 gates



$e^{i5\pi/4}$ HZ

Score $\frac{1601}{250}$ / 3 gates



$e^{i5\pi/4}$ SX_q3

Score $\frac{2601}{250}$ / 5 gates



$e^{i5\pi/4}$ SY_q3

Score $\frac{2601}{250}$ / 5 gates



$e^{i5\pi/4}$ HS

Score $\frac{2601}{250}$ / 5 gates



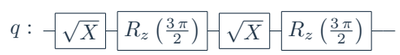
$e^{i5\pi/4}$ HSX_q6

Score $\frac{2601}{250}$ / 5 gates



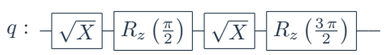
$e^{i5\pi/4}$ HSY

Score $\frac{2101}{250}$ / 4 gates



$e^{i5\pi/4}$ HSdg

Score $\frac{2101}{250}$ / 4 gates



$e^{i5\pi/4}$ SH

Score $\frac{2601}{250}$ / 5 gates



$e^{i5\pi/4}$ SHX

Score $\frac{2101}{250}$ / 4 gates



$e^{i5\pi/4}$ SHY_q2

Score $\frac{2601}{250}$ / 5 gates



$e^{i5\pi/4}$ SHZ

Score $\frac{2101}{250}$ / 4 gates



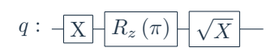
$e^{i5\pi/4}$ SxY_q3

Score $\frac{3001}{500}$ / 3 gates



$e^{i5\pi/4}$ SxZ_q3

Score $\frac{3001}{500}$ / 3 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Sx|X / $k = 6$

ibm_cx, ibm_cz, ibm_ecr, rigetti_cz

$e^{i6\pi/4}$ I

Score $\frac{2001}{250}$ / 4 gates



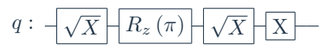
$e^{i6\pi/4}$ X

Score $\frac{2501}{250}$ / 5 gates



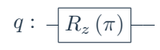
$e^{i6\pi/4}$ Y

Score $\frac{4001}{500}$ / 4 gates



$e^{i6\pi/4}$ Z

Score $\frac{1001}{500}$ / 1 gates



$e^{i6\pi/4}$ H

Score $\frac{2551}{250}$ / 5 gates



$e^{i6\pi/4}$ S

Score $\frac{2601}{250}$ / 5 gates



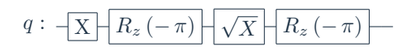
$e^{i6\pi/4}$ Sdg

Score $\frac{3101}{250}$ / 6 gates



$e^{i6\pi/4}$ Sx

Score $\frac{2001}{250}$ / 4 gates



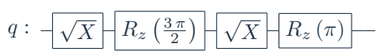
$e^{i6\pi/4}$ Sxdg

Score $\frac{1501}{250}$ / 3 gates



$e^{i6\pi/4}$ HX

Score $\frac{2051}{250}$ / 4 gates



$e^{i6\pi/4}$ HY_q2

Score $\frac{2551}{250}$ / 5 gates



$e^{i6\pi/4}$ HZ

Score $\frac{2051}{250}$ / 4 gates



$e^{i6\pi/4}$ SX_q3

Score $\frac{2551}{250}$ / 5 gates



$e^{i6\pi/4}$ SY_q3

Score $\frac{2551}{250}$ / 5 gates



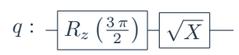
$e^{i6\pi/4}$ HS

Score $\frac{1551}{250}$ / 3 gates



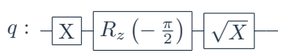
$e^{i6\pi/4}$ HSX_q6

Score $\frac{2101}{500}$ / 2 gates



$e^{i6\pi/4}$ HSY

Score $\frac{3101}{500}$ / 3 gates



$e^{i6\pi/4}$ HSdg

Score $\frac{2051}{250}$ / 4 gates



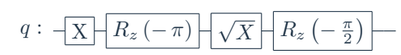
$e^{i6\pi/4}$ SH

Score $\frac{1551}{250}$ / 3 gates



$e^{i6\pi/4}$ SHX

Score $\frac{2051}{250}$ / 4 gates



$e^{i6\pi/4}$ SHY_q2

Score $\frac{2101}{500}$ / 2 gates



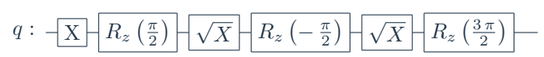
$e^{i6\pi/4}$ SHZ

Score $\frac{3101}{500}$ / 3 gates



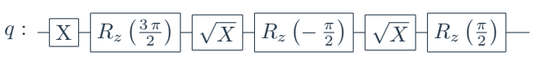
$e^{i6\pi/4}$ SxY_q3

Score $\frac{6303}{500}$ / 6 gates



$e^{i6\pi/4}$ SxZ_q3

Score $\frac{6303}{500}$ / 6 gates



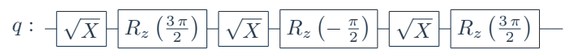
Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

Rz|Sx|X / $k = 7$

ibm_cx, ibm_cz, ibm_ecr, rigetti_cz

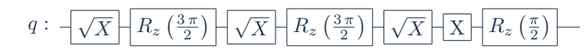
$e^{i7\pi/4}$ I

Score $\frac{6303}{500}$ / 6 gates



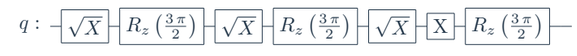
$e^{i7\pi/4}$ X

Score $\frac{7303}{500}$ / 7 gates



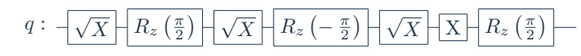
$e^{i7\pi/4}$ Y

Score $\frac{7303}{500}$ / 7 gates



$e^{i7\pi/4}$ Z

Score $\frac{7303}{500}$ / 7 gates



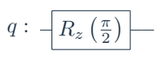
$e^{i7\pi/4}$ H

Score $\frac{1601}{250}$ / 3 gates



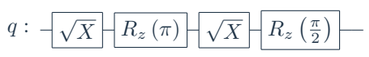
$e^{i7\pi/4}$ S

Score $\frac{1101}{500}$ / 1 gates



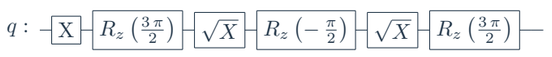
$e^{i7\pi/4}$ Sdg

Score $\frac{2051}{250}$ / 4 gates



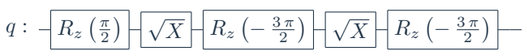
$e^{i7\pi/4}$ Sx

Score $\frac{6303}{500}$ / 6 gates



$e^{i7\pi/4}$ Sxdg

Score $\frac{5303}{500}$ / 5 gates



$e^{i7\pi/4}$ HX

Score $\frac{2101}{250}$ / 4 gates



$e^{i7\pi/4}$ HY_q2

Score $\frac{1601}{250}$ / 3 gates



$e^{i7\pi/4}$ HZ

Score $\frac{2101}{250}$ / 4 gates



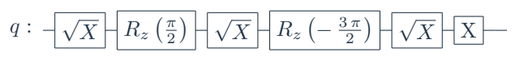
$e^{i7\pi/4}$ SX_q3

Score $\frac{3101}{250}$ / 6 gates



$e^{i7\pi/4}$ SY_q3

Score $\frac{3101}{250}$ / 6 gates



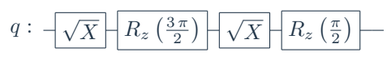
$e^{i7\pi/4}$ HS

Score $\frac{2101}{250}$ / 4 gates



$e^{i7\pi/4}$ HSX_q6

Score $\frac{2101}{250}$ / 4 gates



$e^{i7\pi/4}$ HSY

Score $\frac{2601}{250}$ / 5 gates



$e^{i7\pi/4}$ HSdg

Score $\frac{2601}{250}$ / 5 gates



$e^{i7\pi/4}$ SH

Score $\frac{2101}{250}$ / 4 gates



$e^{i7\pi/4}$ SHX

Score $\frac{2601}{250}$ / 5 gates



$e^{i7\pi/4}$ SHY_q2

Score $\frac{2101}{250}$ / 4 gates



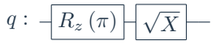
$e^{i7\pi/4}$ SHZ

Score $\frac{2601}{250}$ / 5 gates



$e^{i7\pi/4}$ SxY_q3

Score $\frac{2001}{500}$ / 2 gates



$e^{i7\pi/4}$ SxZ_q3

Score $\frac{2001}{500}$ / 2 gates



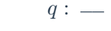
Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

R|Rz / $k = 0$

aqt_rxx

$e^{i0\pi/4}$ **I**

Score 0 / 0 gates



$e^{i0\pi/4}$ **X**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **Y**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **Z**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **H**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **S**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **Sdg**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **Sx**

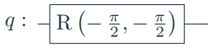
Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **Sxdg**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **HX**

Score $\frac{601}{250}$ / 1 gates

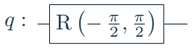


$e^{i0\pi/4}$ **HY_q2**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **HZ**

Score $\frac{601}{250}$ / 1 gates



$e^{i0\pi/4}$ **SX_q3**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **SY_q3**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **HS**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **HSX_q6**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **HSY**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **HSdg**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **SH**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **SHX**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **SHY_q2**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **SHZ**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **SxY_q3**

Impossible: phase outside the local determinant group.

$e^{i0\pi/4}$ **SxZ_q3**

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

R|Rz / $k = 1$

aqt_rxx

$e^{i1\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Sdg

Score $\frac{1101}{500}$ / 1 gates

$$q : \boxed{R_z\left(-\tfrac{\pi}{2}\right)}$$

$e^{i1\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ Sxdg

Score $\frac{551}{250}$ / 1 gates

$$q : \boxed{\mathrm{R}\left(-\tfrac{\pi}{2}, 0\right)}$$

$e^{i1\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HS

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{R_z\left(-\tfrac{\pi}{2}\right)}\boxed{\mathrm{R}\left(-\tfrac{\pi}{2}, 0\right)}$$

$e^{i1\pi/4}$ HSX_q6

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{R_z\left(-\tfrac{\pi}{2}\right)}\boxed{\mathrm{R}\left(-\tfrac{\pi}{2}, -\pi\right)}$$

$e^{i1\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SH

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{\mathrm{R}\left(-\tfrac{\pi}{2}, 0\right)}\boxed{R_z\left(-\tfrac{\pi}{2}\right)}$$

$e^{i1\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SHY_q2

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{\mathrm{R}\left(-\tfrac{3\pi}{2}, 0\right)}\boxed{R_z\left(\tfrac{3\pi}{2}\right)}$$

$e^{i1\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i1\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

R|Rz / $k = 2$

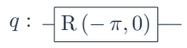
aqt_rxx

$e^{i2\pi/4}$ I

Impossible: phase outside the local determinant group.

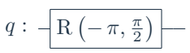
$e^{i2\pi/4}$ X

Score $\frac{501}{250}$ / 1 gates



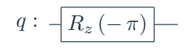
$e^{i2\pi/4}$ Y

Score $\frac{551}{250}$ / 1 gates



$e^{i2\pi/4}$ Z

Score $\frac{1001}{500}$ / 1 gates



$e^{i2\pi/4}$ H

Score $\frac{2203}{500}$ / 2 gates



$e^{i2\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HY_q2

Score $\frac{2203}{500}$ / 2 gates

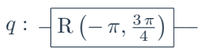


$e^{i2\pi/4}$ HZ

Impossible: phase outside the local determinant group.

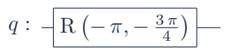
$e^{i2\pi/4}$ SX_q3

Score $\frac{601}{250}$ / 1 gates



$e^{i2\pi/4}$ SY_q3

Score $\frac{601}{250}$ / 1 gates



$e^{i2\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i2\pi/4}$ SxY_q3

Score $\frac{2103}{500}$ / 2 gates



$e^{i2\pi/4}$ SxZ_q3

Score $\frac{2103}{500}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

R|Rz / $k = 3$

aqt_rxx

$e^{i3\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ Z

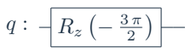
Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ S

Score $\frac{1101}{500}$ / 1 gates

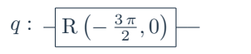


$e^{i3\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ Sx

Score $\frac{551}{250}$ / 1 gates



$e^{i3\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ HSY

Score $\frac{2203}{500}$ / 2 gates



$e^{i3\pi/4}$ HSdg

Score $\frac{2203}{500}$ / 2 gates

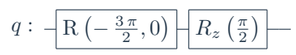


$e^{i3\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SHX

Score $\frac{2203}{500}$ / 2 gates

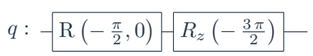


$e^{i3\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SHZ

Score $\frac{2203}{500}$ / 2 gates



$e^{i3\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i3\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

R|Rz / $k = 4$

aqt_rxx

$e^{i4\pi/4}$ I

Score $\frac{1001}{500}$ / 1 gates

$q:$ —$\boxed{R_z\left(-2\,\pi\right)}$—

$e^{i4\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HX

Score $\frac{601}{250}$ / 1 gates

$q:$ —$\boxed{\mathrm{R}\left(-\frac{3\,\pi}{2}, \frac{\pi}{2}\right)}$—

$e^{i4\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HZ

Score $\frac{601}{250}$ / 1 gates

$q:$ —$\boxed{\mathrm{R}\left(-\frac{3\,\pi}{2}, -\frac{\pi}{2}\right)}$—

$e^{i4\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i4\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

R|Rz / $k = 5$

aqt_rxx

$e^{i5\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Sdg

Score $\frac{1101}{500}$ / 1 gates

$$q : \boxed{R_z\left(\frac{3\pi}{2}\right)}$$

$e^{i5\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ Sxdg

Score $\frac{551}{250}$ / 1 gates

$$q : \boxed{\mathrm{R}\left(-\frac{3\pi}{2}, -\pi\right)}$$

$e^{i5\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HS

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{R_z\left(-\frac{\pi}{2}\right)}\boxed{\mathrm{R}\left(-\frac{3\pi}{2}, -\pi\right)}$$

$e^{i5\pi/4}$ HSX_q6

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{R_z\left(-\frac{\pi}{2}\right)}\boxed{\mathrm{R}\left(-\frac{3\pi}{2}, 0\right)}$$

$e^{i5\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SH

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{\mathrm{R}\left(-\frac{\pi}{2}, 0\right)}\boxed{R_z\left(\frac{3\pi}{2}\right)}$$

$e^{i5\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SHY_q2

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{\mathrm{R}\left(-\frac{3\pi}{2}, 0\right)}\boxed{R_z\left(-\frac{\pi}{2}\right)}$$

$e^{i5\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i5\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

R|Rz / $k = 6$

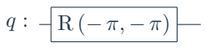
aqt_rxx

$e^{i6\pi/4}$ I

Impossible: phase outside the local determinant group.

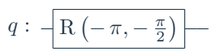
$e^{i6\pi/4}$ X

Score $\frac{501}{250}$ / 1 gates



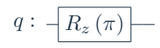
$e^{i6\pi/4}$ Y

Score $\frac{551}{250}$ / 1 gates



$e^{i6\pi/4}$ Z

Score $\frac{1001}{500}$ / 1 gates



$e^{i6\pi/4}$ H

Score $\frac{2203}{500}$ / 2 gates



$e^{i6\pi/4}$ S

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ Sx

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HY_q2

Score $\frac{2203}{500}$ / 2 gates



$e^{i6\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SX_q3

Score $\frac{601}{250}$ / 1 gates



$e^{i6\pi/4}$ SY_q3

Score $\frac{601}{250}$ / 1 gates



$e^{i6\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HSY

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ HSdg

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SHX

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SHZ

Impossible: phase outside the local determinant group.

$e^{i6\pi/4}$ SxY_q3

Score $\frac{2103}{500}$ / 2 gates



$e^{i6\pi/4}$ SxZ_q3

Score $\frac{2103}{500}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

R|Rz / $k = 7$

aqt_rxx

$e^{i7\pi/4}$ I

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ X

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ Y

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ Z

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ H

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ S

Score $\frac{1101}{500}$ / 1 gates

$$q : \boxed{R_z\left(\frac{\pi}{2}\right)}$$

$e^{i7\pi/4}$ Sdg

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ Sx

Score $\frac{551}{250}$ / 1 gates

$$q : \boxed{\mathrm{R}\left(-\frac{\pi}{2}, -\pi\right)}$$

$e^{i7\pi/4}$ Sxdg

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HX

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HY_q2

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HZ

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SX_q3

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SY_q3

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HS

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HSX_q6

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ HSY

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{R_z\left(-\frac{3\pi}{2}\right)}\boxed{\mathrm{R}\left(-\frac{3\pi}{2}, -\pi\right)}$$

$e^{i7\pi/4}$ HSdg

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{R_z\left(-\frac{3\pi}{2}\right)}\boxed{\mathrm{R}\left(-\frac{3\pi}{2}, 0\right)}$$

$e^{i7\pi/4}$ SH

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SHX

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{\mathrm{R}\left(-\frac{3\pi}{2}, 0\right)}\boxed{R_z\left(-\frac{3\pi}{2}\right)}$$

$e^{i7\pi/4}$ SHY_q2

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SHZ

Score $\frac{2203}{500}$ / 2 gates

$$q : \boxed{\mathrm{R}\left(-\frac{\pi}{2}, 0\right)}\boxed{R_z\left(\frac{\pi}{2}\right)}$$

$e^{i7\pi/4}$ SxY_q3

Impossible: phase outside the local determinant group.

$e^{i7\pi/4}$ SxZ_q3

Impossible: phase outside the local determinant group.

Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

U3 / $k = 0$

berkeley_b, cx_u, cz_u, iswap_u, sqrt_iswap_u, sqrt_swap_u, xx90_u, zx_u, zz_u

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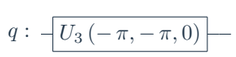
$e^{i0\pi/4}$ I

Score 0 / 0 gates



$e^{i0\pi/4}$ X

Score $\frac{1003}{500}$ / 1 gates



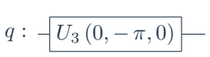
$e^{i0\pi/4}$ Y

Score $\frac{1203}{500}$ / 1 gates



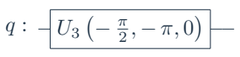
$e^{i0\pi/4}$ Z

Score $\frac{1003}{500}$ / 1 gates



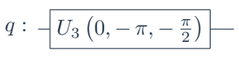
$e^{i0\pi/4}$ H

Score $\frac{1103}{500}$ / 1 gates



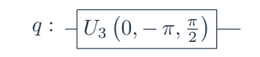
$e^{i0\pi/4}$ S

Score $\frac{1103}{500}$ / 1 gates



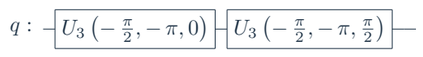
$e^{i0\pi/4}$ Sdg

Score $\frac{1103}{500}$ / 1 gates



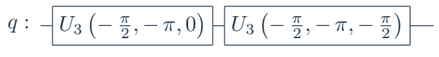
$e^{i0\pi/4}$ Sx

Score $\frac{1153}{250}$ / 2 gates



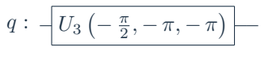
$e^{i0\pi/4}$ Sxdg

Score $\frac{1153}{250}$ / 2 gates



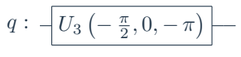
$e^{i0\pi/4}$ HX

Score $\frac{1103}{500}$ / 1 gates



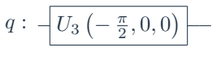
$e^{i0\pi/4}$ HY_q2

Score $\frac{1103}{500}$ / 1 gates



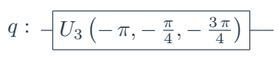
$e^{i0\pi/4}$ HZ

Score $\frac{1103}{500}$ / 1 gates



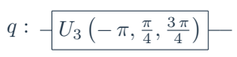
$e^{i0\pi/4}$ SX_q3

Score $\frac{1403}{500}$ / 1 gates



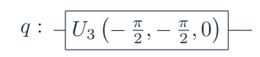
$e^{i0\pi/4}$ SY_q3

Score $\frac{1403}{500}$ / 1 gates



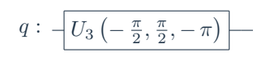
$e^{i0\pi/4}$ HS

Score $\frac{1203}{500}$ / 1 gates



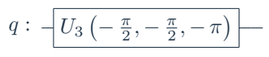
$e^{i0\pi/4}$ HSX_q6

Score $\frac{1203}{500}$ / 1 gates



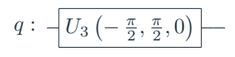
$e^{i0\pi/4}$ HSY

Score $\frac{1203}{500}$ / 1 gates



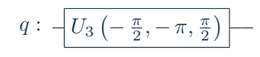
$e^{i0\pi/4}$ HSdg

Score $\frac{1203}{500}$ / 1 gates



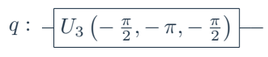
$e^{i0\pi/4}$ SH

Score $\frac{1203}{500}$ / 1 gates



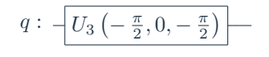
$e^{i0\pi/4}$ SHX

Score $\frac{1203}{500}$ / 1 gates



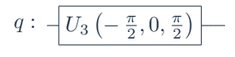
$e^{i0\pi/4}$ SHY_q2

Score $\frac{1203}{500}$ / 1 gates



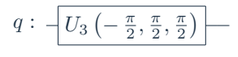
$e^{i0\pi/4}$ SHZ

Score $\frac{1203}{500}$ / 1 gates



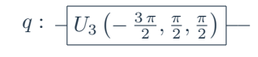
$e^{i0\pi/4}$ SxY_q3

Score $\frac{1303}{500}$ / 1 gates



$e^{i0\pi/4}$ SxZ_q3

Score $\frac{1303}{500}$ / 1 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

U3 / $k = 1$

berkeley_b, cx_u, cz_u, iswap_u, sqrt_iswap_u, sqrt_swap_u, xx90_u, zx_u, zz_u

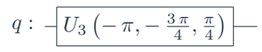
$e^{i1\pi/4}$ I

Score $\frac{1203}{250}$ / 2 gates



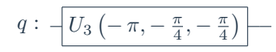
$e^{i1\pi/4}$ X

Score $\frac{1403}{500}$ / 1 gates



$e^{i1\pi/4}$ Y

Score $\frac{1403}{500}$ / 1 gates



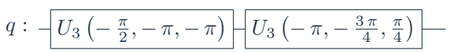
$e^{i1\pi/4}$ Z

Score $\frac{1203}{250}$ / 2 gates



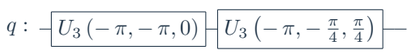
$e^{i1\pi/4}$ H

Score $\frac{1253}{250}$ / 2 gates



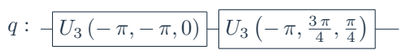
$e^{i1\pi/4}$ S

Score $\frac{1203}{250}$ / 2 gates



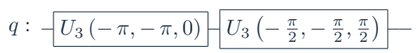
$e^{i1\pi/4}$ Sdg

Score $\frac{1203}{250}$ / 2 gates



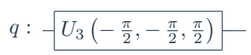
$e^{i1\pi/4}$ Sx

Score $\frac{1153}{250}$ / 2 gates



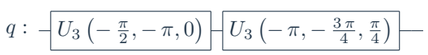
$e^{i1\pi/4}$ Sxdg

Score $\frac{1303}{500}$ / 1 gates



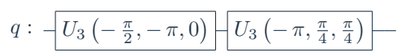
$e^{i1\pi/4}$ HX

Score $\frac{1253}{250}$ / 2 gates



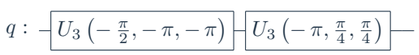
$e^{i1\pi/4}$ HY_q2

Score $\frac{1253}{250}$ / 2 gates



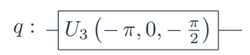
$e^{i1\pi/4}$ HZ

Score $\frac{1253}{250}$ / 2 gates



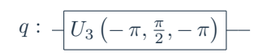
$e^{i1\pi/4}$ SX_q3

Score $\frac{1103}{500}$ / 1 gates



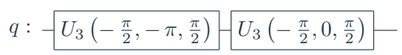
$e^{i1\pi/4}$ SY_q3

Score $\frac{1103}{500}$ / 1 gates



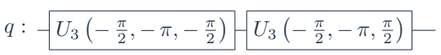
$e^{i1\pi/4}$ HS

Score $\frac{1203}{250}$ / 2 gates



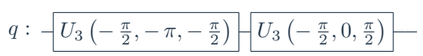
$e^{i1\pi/4}$ HSX_q6

Score $\frac{1203}{250}$ / 2 gates



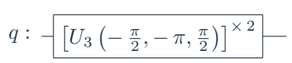
$e^{i1\pi/4}$ HSY

Score $\frac{1203}{250}$ / 2 gates



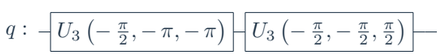
$e^{i1\pi/4}$ HSdg

Score $\frac{1203}{250}$ / 2 gates



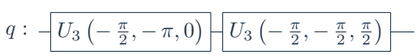
$e^{i1\pi/4}$ SH

Score $\frac{1203}{250}$ / 2 gates



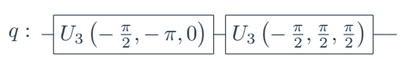
$e^{i1\pi/4}$ SHX

Score $\frac{1203}{250}$ / 2 gates



$e^{i1\pi/4}$ SHY_q2

Score $\frac{1203}{250}$ / 2 gates



$e^{i1\pi/4}$ SHZ

Score $\frac{1203}{250}$ / 2 gates



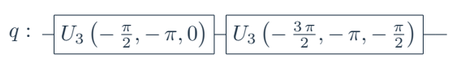
$e^{i1\pi/4}$ SxY_q3

Score $\frac{1153}{250}$ / 2 gates



$e^{i1\pi/4}$ SxZ_q3

Score $\frac{1153}{250}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

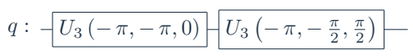
U3 / $k = 2$

berkeley_b, cx_u, cz_u, iswap_u, sqrt_iswap_u, sqrt_swap_u, xx90_u, zx_u, zz_u

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$e^{i2\pi/4}$ I

Score $\frac{1103}{250}$ / 2 gates



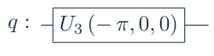
$e^{i2\pi/4}$ X

Score $\frac{1203}{500}$ / 1 gates



$e^{i2\pi/4}$ Y

Score $\frac{1003}{500}$ / 1 gates



$e^{i2\pi/4}$ Z

Score $\frac{1103}{250}$ / 2 gates



$e^{i2\pi/4}$ H

Score $\frac{1153}{250}$ / 2 gates



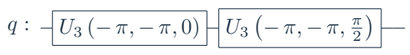
$e^{i2\pi/4}$ S

Score $\frac{1053}{250}$ / 2 gates



$e^{i2\pi/4}$ Sdg

Score $\frac{1053}{250}$ / 2 gates



$e^{i2\pi/4}$ Sx

Score $\frac{1153}{250}$ / 2 gates



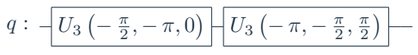
$e^{i2\pi/4}$ Sxdg

Score $\frac{1153}{250}$ / 2 gates



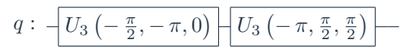
$e^{i2\pi/4}$ HX

Score $\frac{1153}{250}$ / 2 gates



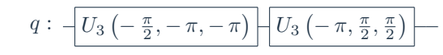
$e^{i2\pi/4}$ HY_q2

Score $\frac{1153}{250}$ / 2 gates



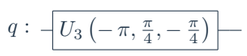
$e^{i2\pi/4}$ HZ

Score $\frac{1153}{250}$ / 2 gates



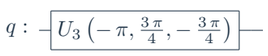
$e^{i2\pi/4}$ SX_q3

Score $\frac{1403}{500}$ / 1 gates



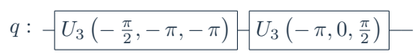
$e^{i2\pi/4}$ SY_q3

Score $\frac{1403}{500}$ / 1 gates



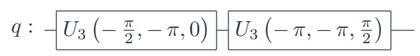
$e^{i2\pi/4}$ HS

Score $\frac{1103}{250}$ / 2 gates



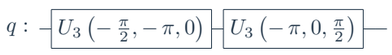
$e^{i2\pi/4}$ HSX_q6

Score $\frac{1103}{250}$ / 2 gates



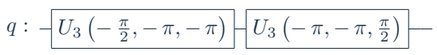
$e^{i2\pi/4}$ HSY

Score $\frac{1103}{250}$ / 2 gates



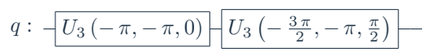
$e^{i2\pi/4}$ HSdg

Score $\frac{1103}{250}$ / 2 gates



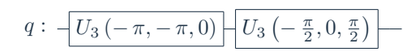
$e^{i2\pi/4}$ SH

Score $\frac{1103}{250}$ / 2 gates



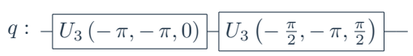
$e^{i2\pi/4}$ SHX

Score $\frac{1103}{250}$ / 2 gates



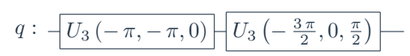
$e^{i2\pi/4}$ SHY_q2

Score $\frac{1103}{250}$ / 2 gates



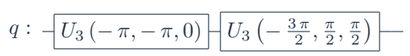
$e^{i2\pi/4}$ SHZ

Score $\frac{1103}{250}$ / 2 gates



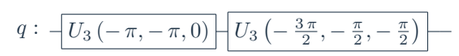
$e^{i2\pi/4}$ SxY_q3

Score $\frac{1153}{250}$ / 2 gates



$e^{i2\pi/4}$ SxZ_q3

Score $\frac{1153}{250}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

U3 / $k = 3$

berkeley_b, cx_u, cz_u, iswap_u, sqrt_iswap_u, sqrt_swap_u, xx90_u, zx_u, zz_u

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$e^{i3\pi/4}$ I

Score $\frac{1203}{250}$ / 2 gates



$e^{i3\pi/4}$ X

Score $\frac{1403}{500}$ / 1 gates



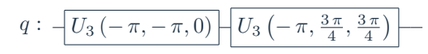
$e^{i3\pi/4}$ Y

Score $\frac{1403}{500}$ / 1 gates



$e^{i3\pi/4}$ Z

Score $\frac{1203}{250}$ / 2 gates



$e^{i3\pi/4}$ H

Score $\frac{1253}{250}$ / 2 gates



$e^{i3\pi/4}$ S

Score $\frac{1203}{250}$ / 2 gates



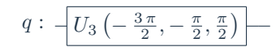
$e^{i3\pi/4}$ Sdg

Score $\frac{1203}{250}$ / 2 gates



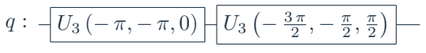
$e^{i3\pi/4}$ Sx

Score $\frac{1303}{500}$ / 1 gates



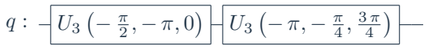
$e^{i3\pi/4}$ Sxdg

Score $\frac{1153}{250}$ / 2 gates



$e^{i3\pi/4}$ HX

Score $\frac{1253}{250}$ / 2 gates



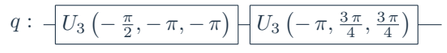
$e^{i3\pi/4}$ HY_q2

Score $\frac{1253}{250}$ / 2 gates



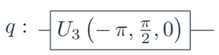
$e^{i3\pi/4}$ HZ

Score $\frac{1253}{250}$ / 2 gates



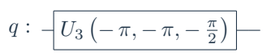
$e^{i3\pi/4}$ SX_q3

Score $\frac{1103}{500}$ / 1 gates



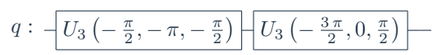
$e^{i3\pi/4}$ SY_q3

Score $\frac{1103}{500}$ / 1 gates



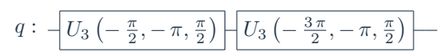
$e^{i3\pi/4}$ HS

Score $\frac{1203}{250}$ / 2 gates



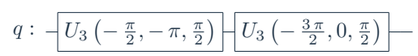
$e^{i3\pi/4}$ HSX_q6

Score $\frac{1203}{250}$ / 2 gates



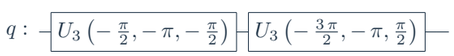
$e^{i3\pi/4}$ HSY

Score $\frac{1203}{250}$ / 2 gates



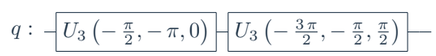
$e^{i3\pi/4}$ HSdg

Score $\frac{1203}{250}$ / 2 gates



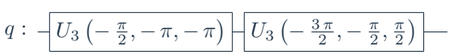
$e^{i3\pi/4}$ SH

Score $\frac{1203}{250}$ / 2 gates



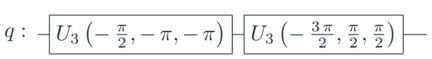
$e^{i3\pi/4}$ SHX

Score $\frac{1203}{250}$ / 2 gates



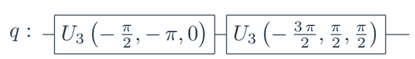
$e^{i3\pi/4}$ SHY_q2

Score $\frac{1203}{250}$ / 2 gates



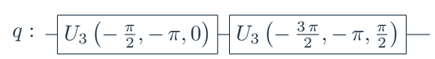
$e^{i3\pi/4}$ SHZ

Score $\frac{1203}{250}$ / 2 gates



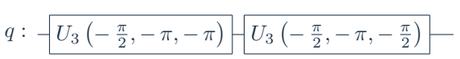
$e^{i3\pi/4}$ SxY_q3

Score $\frac{1153}{250}$ / 2 gates



$e^{i3\pi/4}$ SxZ_q3

Score $\frac{1153}{250}$ / 2 gates



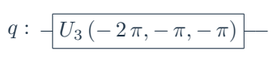
Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

U3 / $k = 4$

berkeley_b, cx_u, cz_u, iswap_u, sqrt_iswap_u, sqrt_swap_u, xx90_u, zx_u, zz_u

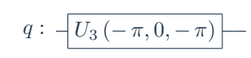
$e^{i4\pi/4}$ I

Score $\frac{1003}{500}$ / 1 gates



$e^{i4\pi/4}$ X

Score $\frac{1003}{500}$ / 1 gates



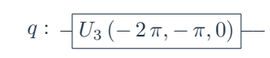
$e^{i4\pi/4}$ Y

Score $\frac{1203}{500}$ / 1 gates



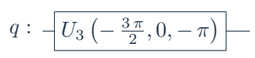
$e^{i4\pi/4}$ Z

Score $\frac{1003}{500}$ / 1 gates



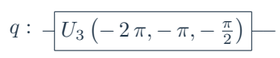
$e^{i4\pi/4}$ H

Score $\frac{1103}{500}$ / 1 gates



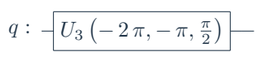
$e^{i4\pi/4}$ S

Score $\frac{1103}{500}$ / 1 gates



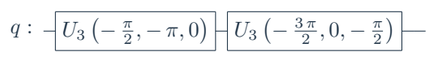
$e^{i4\pi/4}$ Sdg

Score $\frac{1103}{500}$ / 1 gates



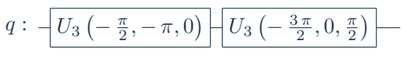
$e^{i4\pi/4}$ Sx

Score $\frac{1153}{250}$ / 2 gates



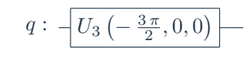
$e^{i4\pi/4}$ Sxdg

Score $\frac{1153}{250}$ / 2 gates



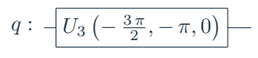
$e^{i4\pi/4}$ HX

Score $\frac{1103}{500}$ / 1 gates



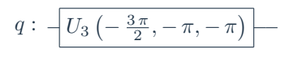
$e^{i4\pi/4}$ HY_q2

Score $\frac{1103}{500}$ / 1 gates



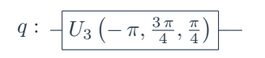
$e^{i4\pi/4}$ HZ

Score $\frac{1103}{500}$ / 1 gates



$e^{i4\pi/4}$ SX_q3

Score $\frac{1403}{500}$ / 1 gates



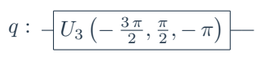
$e^{i4\pi/4}$ SY_q3

Score $\frac{1403}{500}$ / 1 gates



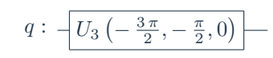
$e^{i4\pi/4}$ HS

Score $\frac{1203}{500}$ / 1 gates



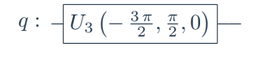
$e^{i4\pi/4}$ HSX_q6

Score $\frac{1203}{500}$ / 1 gates



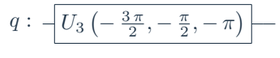
$e^{i4\pi/4}$ HSY

Score $\frac{1203}{500}$ / 1 gates



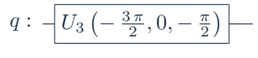
$e^{i4\pi/4}$ HSdg

Score $\frac{1203}{500}$ / 1 gates



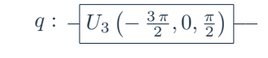
$e^{i4\pi/4}$ SH

Score $\frac{1203}{500}$ / 1 gates



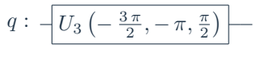
$e^{i4\pi/4}$ SHX

Score $\frac{1203}{500}$ / 1 gates



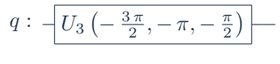
$e^{i4\pi/4}$ SHY_q2

Score $\frac{1203}{500}$ / 1 gates



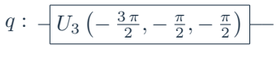
$e^{i4\pi/4}$ SHZ

Score $\frac{1203}{500}$ / 1 gates



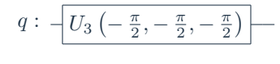
$e^{i4\pi/4}$ SxY_q3

Score $\frac{1303}{500}$ / 1 gates



$e^{i4\pi/4}$ SxZ_q3

Score $\frac{1303}{500}$ / 1 gates



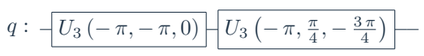
Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

U3 / $k = 5$

berkeley_b, cx_u, cz_u, iswap_u, sqrt_iswap_u, sqrt_swap_u, xx90_u, zx_u, zz_u

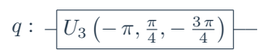
$e^{i5\pi/4}$ I

Score $\frac{1203}{250}$ / 2 gates



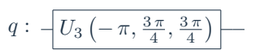
$e^{i5\pi/4}$ X

Score $\frac{1403}{500}$ / 1 gates



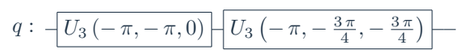
$e^{i5\pi/4}$ Y

Score $\frac{1403}{500}$ / 1 gates



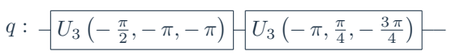
$e^{i5\pi/4}$ Z

Score $\frac{1203}{250}$ / 2 gates



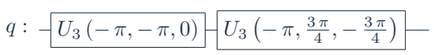
$e^{i5\pi/4}$ H

Score $\frac{1253}{250}$ / 2 gates



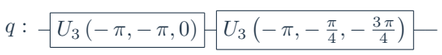
$e^{i5\pi/4}$ S

Score $\frac{1203}{250}$ / 2 gates



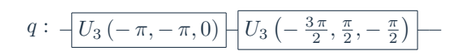
$e^{i5\pi/4}$ Sdg

Score $\frac{1203}{250}$ / 2 gates



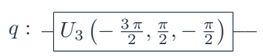
$e^{i5\pi/4}$ Sx

Score $\frac{1153}{250}$ / 2 gates



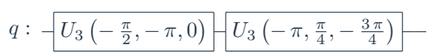
$e^{i5\pi/4}$ Sxdg

Score $\frac{1303}{500}$ / 1 gates



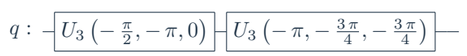
$e^{i5\pi/4}$ HX

Score $\frac{1253}{250}$ / 2 gates



$e^{i5\pi/4}$ HY_q2

Score $\frac{1253}{250}$ / 2 gates



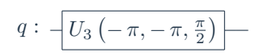
$e^{i5\pi/4}$ HZ

Score $\frac{1253}{250}$ / 2 gates



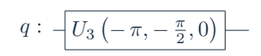
$e^{i5\pi/4}$ SX_q3

Score $\frac{1103}{500}$ / 1 gates



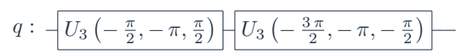
$e^{i5\pi/4}$ SY_q3

Score $\frac{1103}{500}$ / 1 gates



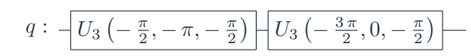
$e^{i5\pi/4}$ HS

Score $\frac{1203}{250}$ / 2 gates



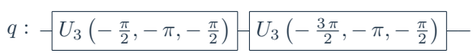
$e^{i5\pi/4}$ HSX_q6

Score $\frac{1203}{250}$ / 2 gates



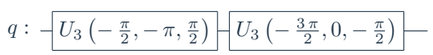
$e^{i5\pi/4}$ HSY

Score $\frac{1203}{250}$ / 2 gates



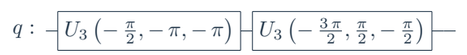
$e^{i5\pi/4}$ HSdg

Score $\frac{1203}{250}$ / 2 gates



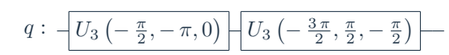
$e^{i5\pi/4}$ SH

Score $\frac{1203}{250}$ / 2 gates



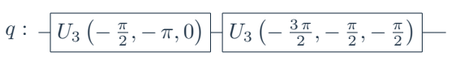
$e^{i5\pi/4}$ SHX

Score $\frac{1203}{250}$ / 2 gates



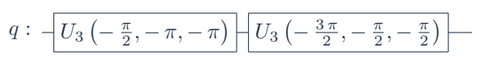
$e^{i5\pi/4}$ SHY_q2

Score $\frac{1203}{250}$ / 2 gates



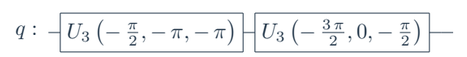
$e^{i5\pi/4}$ SHZ

Score $\frac{1203}{250}$ / 2 gates



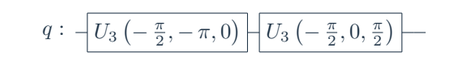
$e^{i5\pi/4}$ SxY_q3

Score $\frac{1153}{250}$ / 2 gates



$e^{i5\pi/4}$ SxZ_q3

Score $\frac{1153}{250}$ / 2 gates



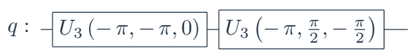
Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

U3 / $k = 6$

berkeley_b, cx_u, cz_u, iswap_u, sqrt_iswap_u, sqrt_swap_u, xx90_u, zx_u, zz_u

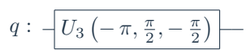
$e^{i6\pi/4}$ I

Score $\frac{1103}{250}$ / 2 gates



$e^{i6\pi/4}$ X

Score $\frac{1203}{500}$ / 1 gates



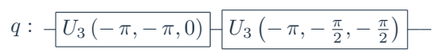
$e^{i6\pi/4}$ Y

Score $\frac{1003}{500}$ / 1 gates



$e^{i6\pi/4}$ Z

Score $\frac{1103}{250}$ / 2 gates



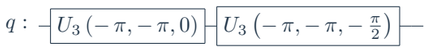
$e^{i6\pi/4}$ H

Score $\frac{1153}{250}$ / 2 gates



$e^{i6\pi/4}$ S

Score $\frac{1053}{250}$ / 2 gates



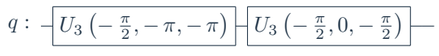
$e^{i6\pi/4}$ Sdg

Score $\frac{1053}{250}$ / 2 gates



$e^{i6\pi/4}$ Sx

Score $\frac{1153}{250}$ / 2 gates



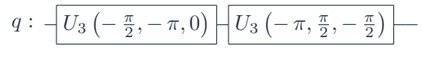
$e^{i6\pi/4}$ Sxdg

Score $\frac{1153}{250}$ / 2 gates



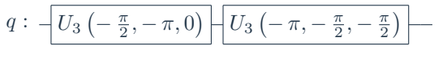
$e^{i6\pi/4}$ HX

Score $\frac{1153}{250}$ / 2 gates



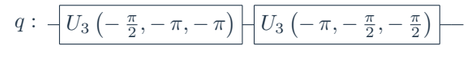
$e^{i6\pi/4}$ HY_q2

Score $\frac{1153}{250}$ / 2 gates



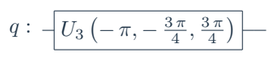
$e^{i6\pi/4}$ HZ

Score $\frac{1153}{250}$ / 2 gates



$e^{i6\pi/4}$ SX_q3

Score $\frac{1403}{500}$ / 1 gates



$e^{i6\pi/4}$ SY_q3

Score $\frac{1403}{500}$ / 1 gates



$e^{i6\pi/4}$ HS

Score $\frac{1103}{250}$ / 2 gates



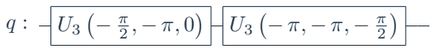
$e^{i6\pi/4}$ HSX_q6

Score $\frac{1103}{250}$ / 2 gates



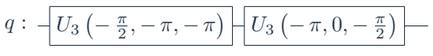
$e^{i6\pi/4}$ HSY

Score $\frac{1103}{250}$ / 2 gates



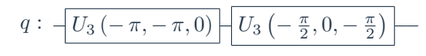
$e^{i6\pi/4}$ HSdg

Score $\frac{1103}{250}$ / 2 gates



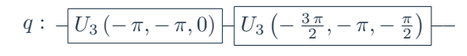
$e^{i6\pi/4}$ SH

Score $\frac{1103}{250}$ / 2 gates



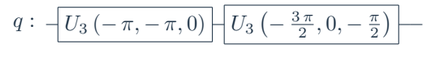
$e^{i6\pi/4}$ SHX

Score $\frac{1103}{250}$ / 2 gates



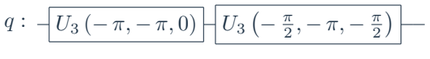
$e^{i6\pi/4}$ SHY_q2

Score $\frac{1103}{250}$ / 2 gates



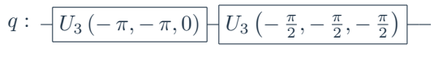
$e^{i6\pi/4}$ SHZ

Score $\frac{1103}{250}$ / 2 gates



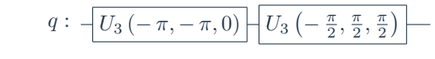
$e^{i6\pi/4}$ SxY_q3

Score $\frac{1153}{250}$ / 2 gates



$e^{i6\pi/4}$ SxZ_q3

Score $\frac{1153}{250}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.

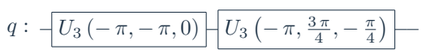
U3 / $k = 7$

berkeley_b, cx_u, cz_u, iswap_u, sqrt_iswap_u, sqrt_swap_u, xx90_u, zx_u, zz_u

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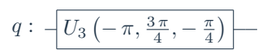
$e^{i7\pi/4}$ I

Score $\frac{1203}{250}$ / 2 gates



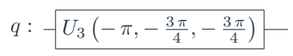
$e^{i7\pi/4}$ X

Score $\frac{1403}{500}$ / 1 gates



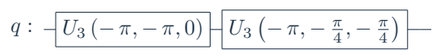
$e^{i7\pi/4}$ Y

Score $\frac{1403}{500}$ / 1 gates



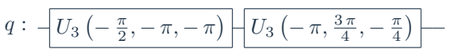
$e^{i7\pi/4}$ Z

Score $\frac{1203}{250}$ / 2 gates



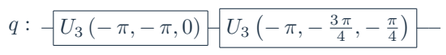
$e^{i7\pi/4}$ H

Score $\frac{1253}{250}$ / 2 gates



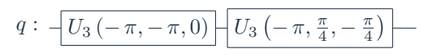
$e^{i7\pi/4}$ S

Score $\frac{1203}{250}$ / 2 gates



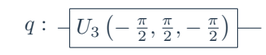
$e^{i7\pi/4}$ Sdg

Score $\frac{1203}{250}$ / 2 gates



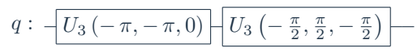
$e^{i7\pi/4}$ Sx

Score $\frac{1303}{500}$ / 1 gates



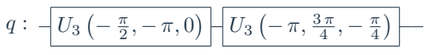
$e^{i7\pi/4}$ Sxdg

Score $\frac{1153}{250}$ / 2 gates



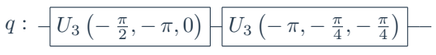
$e^{i7\pi/4}$ HX

Score $\frac{1253}{250}$ / 2 gates



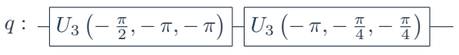
$e^{i7\pi/4}$ HY_q2

Score $\frac{1253}{250}$ / 2 gates



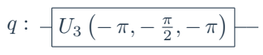
$e^{i7\pi/4}$ HZ

Score $\frac{1253}{250}$ / 2 gates



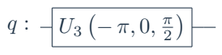
$e^{i7\pi/4}$ SX_q3

Score $\frac{1103}{500}$ / 1 gates



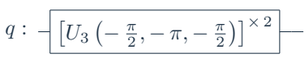
$e^{i7\pi/4}$ SY_q3

Score $\frac{1103}{500}$ / 1 gates



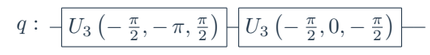
$e^{i7\pi/4}$ HS

Score $\frac{1203}{250}$ / 2 gates



$e^{i7\pi/4}$ HSX_q6

Score $\frac{1203}{250}$ / 2 gates



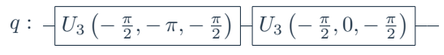
$e^{i7\pi/4}$ HSY

Score $\frac{1203}{250}$ / 2 gates



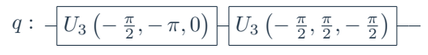
$e^{i7\pi/4}$ HSdg

Score $\frac{1203}{250}$ / 2 gates



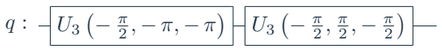
$e^{i7\pi/4}$ SH

Score $\frac{1203}{250}$ / 2 gates



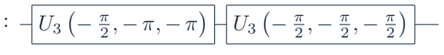
$e^{i7\pi/4}$ SHX

Score $\frac{1203}{250}$ / 2 gates



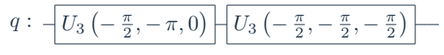
$e^{i7\pi/4}$ SHY_q2

Score $\frac{1203}{250}$ / 2 gates



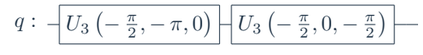
$e^{i7\pi/4}$ SHZ

Score $\frac{1203}{250}$ / 2 gates



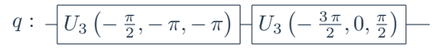
$e^{i7\pi/4}$ SxY_q3

Score $\frac{1153}{250}$ / 2 gates



$e^{i7\pi/4}$ SxZ_q3

Score $\frac{1153}{250}$ / 2 gates



Certificate scope: listed_clifford_preserving_words_only. Phase is part of the target matrix.